30. Ohtori S, Inoue G, Mannoji C et al: Shock wave application to rat skin induces degeneration and reinnervation of sensory nerve fibers, *Neurosci Lett* 315:57-60, 2001.
31. Maier M, Averbeck B, Milz S et al: Substance P and prostaglandin E_2 release after shock wave application to the rabbit femur, *Clin Orthop Rel Res* 406:237-245, 2003.
32. Haake M, Thon A, Bette M: Unchanged c-Fos expression after extracorporeal shock wave therapy: an experimental investigation in rats, *Arch Orthop Trauma Surg* 122:518-521, 2002.
33. Haake M, Thon A, Bette M: Absence of spinal response to extracorporeal shock waves on the endogenous opioid systems in the rat, *Ultrasound Med Biol* 27:279-284, 2001.
34. Haake M, Thon A, Bette M: No influence of low-energy extracorporeal shock waves on spinal nociceptive systems: immune-histochemical analysis of substance P/CGRP, *J Orthop Sci* 7:97-101, 2002.
35. Murata R, Ohtori S, Ochiai N et al: Extracorporeal shockwaves induce the expression of ATF3 and GAP-43 in rat dorsal root ganglion neurons, *Auto Neurosci* 128:96-100, 2006.
36. Takahashi M, Ohtori S, Saisu T et al: Second application of low-energy shock waves has a cumulative effect on free nerve endings, *Clin Orthop Rel Res* 443:315-319, 2006.
37. Abed JM, McClure SR, Yaeger MJ et al: Immunohistochemical evaluation of substance P and calcitonin generelated peptide in skin and periosteum after extracorporeal shock wave therapy and radial pressure wave therapy in sheep, *Am J Vet Res* 68:323-328, 2007.
38. Francis DA, Millis DL, Evans M et al: Clinical evaluation of extracorporeal shock wave therapy for management of canine osteoarthritis of the elbow and hip joint. *Proceedings of the 31st Annual Conference Veterinary Orthopedic Society,* Big Sky, Montana, February 22-27, 2004.
39. Millis DL, Drum M, Whitlock D: Complementary use of extracorporeal shock wave therapy on elbow osteoarthritis in dogs, *Vet Comp Orthop Traumatol* 24(3):A1, 2011.
40. Moretti B, Iannone F, Notarnicola A et al: Extracorporeal shock waves down-regulate the expression of interleukin-10 and tumor necrosis factor-alpha in osteoarthritic chondrocytes, *BMC Musculoskelet Disord* 9:16, 2008.
41. Zhao Z, Ji H, Jing R et al: Extracorporeal shock-wave therapy reduces progression of knee osteoarthritis in rabbits by reducing nitric oxide level and chondrocyte apoptosis, *Arch Orthop Trauma Surg* 132:1147-1153, 2012.
42. Wang CJ, Weng LH, Ko JY et al: Extracorporeal shockwave therapy shows chondroprotective effects in osteoarthritic rat knee, *Arch Orthop Trauma Surg* 131:1153-1158, 2011.
43. Wang CJ, Weng LH, Ko JY et al: Extracorporeal shockwave shows regression of osteoarthritis of the knee in rats, *J Surg Res* 171:601-608, 2011.
44. Wang CJ, Sun YC, Wong T et al: Extracorporeal shockwave therapy shows time-dependent chondroprotective effects in osteoarthritis of the knee in rats, *J Surg Res* 178:196-205, 2012.
45. Kawcak CE, Frisbie DD, McIlwraith CW: Effects of extracorporeal shock wave therapy and polysulfated glycosaminoglycan treatment on subchondral bone, serum biomarkers, and synovial fluid biomarkers in horses with induced osteoarthritis, *Am J Vet Res* 72:772-779, 2011.
46. Ochiai N, Ohtori S, Sasho T et al: Extracorporeal shock wave therapy improves motor dysfunction and pain originating from knee osteoarthritis in rats, *Osteoarthritis Cartilage* 15:1093-1096, 2007.
47. Väterlein N, Lüssenhop S, Hahn M et al: The effect of extracorporeal shock waves on joint cartilage—an in vivo study in rabbits, *Arch Orthop Trauma Surg* 120:403-406, 2000.
48. Dorotka R, Kubista B, Schatz KD et al: Effects of extracorporeal shock waves on human articular chondrocytes and ovine bone marrow stromal cells in vitro, *Arch Orthop Trauma Surg* 123:345-348, 2003.
49. Mayer-Wagner S, Ernst J, Maier M et al: The effect of high-energy extracorporeal shock waves on hyaline cartilage of adult rats in vivo, *J Orthop Res* 28:1050-1056, 2010.
50. Caminoto EH, Alves ALG, Amorim RL et al: Ultrastructural and immune-cytochemical evaluation of the effects of extracorporeal shock wave treatment in the hind limbs of horses with experimentally induced suspensory ligament desmitis, *Am J Vet Res* 66:892-896, 2005.
51. Wang CJ, Huang Hy, Pai HC: Shock wave-enhanced neovascularization at the tendon-bone junction an experiment in dogs, *J Foot Ankle Surgery* 14:17-21, 2001.
52. Wang CJ, Wang FS, Yang KD et al: Shock wave therapy induces neovascularization at the tendon-bone junction—a study in rabbits, *J Orthop Res* 21:984-989, 2003.
53. Hsu RW, Hsu WH, Tai CL: Effect of shock-wave therapy on patellar tendonopathy in a rabbit model, *J Orthop Res* 22:221-227, 2004.
54. Kersh KD, McClure SR, Van Sickle D et al: The evaluation of extracorporeal shock wave therapy on collagenase induced superficial digital flexor tendonitis, *Vet Comp Orthop Tramatol* 19:99-105, 2006.
55. Bosch G, Lin YL, van Schie HTM et al: Effect of extracorporeal shock wave therapy on the biochemical composition and metabolic activity of tenocytes in normal tendinous structures in ponies, *Equine Vet J* 39:226-231, 2007.
56. Rompe JD, Kirkpatrick CJ, Kullmer K et al: Dose-related effects of shock waves on rabbit tendo Achillis—a sonographic and histological study, *J Bone Joint Surg Br* 80:546-552, 1998.
57. Maier M, Tischer T, Milz S et al: Dose-related effects of extracorporeal shock waves on rabbit quadriceps tendon integrity, *Arch Orthop Trauma Surg* 122:436-441, 2002.
58. Maier M, Saisu T, Beckmann J et al: Impaired tensile strength after shock-wave application in an animal model of tendon calcification, *Ultrasound Med Biol* 27:665-671, 2001.
59. Gremion G, Gobelet C, Siegrist O et al: Efficacy of radial shock wave therapy in the chronic patellar tendonitis, *Proceedings, 3rd Biennial Congress of the ISAKOS* 2001.
60. Lohrer H, Schoell J, Arentz S et al: Effectiveness of radial shockwave therapy (RSWT) on tennis elbow and plantar fasciitis, *Proceedings, Annual Symposium* 2001.
61. Danova NA, Muir P: Extracorporeal shock wave therapy for supraspinatus calcifying tendonopathy in two dogs, *Vet Rec* 152:208-209, 2003.

62. Gallagher A, Cross AR, Sepulveda G: The effect of shock wave therapy on patellar ligament desmitis after tibial plateau leveling osteotomy, *Vet Surg* 41:482-485, 2012.
63. Kuo UR, Wu WS, Hsieh YL et al: Extracorporeal shock wave enhanced extended skin flap tissue survival via increase of topical blood perfusion and associated with suppression of tissue pro-inflammation, *J Surg Res* 143:385-392, 2007.
64. Schaden W, Thiele R, Kölpl C et al: Shock wave therapy for acute and chronic soft tissue wounds: a feasibility study, *J Surg Res* 143:1-12, 2007.
65. Haupt G, Haupt A, Ekkernkamp A et al: Influence of shock waves on fracture healing, *Urology* 39:529-532, 1992.
66. O'Leary JM, Knothe Tate ML, Knothe U: Extracorporeal shock waves generate new bone in skeletally mature rats. In *Proceedings, 52nd Annual Meeting of Orthopaedic Research Society*, 2006.
67. Hsu RWW, Tai CL, Chen CYC et al: Enhancing mechanical strength during early fracture healing vial shockwave treatment: an animal study, *Clin Biomech* 18:S33-S39, 2003.
68. Wang FS, Yang KD, Chen RF et al: Extracorporeal shock wave promotes growth and differentiation of bone-marrow stromal cells towards osteoprogenitors associated with induction of TGF-β1, *J Bone Joint Surg Br* 84-B:457-461, 2002.
69. Wang FS, Yang KD, Kuo YR et al: Temporal and spatial expression of bone morphogenetic proteins in extracorporeal shock wave-promoted healing of segmental defect, *Bone* 32:387-396, 2003.
70. Chen YJ, Wurtz T, Wang CJ et al: Recruitment of mesenchymal stem cells and expression of TGF-β_1 and VEGF in the early stage of shock wave-promoted bone regeneration of segmental defect in rats, *J Orthop Res* 22:526-534, 2004.
71. Chen YJ, Kuo YR, Yang KD et al: Shock wave application enhances pertussis toxin protein-sensitive bone formation of segmental femoral defect in rats, *J Bone Miner Res* 18:2169-2179, 2003.
72. Chen YJ, Kuo YR, Yang KD et al: Activation of extracellular signal-regulated kinase (ERK) and p38 kinase in shock wave-promoted bone formation of segmental defect in rats, *Bone* 34:466-477, 2004.
73. Sukul DMKSK, Johannes EJ, Pierik EGJM et al: Effect of high energy shock waves focused on cortical bone: an in vitro study, *J Surg Res* 54:46-51, 1993.
74. Maier M, Milz S, Tischer T et al: Influence of extracorporeal shock-wave application on normal bone in an animal model in vivo, *J Bone Joint Surg Br* 84-B:592-599, 2002.
75. Da Costa Gómez TM, Radtke CL, Kalscheur VL et al: Effect of focused and radial extracorporeal shock wave therapy on equine bone microdamage, *Vet Surg* 33:49-55, 2004.
76. Ringer SK, Lischer CJ, Ueltschi G: Assessment of scintigraphic and thermographic changes after focused extracorporeal shock wave therapy on the origin of the suspensory ligament and the fourth metatarsal bone in horses without lameness, *Am J Vet Res* 66:1836-1842, 2005.
77. Bischofberger AS, Ringer SK, Geyer H et al: Histomorphologic evaluation of extracorporeal shock wave therapy of the fourth metatarsal bone and the origin of the suspensory ligament in horses without lameness, *Am J Vet Res* 67:577-582, 2006.
78. Wang CJ, Huang HY, Chen HH et al: Effect of shock wave therapy on acute fractures of the tibia, *Clin Orthop Rel Res* 387:112-118, 2001.
79. Wang CJ, Yang KD, Wang FS et al: Shock wave treatment shows dose-dependent enhancement of bone mass and bone strength after fracture of femur, *Bone* 34:225-230, 2004.
80. Saisu T, Takahashi K, Kamegaya M et al: Effects of extracorporeal shock waves on immature rabbit femurs, *J Pediatr Orthop B* 13:176-183, 2004.
81. Wang CJ, Wang FS, Huang CC et al: Treatment for osteonecrosis of the femoral head: comparison of extracorporeal shock waves with core decompression and bone-grafting, *J Bone Joint Surg* 87-A:2380-2387, 2005.
82. Ma HZ, Zeng BF, Li XL: Upregulation of VEGF in subchondral bone of necrotic femoral heads in rabbits with use of extracorporeal shock waves, *Calcif Tissue Int* 81:124-131, 2007.
83. Gollwitzer H, Roessner M, Langer R et al: Safety and effectiveness of extracorporeal shockwave therapy: results of a rabbit model of chronic osteomyelitis, *Ultrasound Med Biol* 35(4):595-602, 2009.
84. Uwatoku T, Ito K, Abe K et al: Extracorporeal cardiac shock wave therapy improves left ventricular remodeling after acute myocardial infarction in pigs, *Coron Artery Dis* 18:397-404, 2007.
85. Nishida T, Shimokawa H, Oi K et al: Extracorporeal cardiac shock wave therapy markedly ameliorates ischemia-induced myocardial dysfunction in pigs in vivo, *Circulation* 110: 3055-3061, 2004.
86. Fukomoto Y, Ito A, Uwatoku T et al: Extracorporeal cardiac shock wave therapy ameliorates myocardial ischemia in patients with severe coronary artery disease, *Coron Artery Dis* 17:63-70, 2006.
87. Khattab AA, Brodersen B, Scuermann-Kuchenbrandt D et al: Extracorporeal cardiac shock wave therapy: first experience in the everyday practice for treatment of chronic refractory angina pectoris, *Int J Cardiol* 121:84-85, 2007.
88. Belcaro G, Nicolaides AN, Marlinghaus EH et al: Shock waves in vascular diseases: an in-vitro study, *Angiology* 49:777-788, 1998.
89. Aicher A, Heeschen C, Sasaki K et al: Low-energy shock wave for enhancing recruitment of endothelial progenitor cells: a new modality to increase efficacy of cell therapy in chronic hind limb ischemia, *Circulation* 114:2823-2830, 2006.
90. Rompe, JD, Meurer A, Nafe B et al: Repetitive low-energy shock wave application without local anesthesia is more efficient than repetitive low-energy shock wave application with local anesthesia in the treatment of chronic plantar fasciitis, *J Orthop Res* 23:931-941, 2005.
91. Yeaman LD, Jerome CP, McCullough DL: Effects of shock waves on the structure and growth of the immature rat epiphysis, *J Urol* 141:670-674, 1989.
92. Nassenstein K, Nassenstein I, Schleberger R: Effects of high-energy shock waves on the structure of the immature

epiphysis—a histomorphological study, *Z Orthop Ihre Grenzgeb* 143:652-655, 2005.

93. Van Arsdalen KN, Kruzweil S, Smith J et al: Effect of lithotripsy on immature rabbit bone and kidney development, *J Urol* 146:213-216, 1991.
94. Lüssenhop S, Seemann D, Hahn M et al: The influence of shock waves on epiphyseal growth plates: first results of an in-vivo study with rabbits. In Siebert W, Buch M, editors: *Extracorporeal shockwaves in orthopaedics*, Berlin, 1997, Springer.
95. Mariotto S, Cavalieri E, Amelio E et al: Extracorporeal shock waves: from lithotripsy to anti-inflammatory action by NO production, *Nitric Oxide* 12:89-96, 2005.
96. Lee TC, Huang HY, Yang YL et al: Vulnerability of the spinal cord to injury from extracorporeal shock waves in rabbits, *J Clin Neurosci* 14:873-878, 2007.
97. Gümüş B, Lekili M, Kandiloğlu AR et al: Effects of extracorporeal shockwave lithotripsy at different stages of pregnancy in the rabbit, *J Endourol* 11:323-326, 1997.
98. Miller D, Dou C, Song J: Lithotripter shockwave-induced enhancement of mouse melanoma lung metastasis: dependence on cavitation nucleation, *J Endourol* 18:925-929, 2004.
99. Oosterhof GON, Cornel EB, Smits GAHJ et al: The influence of high-energy shock waves on the development of metastases, *Ultrasound Med Biol* 22:339-344, 1996.
100. Kambe M, Ioritani N, Shirai S et al: Enhancement of chemotherapeutic effects with focused shock waves: extracorporeal shock wave chemotherapy (ESWC), *In Vivo* 10: 369-376, 1996.
101. Holmes RP, Yeaman LI, Li WJ et al: The combined effects of shock waves and Cisplatin therapy on rat prostate tumors, *J Urol* 144:159-163, 1990.
102. Miller DL, Song J: Lithotripter shock waves with cavitation nucleation agents produce tumor growth reduction and gene transfer in vivo, *Ultrasound Med Biol* 28:1343-1348, 2002.

23 Other Modalities in Veterinary Rehabilitation

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Several alternative and complementary modalities have been used in veterinary rehabilitation, including static magnetic fields, pulsed electromagnetic fields (PEMFs), and shortwave diathermy. Other modalities that have been used in rehabilitation include acupuncture and chiropractic. Acupuncture and chiropractic have been described in detail elsewhere, have their own training programs, and are stand-alone disciplines and are not described here.

Static Magnet Therapy

Magnet therapy has been used as an alternative therapy in human medicine for many years and has gained popularity more recently in veterinary medicine. Magnets used for therapeutic purposes are either static in nature or electromagnets. Static magnets have magnetic fields that do not change, whereas electromagnets generate magnetic fields only when electrical current flows through them. Magnet strength is measured in units of gauss (G) or Tesla (T) ($1\ T = 10^4\ G$). The earth's magnetic field is approximately 0.5 G, whereas magnetic resonance imaging machines use magnets of 15,000 G and higher. Static magnets for therapeutic purposes are usually in the 300- to 5000-G range.

History

Magnet therapy in the treatment of various medical conditions dates back as early as 200 AD with Greek healers reportedly using magnetic rings as a treatment for arthritis.[1] During the Middle Ages, magnets were thought to be effective in the treatment of baldness, wounds, arthritis, and gout.[1] In the early 1600s, an English physician, Thomas Browne, may have been one of the first to scientifically evaluate magnets. He concluded that the use of magnets in the treatment of such diseases as gout, headache, and venereal disease amounted to wishful thinking.[1] Because of its continued popularity as an alternative therapeutic modality and the business that it generates, therapeutic efficacy has been studied in various models.

Magnets continue to be in use today, with a myriad of products available claiming various therapeutic values. An Internet search[2] of magnetic therapy for dogs yields products such as beds, collars, and jackets, among others, that claim to have beneficial effects on pain, blood flow, tissue oxygenation, bone and tissue regeneration, inflammation, and sleep. The current literature is sparse, however, and little convincing evidence exists for these claims. For all of these applications, increased blood flow is one theory provided to explain why magnets may have an effect. At this time no solid explanation for why magnets may have a physiologic effect seems to be available. Unfortunately, only weak evidence exists, not only for the effects of pain alleviation and bone or wound healing, but also for the theorized cause, blood flow changes. When studies have shown a potential physiologic effect, the mechanism causing the effect is often poorly explained or may be more strongly associated with a placebo effect.

Effects on Blood Flow

The Lorentz force law states that a magnetic field exerts a force on a moving ionic current.[3] The Hall effect further states that when a magnetic field is placed perpendicular to the direction of flow of an electric current, it will tend to deflect and separate the charged ions.[3] Ions will be deflected in opposite directions depending on the magnetic pole encountered and the charge of the ion.[3] In theory, when a magnetic field with a series of alternating north and south poles is placed over a blood vessel, the influence of the field may cause positive and negative ions to bounce back and forth between the sides of the vessel, creating currents in moving blood similar to those in a river.[3] The combination of the electromotive force, altered ionic pattern, and currents may potentially cause blood vessel dilation with a corresponding increase in blood flow.[4] However, magnetic forces are smaller than the physical forces already present caused by the heart muscle.[3]

The iron in hemoglobin causes blood to be only weakly repelled or attracted by magnets.[5] If they were attracted to a magnet this could have a clotting effect negatively affecting circulation.[3] Static magnets, to be effective, must overcome the strong physical forces of blood flow propelled by the heart and normal thermal induced Brownian random movement of the particles suspended in the blood, which is unlikely.[3]

Increased flow rates of concentrated saline have been observed within a glass capillary tube under the influence of a magnetic field.[3] The reason for the increased flow is not explained but because the capillary cannot expand, it cannot be related to vasodilation.[3] However, the investigator of this study warns that these findings should not be extrapolated to tissue perfusion.[6] Many studies have shown static magnets to be ineffective in altering circulation both in animal and human studies regardless of the level of magnetic field.[3]

In a study performed on six horses, the effect of static magnetic therapy on blood flow was assessed in the metacarpal region.[7] Red blood cells were labeled with Tc 99m phosphate and dorsal images of both metacarpi were acquired with a gamma camera. One metacarpus was treated with a magnetic wrap with 270 G at its surface, and the other metacarpus was treated with an identical wrap in which the magnets had been replaced with identical size and weighted polytetrafluoroethylene sheet. The metacarpi were treated for 48 hours and the scintigraphy was repeated. No significant difference was noted in perfusion of the metacarpal region between groups. The authors noted that in the commercially available wraps used in this study, the strength of the magnetic field dropped off to 0.5 G at 7 mm from the surface of the magnet, making it unlikely that the underlying blood vessels were actually exposed to a clinically relevant magnetic field. Static magnetic fields have also been shown to have no significant effect on mucosal blood flow in humans,[8] soft tissue healing in rats,[9] or blood flow to the skin of rats.[10]

Pain Alleviation

A variety of theories exist attempting to explain how magnets may provide pain relief. Likewise, a wide variety of protocols with varying magnetic strengths is observed in the literature. One theory to explain pain relief is associated with increased or altered blood flow. As discussed previously, little to no evidence supports the contention that magnets have a measureable effect on blood flow. A second theory to explain pain relief is reduced nerve conductivity. This theory proposes that nociceptive C fibers have a lower threshold potential and that magnetic fields selectively attenuate neuronal depolarization by shifting the membrane resting potential.[11] However, this is an unlikely theory as a 10% reduction in nerve conductivity would require 24 T,[12] a field strength obtained only in the strongest superconducting magnets used for research.[3]

Clinical Effects

Perhaps one of the most successful studies conducted on the use of static magnets to treat pain was done by Vallbona and colleagues[13] on 50 patients with postpolio syndrome who reported muscular or arthritic pain. Based on a hypothesis that magnets applied over trigger points would have the greatest effect, they applied static magnets ranging in strength from 300-500 G to palpated trigger points for 45 minutes while patients remained in the experimental setting.[13] This study was double-blinded and appears to be one of the best-controlled studies using magnets because patients had little opportunity to test for magnetism. Pain levels were assessed by palpation of trigger points before and after application of the magnetic device. The McGill pain questionnaire was used to evaluate subjective pain, elicited by palpation of the single most painful trigger point, even if multiple sites were present. Following application of the magnets, trigger points were again palpated and pain was reassessed by the subjects. Twenty-two patients in the active group showed improvement as compared with four in the placebo group, a significant difference.

In 2007, Pittler and colleagues[14] published a systematic review and metaanalysis of 29 randomized trials in humans on the effect of static magnets on pain and concluded that the evidence does not support the use of magnets for pain relief for low back pain, delayed-onset muscle soreness, or foot pain.[14] The exceptions to this finding were three double-blinded randomized controlled trials assessing pain relief in osteoarthritis (OA) that found pain reductions relative to placebo when treatments lasted 2 to 12 weeks.[14] A separate study evaluated the use of magnets continuously for 24 hours and found that there was no pain relief. The authors concluded that for the condition of OA, there is insufficient evidence to exclude a clinically beneficial effect and believed that further investigation might be warranted.[14]

In a large randomized double-blind study assessing the effectiveness of magnet therapy on pain intensity and opioid requirements in human patients with acute postoperative pain, investigators found no difference in pain intensity level or opioid requirement between groups treated with magnets or sham magnets.[15] Patients with various abdominal surgeries or lipoma excision were treated with quadrapolar static magnets of 1900 G strength placed at the ends of the incisions and around them and left in place for 2 hours. The control group was treated with sham magnets of equal weight and size. Patients rated their pain via a numerical rating scale and the number of rescue doses of morphine was recorded. The authors concluded that magnetic therapy should not be used for treatment of postoperative acute pain or any pain that results from tissue injury.[15]

A comparison of commercially available static bipolar magnetic shoe insoles with identical sham insoles in 101 human subjects diagnosed with plantar fasciitis (heel pain) showed no significant difference in outcome variables studied.[16] Insoles were worn for at least 4 hours a day, 4 days a week for 8 weeks. The sham group reported significant improvement in morning foot pain intensity, and both groups reported an improvement in how much foot pain interfered with their employment enjoyment; however, magnets did not induce additional benefit over nonmagnetized insoles. The sham insoles used in this study averaged a surface reading of 2.2 G and active magnetic insoles averaged 192.1 G.[16]

In a multicenter, double-blinded, placebo-controlled study investigating the role of static magnetic insoles on diabetic peripheral neuropathy–caused neuropathic foot pain, investigators found that chronic use of 450 G multipolar, static magnetic shoe insoles significantly reduced symptoms of burning, numbness, tingling, and exercise-induced foot pain compared with sham insoles.[17] Subjects wore magnetic insoles or sham insoles for 24 hours per day for 4 months and tabulated validated daily pain scores and unvalidated quality of life scores. No new analgesic drugs were allowed during the study. Significant differences between groups were not apparent until after the second month of treatment. Of note was that the magnitude of the reduction of burning, numbness, tingling, and exercise-induced foot pain, especially in the severe and extreme cases, was comparable or better than those previously reported for gabapentin, tramadol, and lamotrigine studies.[17] The mechanism of action of the magnets is unknown, but this study suggests that long-term use may be required to see improvement in symptoms.

The effectiveness of commercially available magnetic bracelets for pain control in OA of the hip and knee in people was evaluated in a multicenter, randomized, placebo-controlled trial.[18] Three groups were evaluated over a 12-week period: (1) standard strength static bipolar magnetic bracelet, (2) a weak magnetic bracelet, or (3) a nonmagnetic (dummy) bracelet. Mean pain scores using the standardized validated Western Ontario and McMasters Universities score were reduced more in the standard magnet group than in the dummy group (27% change from baseline). However, some patients discovered that the bracelets had a magnetic field, resulting in unblinding, although the analysis indicated that this had a small effect on the results. The authors concluded that pain from OA of the hip and knee decreases when wearing magnetic bracelets, but it remains uncertain whether the effect is due to specific or nonspecific (placebo) effects.

Studies on magnets for therapeutic purposes in animals are limited. Two studies by workers from Hungary demonstrated a reduction of writhing from visceral pain in mice induced by intraperitoneal injection of acetic acid. Mice were treated by placing them in a cage with magnets with nonhomogeneous 2-754 mT strength. The pain reaction was significantly reduced in the treated mice compared with mice injected and placed in cages without the magnetic field.[19,20] The authors hypothesized that the observed effect was somehow mediated by the peripheral opioid system because naloxone antagonized the static magnetic field–induced analgesic response if given peripherally but not centrally.[20]

One study determined the effect of a permanent magnetic field on the progression of OA in a canine model.[21] OA was created by transecting a cranial cruciate ligament. Dogs were subjected to no floor mattress, a floor mattress with ceramic pieces placed between two layers of foam but no magnetic field (sham control), or a floor mattress with ceramic permanent magnets placed between two layers of foam. The magnetic field had a surface field strength of 1100 G, whereas the magnetic field strength at the surface of the mattress was 450-500 G and was equally distributed over the surface. The magnetic mattress group appeared to have less synovitis, less synovial effusion, less disruption of the cartilage surface, and less cartilage ulceration than did the other groups. Histologic scores for OA were less in the magnetic mattress group as compared with the other groups, but the differences were not statistically significant. There were similar trends in immunohistochemical studies for matrix metalloproteinase (MMP)-1 and MMP-3, with the magnetic mattress group having less staining; this group had similar MMP-3 to normal cartilage. The mattress and magnetic field group did not differ from the normal group in MMP-3 as determined by Western blot analysis. The results of this study suggest that cartilage exposed to a magnetic field may have reduced progression of OA early in the disease process. Further studies are needed to confirm this and to delineate the mechanisms of action that magnetic fields may have in reducing the progression of OA.

Use

Although the evidence supporting use of magnet therapy is limited at best, and at worst not generally supported, they continue to be used by owners on their pets. Fortunately there are no known side effects of their use; however, it is recommended that they not be used in patients with pacemakers or any other device that may be adversely affected by the magnetic field. If used, magnets should be placed directly over the affected target area to avoid the rapid decrease in magnetic field over short distances. In addition, any positive effects seem to be associated with the use of strong magnets (50-120 mT). Treatment time should be over prolonged periods if tolerated and magnets should not be considered a sole treatment modality but rather an adjunct to more conventional, proven therapy.

Pulsed Electromagnetic Fields

PEMFs have been used for a variety of purposes, but their main therapeutic purpose is for enhancement of bone or tissue healing and pain control.[22] Piezoelectric potentials are created when bone is stressed as the result of electrolyte-rich fluid moving through bone channels. These electrical currents may act as a stimulus for new bone growth. A similar mechanism occurs with cartilage compression leading to current transduction and subsequent cartilage matrix synthesis. Streaming potentials are created within cartilage when electrolyte-rich fluid flows past areas of compression and fixed negatively charged sulfated proteoglycans.[23] The maximal production of glycosaminoglycan occurs at a low frequency (less than 15 Hz), and intermittent compression is better than continuous compression. At 15-16 Hz, calcium ion transport occurs and cartilage growth is stimulated, perhaps because of increased transcription and DNA synthesis. The effect occurs quickly, with peak transcription in 20 minutes.

Electromagnetism is generated by running an electric current through a coiled wire (electromagnetic induction).[3] PEMFs are hypothesized to have electric rather than magnetic effects on tissue.[3] PEMF stimulation may act on cell surface receptors, such as parathyroid hormone and transforming growth factor (TGF)-β, and secondary messengers, such as calcium or cyclic adenosine monophosphate. PEMF may have an effect on stabilizing intracellular calcium stores by altering the conformation of calcium channel proteins that may ultimately reduce the production of free radicals by mitochondria and subsequent inflammation.[24]

Wound Healing

Healing may be promoted by using PEMF in chronic wounds,[25] in soft tissue, and in neuronal regeneration.[26] The effects of PEMF on healing of open and sutured wounds was evaluated in dogs.[27] Open and sutured wounds were created in the skin of the trunk of the dogs. The PEMF-treated dogs received treatment twice a day starting the day before surgery and lasting through day 21 after surgery. PEMF treatment resulted in significantly enhanced epithelialization of open wounds 10 and 15 days after surgery. PEMF treatment also shortened wound contraction time. Conversely, PEMF have not been found to have any significant effects in other studies of soft-tissue wound healing in rats[9] or on blood flow in the brains of mice.[28]

Bone Healing

Mechanical and electrical signals may regulate synthesis of extracellular matrix.[29] Endogenous electric current densities produced by mechanical loading of bone under physiologic conditions approximate 1 Hz and 0.1 to 1 uA/cm^2.[6] PEMF therapy stimulates biologic processes pertinent to osteogenesis[30] and bone graft incorporation.[31] In fact, several devices are approved for the treatment of delayed and nonunion fractures. PEMF may also attenuate bone loss during disuse.[32] PEMF therapy study results suggest that therapy may delay healing of fresh fractures, however.[33] One recent metaanalysis suggested that the proportion of delayed and nonunion fractures that united with PEMF was small and not statistically significant, and there was no reduction in pain.[34]

The effect of PEMF on the healing of experimental lumbar spinal fusions was evaluated in dogs. Bilateral posterior facet fusions were performed at L1-2 and L4-5 in adult dogs.[35] After surgery, one group was stimulated with a pulse burst–type signal PEMF for 30 minutes a day, another group was stimulated with the same PEMF for 60 minutes a day, and a third group received no active PEMF stimulation. No difference in the radiographic or histologic appearance of the fusion mass could be detected between the stimulated and control groups, suggesting that PEMF stimulation had no effect on the healing of posterior spinal fusions in this canine model. However, the use of PEMF in the late phase of bone healing in a stable tibial defect model did stimulate bone healing in one study in which dogs were stimulated 1 hour per day for 8 weeks.[36] Load bearing in the PEMF group at 8 weeks was greater than in the control group. In addition, callus area increased earlier in the PEMF group, and maximum torque, torsional stiffness, new bone formation in the osteotomy gap, and mineral apposition rate were greater in the PEMF group.

Osteoarthritis

One of the most common uses for PEMF is the treatment of OA. PEMF treatment has a number of physiologic effects on cells and tissues, including the upregulation of gene expression of members of the TGF-β family, an increase in glycosaminoglycan levels, and perhaps an anti-inflammatory action.[37]

Pipitone and Scott[38] looked at the effects of magnetic pulse treatment for knee OA in human patients. They used a unipolar magnetic device that was used to treat both knees simultaneously for 10 minutes three times a day for 6 weeks. The devices were set at 7.8 Hz for the morning and afternoon treatment sessions and at 3 Hz for the evening session. Outcome measures included evaluation by a rheumatologist, and several subjective pain assessment scales, two of which are disease-specific measures validated for use in patients with knee or hip OA. There were no significant differences between the active- and sham-treated groups in any outcome measures after treatment.[38]

Another review article of PEMF therapy for people with knee OA concluded that PEMF has little benefit in the treatment of OA in people, and that resources might be better directed to other rehabilitation efforts, such as exercise.[39]

Another randomized, double-blind, placebo-controlled clinical trial investigated the effectiveness of PEMF for the treatment of OA of the knee in people.[40] Patients were treated 2 hours daily, 5 days per week for 6 weeks. Comparison of all treated patients with an untreated control group revealed no significant improvements over time. Analysis of patients younger than 65 years old, however, revealed a significant improvement for stiffness on the treated knee. Analysis of activities of daily living score for the PEMF-treated group revealed a significant correlation between less improvement and increasing age.

A Cochrane review evaluated PEMF for OA treatment.[41] Two trials used PEMF, using a noncontact device that delivered three signals in stepwise fashion, ranging from a 5 Hz to 12 Hz frequency at 10 G to 25 G of magnetic energy. The affected knee was treated for 9 hours over a 1-month period. There was a small to moderate effect on outcomes for knee OA, with clinical benefit ranging from 13-23% greater with active treatment than with placebo. The main outcome measures were for pain, ability to conduct activities of daily living in terms of pain or difficulty, joint pain on motion, and tenderness. Although these results suggest that PEMF result in improvement for knee OA, they suggested that further studies should be performed to confirm these results and to establish the clinical relevancy of treatment.

In another Cochrane review of electrotherapies used for mechanical neck disorders, 5 of 11 studies included used magnets as an intervention.[22] Four of the five used pulsed magnets, whereas the fifth used static magnets. Two of the five had significant results and both of these studies used pulsed magnets, one using low-frequency 5-25 Hz 10-25 G, and one using high-frequency 27 Hz.[22] Interestingly, two of the five studies were led by the same author, using a similar intervention with reported significance in the first study followed by no significant difference 2 years later in the second study. Furthermore, another group used a similar intervention and had no significant results. Finally, the subjects of the two studies with positive results retained these benefits for only a short duration.[22]

Another Cochrane review assessed the effectiveness of PEMF compared with placebo for the treatment of knee OA in randomized, controlled trials.[42] Nine studies, including 483 patients, were pooled. No significant difference could be shown for pain or stiffness. There was a significant effect on activities of daily living and scores. The authors concluded that PEMF improve clinical scores and function as an adjuvant (not stand-alone) treatment in patients with OA of the knee, but there is no evidence for an effect on pain. Another study indicated that PEMF does not add additional benefit to patients already receiving treatment for knee OA, including hot packs, therapeutic ultrasound (US), and exercise.[43] Another study also found no added benefit to PEMF therapy for knee OA when patients received hot packs and transcutaneous electrical nerve stimulation and exercise.[44]

A randomized, controlled, blinded, clinical trial consisting of 60 dogs with OA indicated some positive effects of pulsed signal therapy (PST).[45] Dogs in the treatment group were given nine treatments over 11 days. There were 24 dogs in the treatment group and 35 dogs in the control group. Peak vertical force was significantly more improved in the treatment group from day 0 to day 11, but not from day 0 to day 42. The improvement in the canine brief pain inventory (CBPI) score and measurement of joint extension was significantly better in the treatment group for both time points, and the function portion of the CBPI and thigh circumference was significantly improved in the treatment group from day 0 to day 42. This study suggested that PST was useful in treating canine patients with OA.

Pain

Some investigators have evaluated the use of PEMF for pain management after surgery. A randomized, prospective, double-blind study was performed to evaluate the long-term effects of PEMF in patients undergoing arthroscopic treatment of knee cartilage.[46] Patients with knee pain were recruited and treated by arthroscopy with chondroabrasion and/or perforations and/or radiofrequencies. They were randomized into two groups: a control group (magnetic field at 0.05 mT) and an active group (magnetic field of 1.5 mT). Patients were instructed to use PEMF therapy for 90 days, 6 hours per day. The patients were evaluated with a knee injury and osteoarthritis outcome score before arthroscopy, and 45 and 90 days after treatment. Patients were also interviewed for long-term outcome 3 years after surgery. Differences were significant at 90 days. The percentage of patients who used nonsteroidal antiinflammatory drugs was 26% in the active group and 75% in the control group. At 3 years follow-up, the number of patients who completely recovered was higher in the active group compared with the control group.

A randomized controlled clinical trial evaluated PEMF therapy on postoperative pain in dogs following ovariohysterectomy, and the interaction with postoperative morphine analgesia.[47] Following extubation, dogs were divided into four groups: a control group that received no PEMF, a PEMF group, a morphine group with no PEMF, and a PEMF and morphine group. Although no clear benefit was seen in this study, the results suggested that PEMF may augment morphine analgesia following ovariohysterectomy in dogs.

Use

PEMF therapy appears to have benefits for the treatment of certain conditions. Many of the biophysical effects on cells and tissues have been confirmed. However, its use is still limited likely because of confusing and often conflicting results. As with many modalities used in human and veterinary medicine, dose, frequency, and characteristics of the modality used in studies are bewildering, making the meaningful interpretation of studies difficult. In addition, dogs must remain still when in a PEMF unit. This can provide a challenge for many dogs. Although the modality may show promise for certain conditions, its common use in veterinary medicine is hampered by the lack of prospective, blinded, placebo-controlled studies; the expense of the equipment; and the need to maintain dogs in a relatively quiet state during treatment.

Pulsed Shortwave Diathermy

Pulsed shortwave diathermy (PSWD) operates at approximately 27.12 MHz and creates an electrical field, which heats the tissues in its field. It is also called *pulsed shortwave therapy* and *pulsed electromagnetic energy.* It has some advantages over therapeutic ultrasound (US), which include:

- Larger treatment area than US
- Less reflection off bone
- Use over irregular areas such as the paw, where US use is difficult

The disadvantages include a few additional contraindications not seen in US, including metal implants in the field because metal heats up in response to PSWD.

Contraindications and Precautions

- Over the eyes
- Over or near malignancies and tumors
- Over a pregnant uterus
- Over epiphyses of growing bone
- Over the spinal cord after laminectomy
- Vascular thrombus and embolus
- Over the heart
- Electromedical devices (e.g., pacemakers)
- Plastic and metal implants
- Metal in the treatment field and immediate area including collars, or performed on a metal treatment table
- Decreased or absent sensation
- Over acute inflammation
- Over areas of active infection

Another disadvantage is that dogs must remain still during diathermy, which can provide a challenge for many dogs. Most PSWD units involve placing a drum with a built in spacer over a body part and having the patient stay in one place. Newer technology that attaches a sleeve over a body part may have implications for veterinary rehabilitation. Although diathermy has evidence for tissue heating and increases in muscle flexibility in humans, its use in veterinary medicine is hampered by the lack of prospective, blinded, placebo-controlled studies; the expense of the equipment; and the need to maintain dogs in a relatively quiet state during treatment.

In humans, PSWD has been shown to be as effective as US in heating muscle and more effective than US in heating large areas of muscle.[48,49] A well-controlled study compared PSWD and stretch to sham PSWD and stretch in hamstring flexibility. After 5 days of treatment the diathermy/stretch group had an increase in knee extension of 15.8 degrees ± 2.2 degrees, compared with the sham diathermy/stretch, which had an increase in knee extension of 5.2 degrees ± 2.2 degrees. Diathermy with stretch significantly improved hamstring flexibility in comparison with stretching only in this study.[50]

Diathermy has also been investigated for tissue healing. An in vitro study found PSWD given at mean power of 48 W for 10 minutes increased fibroblast proliferation compared with control groups ($P < 0.001$), and chondrocyte proliferation also increased with PSWD exposure of 6 W at 10 minutes' duration ($P = 0.015$).[51]

Summary

Static magnets, PEMF, and diathermy are some of the modalities that are emerging in veterinary rehabilitation. Unfortunately, despite some evidence that there are biologic effects in tissue culture or laboratory experiments, there are few published well-designed clinical studies in veterinary medicine to make firm recommendations for their use. In particular, questions regarding dosage, the length of treatment, and the most effective treatment protocols for specific conditions are lacking. Users should demand proper studies of products before purchasing them, and veterinary funding agencies should provide adequate resources for investigators to test such devices independently.

REFERENCES

1. Basford JR: A historical perspective of the popular use of electric and magnetic therapy, *Arch Phys Med Rehabil* 82:1261-1269, 2001.
2. www.google.com, search terms “magnetic therapy for dogs.”
3. Ramey D: Magnetic and electromagnetic therapy in horses, *Compendium Cont Ed Pract Vet* 553-560, 1999.
4. Porter M: Magnetic therapy, *Equine Vet Data* 17(7):371, 1997.
5. Livingston J: Magnetic therapy: plausible attraction? Acemagnetics.com, 2011. Retrieved March 29, 2012, from http://www.acemagnetics.com/eduarticles-magsportsbracelets-plausibleatt.html.
6. MacGinitie L, Gluzband Y, Grodzinsky A: Electric field stimulation can increase protein synthesis in articular cartilage explants, *J Orthop Res* 12(2):151-160, 1994.
7. Steyn PF, Ramey DW, Kirschvink J et al: Effect of a static magnetic field on blood flow to the metacarpus in horses, *J Am Vet Med Assoc* 217:874-877, 2000.
8. Saygili G, Aydinlik D, Ercan M et al: Investigation of the effect of magnetic retention systems used in prostheses on buccal mucosal blood flow, *Int J Prosthodont* 5(4):326-332, 1992.
9. Leaper D, Foster S, Brennan P et al: Do magnetic fields influence soft tissue wound healing? A preliminary communication, *Equine Vet J* 17(3):178-180, 1985.
10. Barker A, Cain M: The claimed vasodilatory effect of a commercial permanent magnet foil: results of a double-blind trial, *Clin Phys Physiol Meas* 6(3):261-263, 1985.
11. Lednev V: Possible mechanism for the influence of weak magnetic fields on biological systems, *Bioelectromagnetics* 12(2):71-75, 1991.
12. Wikswo J, Barach J: An estimate of the steady magnetic field strength required to influence nerve conduction, *IEEE Trans Biomed Eng* 27(12):722-723, 1980.

13. Vallbona C, Hazlewood C, Jurida G: Response of pain to static magnetic fields in postpolio patients: a double-blind pilot study, *Arch Phys Med Rehabil* 78(11):1200-1203, 1997.
14. Pittler MH, Brown EM, Ernst E: Static magnets for reducing pain: a systematic review and meta-analysis of randomized trials, *Can Med Assoc J* 177(7):736-742, 2007.
15. Cepeda MS, Carr DB, Sarquis T et al: Static magnetic therapy does not decrease pain or opioid requirements: a randomized double-blind trial, *Ambul Anesth* 104(2):290-294, 2007.
16. Winemiller MH et al: Effects of magnetic vs sham-magnetic insoles on plantar heel pain, a randomized controlled trial, *JAMA* 290(11):1474-1478, 2003.
17. Weintraub MI, Wolfe GI, Barohn RA et al: Static magnetic field therapy for symptomatic diabetic neuropathy: a randomized, double-blind, placebo-controlled trial, *Arch Phys Med Rehabil* 84:736-746, 2003.
18. Harlow T, Greaves C, White A et al: Randomised controlled trial of magnetic bracelets for relieving pain in osteoarthritis of the hip and knee, *BMJ* 329:18-25, 2004.
19. Laszio J, Reiczigel J, Szekely L et al: Optimization of static magnetic field parameters improves analgesic effect in mice, *Bioelectromagnetics* 28:615-627, 2007.
20. Gyires K, Zadori ZS, Racz B et al: Pharmacological analysis of inhomogeneous static magnetic field-induced antinociceptive action in the mouse, *Bioelectromagnetics* 29:456-462, 2008.
21. Rogachefsky RA, Altman RD, Markov MS et al: Use of a permanent magnetic field to inhibit the development of canine osteoarthritis, *Bioelectromagnetics* 25:260-270, 2004.
22. Kroeling P, Gross A, Goldsmith C, Cervical Overview Group: A Cochrane review of electrotherapy for mechanical neck disorders, *Spine* 30(21):E641-E648, 2005.
23. Maroudas A, Muir H, Wingham J: The correlation of fixed negative charge with glycosaminoglycan content of human articular cartilage, *Biochim Biophys Acta* 177(3):492-500, 1969.
24. Gordon GA: Designed electromagnetic pulsed therapy: clinical applications, *J Cell Physiol* 212:579-582, 2007.
25. Ieran M, Zaffuto S, Bagnacani M et al: Effect of low frequency pulsing electromagnetic fields on skin ulcers of venous origin in humans: a double-blind study, *J Orthop Res* 8(2):276-282, 1990.
26. Ito H, Bassett C: Effect of weak, pulsing electromagnetic fields on neural regeneration in the rat, *Clin Orthop Relat Res* 181:283-290, 1983.
27. Scardino MS, Swaim SF, Sartin EA et al: Evaluation of treatment with a pulsed electromagnetic field on wound healing, clinicopathologic variables, and central nervous system activity of dogs, *Am J Vet Res* 59(9):1177-1181, 1998.
28. Bellossi A, Nicol L, Dazord L: No effect of a low-frequency pulsed magnetic field on the brain blood flow among mice, *Panminerva Med* 35(1):57-59, 1993.
29. Aaron R, Ciombor D: Acceleration of experimental endochondral ossification by biophysical stimulation of the progenitor cell pool, *J Orthop Res* 14(4):582-589, 1996.
30. Shimizu T, Zerwekh J, Videman T et al: Bone ingrowth into porous calcium phosphate ceramics: influence of pulsing electromagnetic field, *J Orthop Res* 6(2):248-258, 1988.
31. Miller G, Burchardt H, Enneking W et al: Electromagnetic stimulation of canine bone grafts, *J Bone Joint Surg Am* 66 (5):693-698, 1984.
32. Skerry TM, Pead MJ, Lanyon LE: Modulation of bone loss during disuse by pulsed electromagnetic fields, *J Orthop Res* 9:600-608, 1991.
33. De Haas W, Lazarovici M, Morrison D: The effect of low frequency magnetic fields on the healing of the osteotomized rabbit radius, *Clin Orthop Relat Res* 145:245-251, 1979.
34. Griffin XL, Costa ML, Parsons N et al: Electromagnetic field stimulation for treating delayed union or non-union of long bone fractures in adults, *Cochrane Database of Systematic Reviews* 2011, Issue 4. Art. No.: CD008471. DOI: 10.1002/14651858.CD008471.pub2.
35. Khanaovitz N, Arnoczky SP, Nemzek J et al: The effect of electromagnetic pulsing on posterior lumbar spinal fusions in dogs, *Spine* 19(6):705-709, 1994.
36. Inoue N, Ohnishi I, Chen D et al: Effect of pulsed electromagnetic fields (PEMF) on late-phase osteotomy gap healing in a canine tibial model, *J Orthop Res* 20:1106-1114, 2002.
37. Fini M, Giavaresi G, Carpi A et al: Effects of pulsed electromagnetic fields on articular hyaline cartilage: review of experimental and clinical studies, *Biomed Pharmacother* 59(7):388-394, 2005.
38. Pipitone N, Scott DL: Magnetic pulse treatment for knee osteoarthritis: a randomized, double-blind, placebo-controlled study, *Curr Med Res Opin* 17(3):190-196, 2001.
39. McCarthy CJ, Callaghan MJ, Oldham JA: Pulsed electromagnetic energy treatment offers no clinical benefit in reducing the pain of knee osteoarthritis: a systematic review, *BMC Musculoskeletal Disorders* 7:51, 2006.
40. Thamsborg G, Florescu A, Oturai P et al: Treatment of knee osteoarthritis with pulsed electromagnetic fields: a randomized, double-blind, placebo-controlled study, *Osteoarthr Cart* 13(7):575-581, 2005.
41. Hulme JM, Welch V, de Bie R et al: Electromagnetic fields for the treatment of osteoarthritis, *Cochrane Database of Systematic Reviews* 2009, Issue 1. Art. No.: CD003523. DOI: 10.1002/14651858.CD003523.
42. Vavken P, Arrich F, Schuhfried O et al: Effectiveness of pulsed electromagnetic field therapy in the management of osteoarthritis of the knee: a meta-analysis of randomized controlled trials, *J Rehabil Med* 41:406-411, 2009.
43. Özgüçlü E, Çetin A, Çetin M et al: Additional effect of pulsed electromagnetic field therapy on knee osteoarthritis treatment: a randomized, placebo-controlled study, *Clin Rheumatol* 29:927-931, 2010.
44. Ay S, Evcik D: The effects of pulsed electromagnetic fields in the treatment of knee osteoarthritis: a randomized, placebo-controlled trial, *Rheumatol Int* 29:663-666, 2009.
45. Gordon-Evans WJ, Knap K: Randomized, controlled, blinded clinical trial evaluating pulsed signal therapy for the treatment of osteoarthritis in the treatment of osteoarthritis in dogs, *Proc Ann Meeting Am Coll Vet Surg*, 2010.
46. Zorzi C, Dall'oca C, Cadossi R et al: Effects of pulsed electromagnetic fields on patients' recovery after arthroscopic surgery: prospective, randomized and double-blind study, *Knee Surg Sports Traumatol Arthrosc* 15:830-834, 2007.
47. Shafford HL, Hellyer PW, Crump KT et al: Use of a pulsed electromagnetic field for treatment of post-operative

pain in dogs: a pilot study, *Vet Anaesth Analg* 29(1):43-48, 2002.

48. Garrett CL, Draper DO, Knight KL: Heat distribution in the lower leg from pulsed short-wave diathermy and ultrasound treatments, *J Athl Train* 35:50-55, 2000.
49. Draper DO, Knight K, Fujiwara T et al: Temperature changes in human muscle during and after pulsed short-wave diathermy, *J Orthop Sports Phys Ther* 29:13-22, 1999.
50. Draper DO, Castro JL, Feland B et al: Shortwave diathermy and prolonged stretching increase hamstring flexibility more than prolonged stretching alone, *J Orthop Sports Phys Ther* 34:13-20, 2004.
51. Hill J, Lewis M, Mills P et al: Pulsed short-wave diathermy effects on human fibroblast proliferation, *Arch Phys Med Rehabil* 83:832-836, 2002.

SECTION V

Therapeutic Exercise and Manual Therapy

24 Biomechanics of Physical Rehabilitation and Kinematics of Exercise

Joseph P. Weigel and Darryl Millis

Mechanics as developed by Sir Isaac Newton describes how force and mass interact in three-dimensional space and time on the surface of the earth. Biomechanics is the application of Newtonian mechanics to biologic systems. Newtonian mechanics is founded on three classic laws of motion. The first law describes the static state of a body under the influence of balanced force. The second law describes the motion or dynamic state of a body under the influence of unbalanced force. The third law describes the interaction between multiple bodies.

Law I (Law of Statics)

When the sum of all the forces acting on a body equals zero, the body is in a static state (i.e., the body is at rest or in unchanging, constant motion with no acceleration).

$$\Sigma F = 0 F = 0$$

Under certain conditions, the static state is desirable. For example, to enhance the conditions for healing, fractured bones are managed by fixation systems designed to place fracture segments in a static state or at rest. It is therefore the goal of the surgeon to balance, both in direction and magnitude, all the forces that tend to cause motion of the fragments. In the case of a fractured tuber calcis the proximal fragment is under a large tension load caused by the pull of the attached gastrocnemius muscle. Without fixation this pull is unopposed, causing motion of the bone fragments, which disrupts the body's attempt to bridge and stabilize the fracture. When a system of pins and a tension band wire are placed across the fracture, the wire, by virtue of its position and strength, counters the tension load of the gastrocnemius muscle with a corresponding compression load provided by the tension band wire. When the sum of the forces acting across the fracture equals zero, the fracture is in static equilibrium, a condition that fosters bone bridging. It is important to exercise caution when applying physical rehabilitation to animals with healing fractures such that these activities do not result in creating an imbalance of force, causing motion of the fracture. The static state of the fracture must be maintained even though the therapist is focused on motion of joints and the strength of muscles.

Static exercise can be an effective way of stretching and enhancing flexibility by the strategic and controlled application of force without producing motion. Static stretching may improve soft tissue extensibility, which increases the capacity for energy absorption and transfer when running, for example. Practical methods for preexercise preparation in animals have not been established. Dogs are often left sedentary for long periods and then are suddenly released for unrestrained activity, often at high performance levels. After long periods of relative inactivity, ligaments, muscle, tendons, and fascia are "tight," less ductile and less able to tolerate acute strain. Stiff soft tissue has less capacity to absorb impact, lending itself to tearing or rupture. A common thread in the clinical history of cranial cruciate injury in the dog is the occurrence of sudden lameness during vigorous activity following long periods of inactivity.

Law II (Law of Dynamics)

When the sum of all the forces (resultant force) acting on a body is not equal to zero, the body is in a dynamic state (i.e., the body is not at rest and is in a state of changing motion or accelerating).

$$\vec{F} = m\vec{a}$$

Force (F), is a vector and therefore has not only intensity (magnitude) but also direction. Acceleration (a), the change in velocity over time, also has intensity as well as direction. Newton's second law directly relates the cause of motion to the motion itself. In terms of magnitude, the harder a body is pulled or pushed by an increasing force, the faster it moves as seen with increasing acceleration and velocity. In terms of direction, the body will move or accelerate in the same direction as the applied force that has caused the motion. Therefore motion analysis or dynamics involves the application of arithmetic functions but within three-dimensional space, which also necessitates the use of trigonometric functions. Newton's second law of motion can be expanded to describe the associated concepts of motion such as impulse, momentum, energy, work, and power.

The physical therapist applies Newton's second law of motion when the speed or acceleration (change in velocity over time) of a therapeutic exercise is increased. When speed or acceleration increase, the load (force) on the limb must also be increased to achieve the faster motion. For example, hitting a wall with an automobile at 5 miles an hour might dimple a bumper, but hitting the same wall with the same automobile at 100 miles an hour could result in irreparable damage. Going faster means greater force, which then increases the risk for exacerbating an injury or inducing new injury. For example, repeated rapid, forceful flexion of the stifle in the case of a contracted quadriceps muscle can result in avulsion of the tibial tuberosity. Slow range of motion (ROM) exercises, slow stretching, and slow walking keep the applied forces low, preserving and protecting healing tissues.

Law III (Law of Action and Reaction)

When several bodies interact, the force resulting from the interaction is equal in intensity to the force causing the interaction and will act along the same line but opposite in direction.

Newton's third law describes the event when one body impacts another. Newton observed that when one mass, moving under the influence of a causative force, impacts a second mass, a reaction force occurs. This reaction force is equal to the magnitude of the impacting force (original causative force) but acts in the opposite direction. This is easily observed in the case of a bouncing ball. When a ball is dropped to the floor, it impacts the floor and then reverses direction, moving away from the floor. The initial throwing force is reversed after the impact into a reaction or "bouncing" force. If the bouncing ball is replaced by a bag of sand, the bag will not bounce; however, reaction forces are still created but are absorbed by the sand in the bag. When an animal moves across the floor there is both an elastic response (like the bouncing ball) where force is reflected and an inelastic response (like the bag of sand) where force is absorbed.

Newton's third law is the basis for the use of a force plate in analyzing an animal's gait. When in motion, the animal impacts the plate, which results in a reaction force that is recognized by the plate and analyzed by a computer program. The reaction force, recorded at the point of foot impact (foot strike), is directly related to the force of weight bearing. This reaction force, called the *ground reaction force* (GRF), is transmitted up the limb in part as an elastic reaction where it is reflected by the stiffer tissues, and in a large part as an inelastic response where it is absorbed by the softer tissues. This absorption of force is an important function of the limb and all its components, and as a result, is an important consideration in physical rehabilitation. The therapist can manipulate the reaction force by applying various forms of exercise. For example, the aquatic treadmill mitigates the reaction force by using the buoyancy of the animal in water to reduce its weight. Slinging the abdomen during ambulation also reduces the impact of weight, thereby decreasing the reaction force. Sometimes a reaction force is magnified so it can be a means of "active therapy" during which it can stretch contracted tissues. Exercising down an inclined plane magnifies the reaction force by increasing the magnitude of impact on the front limbs. Also, a reaction force can be magnified by applying a syringe cap to the foot of the sound limb to encourage a shift of weight bearing to the contralateral limb, accelerating its rehabilitation. Conditions of the terrain, such as the stiffness of the floor, obstructions in the gait pathway, or unstable flooring can alter the magnitude and direction of reaction forces. When devising therapeutic plans, the therapist should consider the nature of the reaction forces and how they can be modified for the patient's benefit.

In the case of animal physical rehabilitation, the application of the Newtonian laws of motion can be an ominous challenge. The animal is a complex and sophisticated system of interconnected objects having complex patterns of motion. This chapter attempts to incorporate the fundamental concepts of biomechanics to the practice of animal rehabilitation.

The Body in Equilibrium: Statics

Force is intuitively understood as a measure of intensity but also it must be viewed with respect to direction. For example pulling on a car door causes it to open while pushing on it causes it to close. To know only how hard to pull or push is insufficient in understanding the design, operation, and composition of a car door. Similarly, in biomechanics it is not only beneficial to know how hard the leg impacts during an exercise but also from which direction the force is acting. For example, living bone responds to weight bearing by increasing its mass and this response is not only caused by the intensity of the weight but also by where on the bone the weight load is localized. The classic Wolff's Law can be described as the case where bone increases in mass in response to force that tends to push it together, whereas it decreases in mass where the force tends to pull it apart. In the case of corrective osteotomies, the goal is to help improve limb alignment, distribution of weight bearing forces, and joint function by altering the anatomy and redirecting weight bearing, creating an improved mechanical environment.

When the animal is at rest (standing still), the forces acting on the body have not disappeared but have been counterbalanced by forces acting in opposite directions. Mechanical analysis of the body in this static state facilitates understanding of the types of forces at play. There are four basic forces: compression, tension, shear, and moment or torque. The kind of force depends on its direction. A force that tends to push on a body is referred to as *compression*; if it tends to pull on the body, it is called *tension*; if it tears or shreds the body, it is *shear*; and finally, if it tends to rotate or turn the body, it is a *moment* or *torque*.

To evaluate direction, a reference or coordinate structure is necessary. Commonly a three-axis rectangular system representing the three dimensions of space (height, width, and length) is used. By convention this axis system is composed of three perpendicular axes, labeled z for the vertical direction, y for the horizontal direction, and x for the transverse direction.

Compression (Push)

When the vertical dimension of the limb is oriented along the z axis, a force parallel to the z axis and directed toward the center of the limb is called *compression.* Compression is the major load on the hard tissue elements, bone and cartilage, as it tends to push them together.

Tension (Pull)

With the vertical dimension of the limb oriented along the z axis, a force parallel to the z axis and directed away from the center of the limb is called *tension.* Tension is the major load on the soft tissue elements, muscles, tendons, and ligaments, as it tends to pull them apart.

Shear (Tear or Shred)

With the vertical dimension of the limb oriented along the z axis, a force perpendicular to the z axis is called *shear.* Shear is a significant force causing the elements of the system to tear or slide apart. For example, the cranial cruciate ligament (CCL) encounters a shear load that occurs across the stifle during weight bearing. When this ligament is torn, the shear is unopposed and cranial drawer occurs. Redirection of the weight-bearing forces by a tibial plateau leveling osteotomy eliminates the shear component, thereby preventing the occurrence of cranial drawer while in weight bearing.

Rotation (Moment of a Force or Torque)

A force can also tend to cause a mass to rotate about a point called the *center of rotation.* The farther away from the center of rotation the force is applied, the greater is the tendency for rotation. This tendency is called "moment of a force," and the distance from the center of rotation to the point of force application is called the *moment arm.* Mathematically, moment is represented by multiplying the force by the moment arm:

$$\overrightarrow{M} = \vec{F}\vec{d}$$

where M is the moment of the force, F is the force, and d is the moment arm.

Moment is critical to the understanding of how force acts in the musculoskeletal system. Although forces of body weight and muscle contraction push and pull at the body, they also cause rotation at each of the joints when these forces do not pass through the center of rotation of the joint. For example, there is a tendency toward flexion (rotation in one direction) of the joints in the front limb of a standing animal, which is caused by the body's weight pushing down at a location that is off center of the respective joint. The animal is able to stand because the tendency toward flexion is counterbalanced by a tendency toward extension (counter rotation in the opposite direction) caused by contraction of several muscles, including the supraspinatus, triceps brachii, and extensor carpi radialis muscles. In the rear limb, flexion caused by the body's weight at the hip, stifle, and tarsal joints is counterbalanced by contraction of the gluteal, quadriceps, and gastrocnemius muscles, which causes extension at each of these joints. The actions of these forces of weight and muscle contraction are defined as *moments,* which are very important, especially in the analysis of gait.

Moments of force are important in the effective function of muscle in sustaining weight bearing. For example, in the stifle the normal position of the patella creates an effective moment by the quadriceps about the center of rotation of the stifle such that extension of the stifle is maintained enough to bear weight. However, the function is altered when the patella is not in its normal position and is dislocated medially. In the dislocated position the moment arm of the tension force in the quadriceps is reduced or eliminated (Figure 24-1). The closer the tension force is to the center of rotation, the less extension is possible. Therefore a dog with luxating patellas is weak in the rear limbs and unable to bear weight normally or perform activities, such as jumping.

Lever Systems in the Body (Application of Moments of Force)

A lever is a simple machine using the principle of moments to overcome a resistance load. There are three classes of levers: Class I, in which the fulcrum (point of rotation) is between the effort force (applied force usually resulting from muscle contraction) and the resistance force (load force, which is generally the weight of the body or limb); Class II, in which the resistance is between the effort and

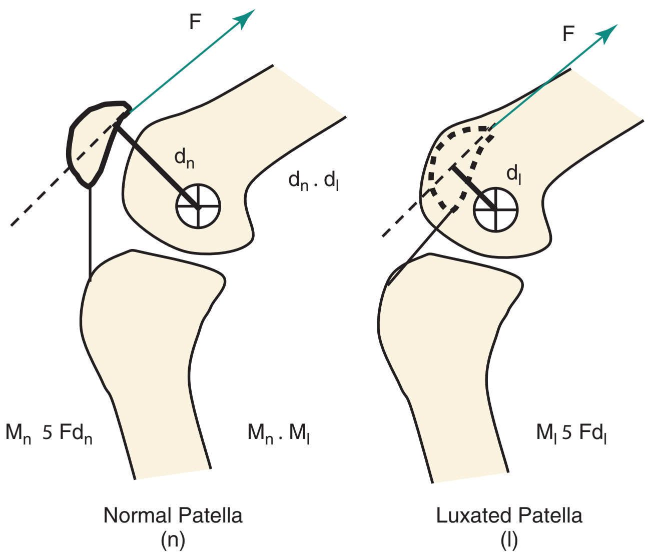


Figure 24-1 The moment arm of the tension force in the quadriceps is reduced or eliminated when the patella is luxated medially. Luxation of the patella compromises weight bearing by lessening the moment (M_l) generated by quadriceps contraction by shortening the moment arm (d_l).

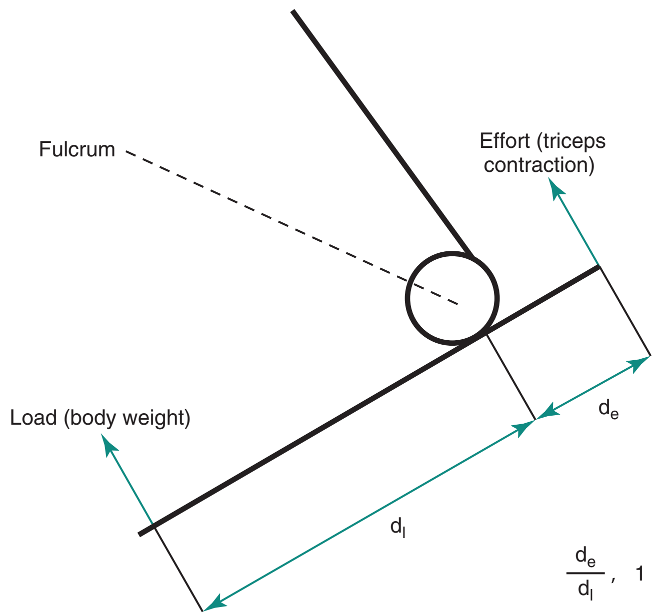


Figure 24-2 Extension of the elbow is a Class I lever system. The fulcrum is located at the center of rotation of the joint, the resistance or load force is located at the point where the foot contacts the ground and the effort or action force produced in the triceps muscle is located at its insertion point on the olecranon.

fulcrum; and Class III, in which the effort is between the resistance and the fulcrum.

A variety of levers are used by the animal to extend and flex the various joints of the limbs. For example, extension of the elbow is accomplished with a Class I lever in which the fulcrum is located at the center of rotation of the joint, the resistance or load force is located at the point where the foot contacts the ground, and the effort or action force, which is produced in the triceps muscle, is located at its insertion point on the olecranon (Figure 24-2). With flexion of the elbow, a Class III lever is in operation in which the effort is located at the insertion of the biceps brachii, which is between the fulcrum (elbow joint) and the resistance force, which is located at the end of the antebrachium (Figure 24-3).

Class II levers are uncommon. An example is the rear foot, in which the weightbearing toes act as the fulcrum, the resistance or load force is the body weight just caudal to and at the foot, and the effort force is the gastrocnemius muscle (Figure 24-4).

In contrast, both extension and flexion of the stifle joint are accomplished with a Class III lever, in which the effort or action load is located at the insertion of the quadriceps muscle on the tibial tuberosity (for extension) or the insertion of the semitendinosus muscle on the proximal tibia (for flexion), which is located between the fulcrum (stifle joint) and the resistance force, which is located at the distal end of the limb.

The relative magnitudes of the effort versus the resistance is the value of understanding lever systems and their

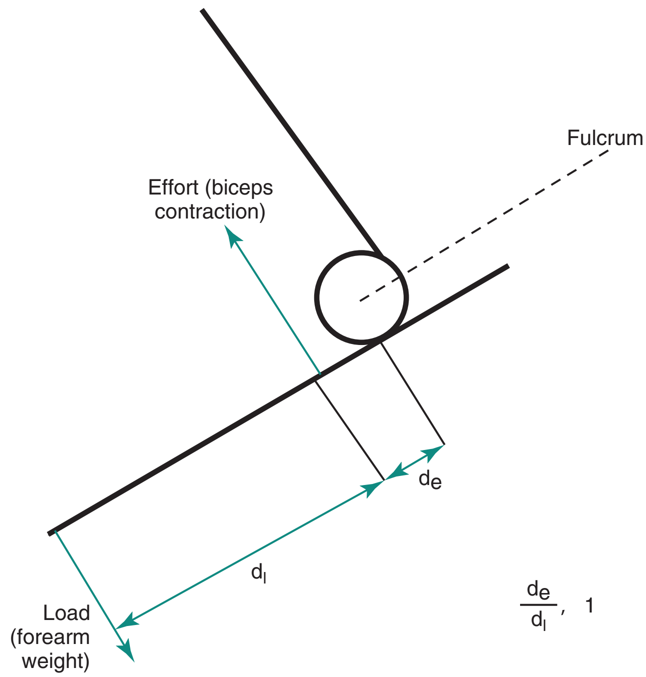


Figure 24-3 A Class III lever occurs with flexion of the elbow where the effort is located at the insertion of the biceps brachii, which is between the fulcrum (elbow joint) and the resistance force, which is located at the end of the antebrachium.

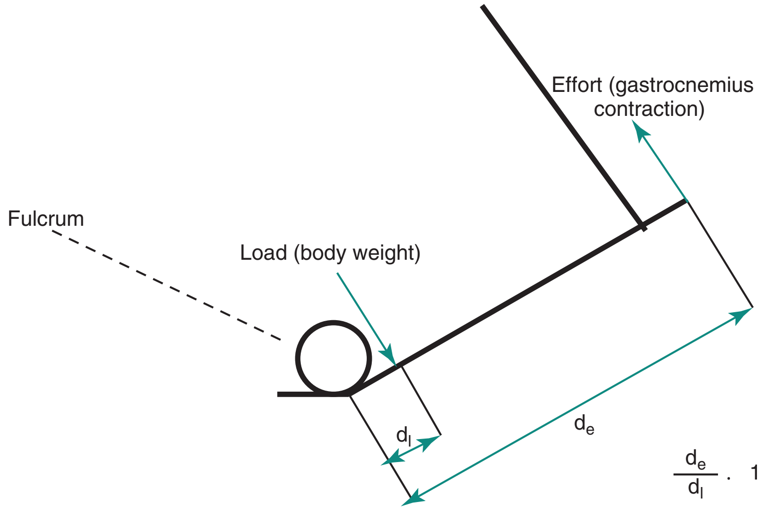


Figure 24-4 An example of a Class II lever is the rear foot, where the weight-bearing toes act as the fulcrum, the resistance or load force is the body weight just caudal to and at the foot, and the effort force is the gastrocnemius muscle.

function in the musculoskeletal system. The effort force multiplied by its distance (moment arm, d_e) to the fulcrum is the effort moment. The resistance force multiplied by its perpendicular distance (moment arm, d_o) to the fulcrum is the load moment. In the balanced state in which the lever is not moving or moving without acceleration, the effort moment must equal the load moment. This Law of the Lever system establishes the relationship between the effort and resistance loads. When comparing the magnitude of required effort to overcome the magnitude of a specified resistance, compare the respective moment arms. If the effort moment arm is longer than the resistance moment arm, then the effort force necessary to overcome the resistance will be relatively less than the magnitude of the resistance.

$$(E)(d_e) = (R)(d_o)$$

In the case in which rotation is inhibited by contracture or fibrosis leading to decreased ROM, therapists use "passive" ROM exercise by pushing or pulling on the lower part of the limb to induce flexion or extension in a target joint. Passive ROM is often used to reduce the possibility of stiffness or to improve the mobility of stiff joints. In some of these exercises the lever action is altered. For example, stretching in flexion of a dog with hyperextension stiffness of the stifle presents the conditions for a Class II lever (Figure 24-5). The point of resistance is the insertion of the quadriceps muscle on the tibial tuberosity, which is located between the fulcrum (stifle joint), and the effort, which is the location of the therapist's hold on the distal crus of the leg. In this case, the effort is relatively far from the stifle when compared with the resistance that is close to the joint. When the quadriceps is adhered to the underlying bone and joint capsule and has lost its elasticity, it has also lost its capacity to absorb energy and load. As the therapist pushes on the tibia to produce flexion in the stifle, the resistance load in the stiff quadriceps quickly rises, creating a magnitude of force that has the ability to result in tearing of the quadriceps or even avulsion of the tibial tuberosity. Because the moment arm length of the effort force is great, a large moment can be generated, potentially causing damage. It is important for the therapist to understand the relationship between the effort or action load and the reactive or resistance load when employing certain therapeutic exercises. In the previous example,

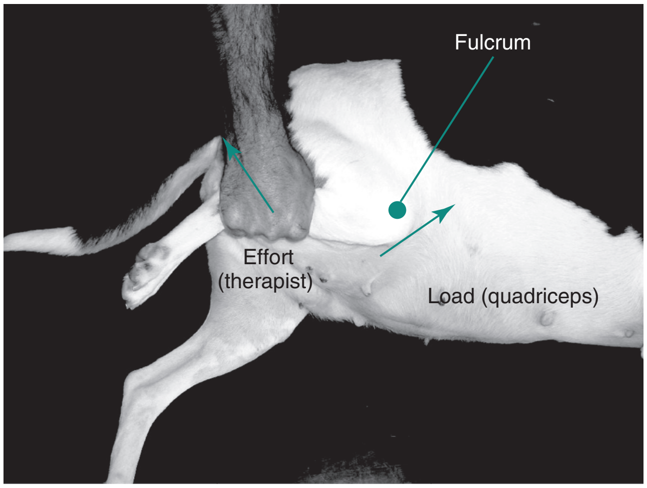


Figure 24-5 Stretching in flexion of a dog with hyperextension stiffness of the stifle presents the conditions for a Class II lever. The point of resistance is the insertion of the quadriceps muscle on the tibial tuberosity, which is located between the fulcrum (stifle joint), and the effort, which is the location of the therapist's hold on the distal shin or crus of the leg. In this case the effort is relatively far from the stifle when compared with the resistance, which is close to the joint.

Figure 24-6 Wheelbarrowing to increase weight bearing on the front limbs results in the spine becoming a Class II lever in which the resistance (center of gravity of the body) is located between the fulcrum at the shoulder joint and the effort where the therapist is lifting the caudal end of the body. As the spine is raised from the rear, the center of gravity of the body moves to a location that is more dorsal to the shoulder.

moving the hands closer to the stifle joint reduces the moment arm length, and therefore makes it more difficult to generate a large moment, reducing the possibility of tissue damage.

Other exercises such as "wheel barrowing" on the front limbs or "dancing" on the rear limbs employ the use of lever systems. In the case of wheel barrowing or walking down an incline (to increase weight bearing on the front limbs), the spine becomes a Class II lever in which the resistance (center of gravity [COG] of the body) is located between the fulcrum at the shoulder joint and the effort in which the therapist is lifting the caudal end of the body. It is also important to note that as the spine is raised from the rear, the COG of the body moves to a location that is more dorsal to the shoulder, introducing a horizontal component of the weight, which acts as a shear force across the shoulder joint (Figure 24-6). As a result, increased stress in the joint capsule, collateral ligaments, and tendons will likely occur. This horizontal shear force may also cause an additional moment to occur at the level of the elbow joint, which is likely resisted by additional contraction of the triceps muscle. It has been observed that the angle of the shoulder joint becomes more extended with wheelbarrowing,[1] which moves weight bearing from the center of the humeral head to a more cranial location where there is less articular cartilage and bone. Dancing is the reverse of this example; in dancing the spine is raised from the front and used as a Class II lever. As with the effects on the shoulders, the same likely occurs on the hips. For example, the addition of shear force across the hip joint has the potential of increasing mechanical stresses in the articular cartilage, which may be deleterious to a dysplastic joint. Increased moment across the stifle would likely be resisted by increased contraction of the quadriceps muscles, strengthening this muscle group. The effectiveness of physical rehabilitation exercises could be enhanced by detailed mechanical evaluation looking at not only the magnitudes of weight-bearing forces but also the altered direction of these forces.

Tissue Response to Force

To understand how force affects the body, the response of the various tissues to force is important to analyze. Each of the tissues involved respond differently because the composition of each is different. These responses are unique to the tissue and can be called the *material properties of the tissue.* These properties include measures of stiffness, elasticity, plasticity, strength, failure limits, and energy absorption. Basically, a tissue responds to force in two ways: first, an intrinsic resistance to the applied force is called *stress*; second, an accommodation to the applied force by a change in the shape of the tissue is called *strain.*

Stress

Stress is directly related to the magnitude of the applied force (i.e., the greater the applied force, the greater the stress within the tissue). Stress is also inversely related to the area over which the total force is acting (i.e. the larger the area, the lower the stress). This understanding of the nature of stress explains how a scalpel cuts the skin. A sharp blade has an extremely thin edge so that with light pressure the force is concentrated at the edge of the blade and the stress in the skin is very high, causing the skin to break away and separate. A dull blade with a wider edge requires more force to increase the stress to the skin and cause it to separate.

Stress is particularly important in the understanding of how articular cartilage responds to force applied across a joint surface. Under conditions in which the anatomy is normal and the joint is physiologically and mechanically normal, the surface of the joint spreads the applied forces over a wide area such that the stress at any specific point on the cartilage is low. However, if the joint is anatomically deranged, such as with a growth deformity, or is mechanically unstable, such as with hip dysplasia, joint forces are concentrated on smaller surface areas, causing high degrees of stress and subsequent breakdown of the articular cartilage (Figure 24-7).

Motion of a joint is not only important for the physiologic health of the articular cartilage, but also for the mechanical integrity of the cartilage. For example,

Figure 24-7 In a mechanically unstable joint, such as in a dog with hip dysplasia, joint forces are concentrated on small surface areas, causing high degrees of stress and subsequent breakdown of the articular cartilage.

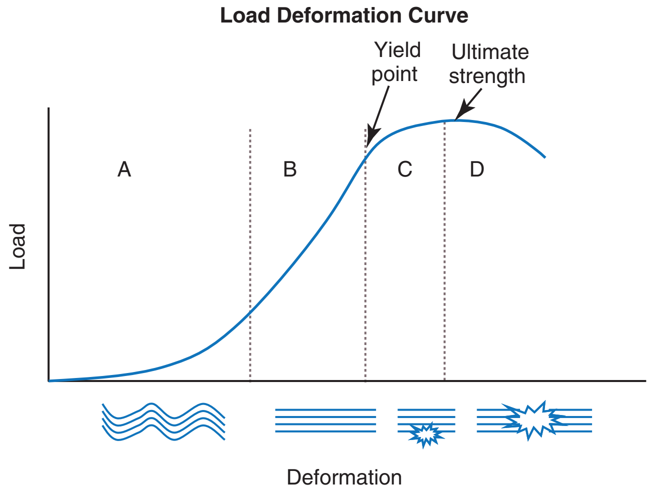


Figure 24-8 A stress versus strain curve demonstrates additional properties of the tissue response to force. **A,** The initial response in the graph is termed the *toe region* of the graph. The collagen fibers are in a completely relaxed state with a crimped or wavy pattern. **B,** The linear portion of the graph describes the elasticity of the tissue. The slope of the graph gives a measure of stiffness of the tissue. **C,** The upper curved portion of the graph describes the plasticity of the tissue. In this range the tissue is deforming, but when the load is removed, the tissue does not return to its original shape, meaning that a permanent change has taken place. **D,** The peak of the curve is the highest level of stress the tissue tested can withstand and is referred to as the ultimate strength of the tissue.

long-term immobilization of a normal joint can induce osteoarthritis. The lack of motion results in a concentration of force over a small area of articular cartilage. This increases the stress, causing a breakdown in the cartilage structure, inducing an inflammatory response and osteoarthritis. Distributing weight-bearing forces over a wider area of the joint surface is one way to minimize articular cartilage stress, but also reducing the weight-bearing force itself while maintaining joint motion should reduce the stress. This may be accomplished with an aquatic treadmill in which the water tempers the weight-bearing force but allows normal ROM.

Strain

As a result of an applied force, a tissue can change shape. For example, a ligament stretched by a tension force will lengthen, and this increase in length is termed *tensile strain.* Likewise, compressive force compacts tissue, creating a compressive strain, and shear force translates portions of tissue, creating a shear strain. For some tissues, strain and strain rates may depend on the direction or orientation of the force causing the strain. Such a property is referred to as *anisotropy.* For example, depending on which direction the force is acting on a section of bone, the strain value is different. Bone is stronger under compression than it is under tension, meaning that the strain is less (i.e., the bone deforms less under a compressive load than it does under the same intensity load in tension). The opposite is true in the case of a ligament: the ligament deforms less under tension than it does under compression.

Stress versus Strain

For any tissue under the influence of force, its response can be graphed as stress versus strain, which demonstrates additional properties of the tissue response to force (Figure 24-8). The initial response in the graph is termed the *toe region* of the graph. This phenomenon is more typical with soft tissue such as a tendon or ligament as opposed to bone. A tendon in its completely relaxed state has a crimped or wavy pattern of the collagen fibers. When the tendon is mounted in the testing machine and then pulled, the initial recording represents a straightening of the crimp and does not represent any part of the strength of the tendon in tension. It is also helpful to pass through this "toe" region by "prestressing" the tendon (i.e., stretching the tendon to eliminate the crimp). The slope of the graph gives a measure of stiffness of the tissue. Steep slopes occur when the tissue is inflexible and brittle like that of bone, whereas a shallow slope represents tissue with flexibility, like that of muscle or tendon. The linear portion of the graph describes the elasticity of the tissue. In this range the tissue is deforming or undergoing strain, but when the load is removed, the tissue returns to its original length. The upper curved portion of the graph describes the plasticity of the tissue. In this range the tissue is deforming, but when the load is removed the tissue does not return to its original shape, meaning that a permanent change has taken place; therefore the tissue has undergone a "plastic deformation." The peak of the curve is the highest level of stress the tissue tested can withstand and is referred to as the *ultimate strength* of the tissue. Failure under stress is demonstrated as a decreasing capability to maintain the stress and the value begins to drop as the load increases, or complete failure as the stress drops to zero, indicating that the tissue can no longer hold up under any applied load. In addition to these features, the area under the stress versus strain curve represents the amount of energy that is absorbed or stored in the material as it is loaded. Once failure occurs, the stored energy is released in the form of a fracture or tearing event.

For bone, the response is more predictable and consistent as compared with muscle, tendon, and ligament. Bone behaves in a linear way as a elastic solid, but for the soft tissues where the response is more variable and less predictable, the response is nonlinear. As opposed to bone, these tissues behave as a viscoelastic material. A viscoelastic solid's properties, like stiffness, change over time. The response of a viscoelastic material additionally depends on factors like moisture content and temperature. For example, under the condition of cold temperatures, the tissue is stiffer so the stress is greater. When exercise is planned, often the tissues are physically warmed, which has the effect of reducing the stiffness and increasing its flexibility and elasticity. If the ligaments are cold and therefore stiff, a normal load could result in a higher than normal stress and in some cases the stress may be high enough to supersede the capacity of the ligament to remain intact, resulting in ligament damage.

Stress relaxation and creep are also additional phenomena that occur in viscoelastic materials. Stress relaxation is the condition when a tissue is deformed to a consistent state and the stress gradually diminishes over time. Although bone is more like an elastic solid, it exhibits some properties of viscoelasticity when compared with other solids, such as metal. The moisture in bone accounts for this response. In fracture repair, bolts and nuts are not used because of the viscoelasticity of bone. When a bolt and nut are used, the material in between is compressed. This material resists the compression by pushing back on the bolt and nut. This interaction of the applied force by the bolt and nut and the reactive force in the material stabilizes the bolt and nut. In a material that retains its stiffness the fixation remains stable; however, if the material is viscoelastic and its properties change, such as with bone, the stiffness decreases as water is squeezed from the bone. As the stiffness decreases, the push back decreases and the bolt and nut become loose and the fixation is lost. This phenomenon is referred to as *stress relaxation* in which the strain is constant but the stress decreases over time.

The converse event, in which the stress is constant and the strain changes over time, is referred to as *creep*. Creep is used frequently in passive ROM exercises in which a muscle tendon unit is stretched with a constant force and held for a period. During this time the strain in the muscle changes without increasing the force. The therapist can then relax the muscle, rest it, and then repeat the exercise, stretching the tissue to a new length while maintaining the pressure, allowing the tissue to stretch more without increasing the force. Understanding how creep functions, and that force does not have to be continuously increased to achieve lengthening, helps to make injuries, such as avulsion fractures, less likely to occur during therapy sessions.

The Body in Motion: Dynamics

When the forces acting on a body become unbalanced and the sum of these forces is not equal to zero (resultant force), the body will move in the same direction as the resultant force. The displacement, in terms of acceleration, is directly proportional to the resultant force by Newton's second law:

$$\vec{F} = \vec{m}\vec{a}$$

where F is the resultant force, m is the mass, and a is the acceleration.

Evaluation of the left side, or the force side, of this equation is the science of kinetics, whereas evaluation of the right side, or the displacement side, is the science of kinematics.

Kinetics

Kinetic gait analysis can be used to evaluate normal weight bearing, identify alterations in weight bearing, aid in the diagnosis of disorders of locomotion, and evaluate treatment effects. The typical format for evaluating the forces involved in the motion of an animal is the use of the force platform. The force platform is a plate instrumented with a series of strain gauges or crystals and mounted into the floor. Because of their symmetry and convenient speed, the walk and trot are the conventional gaits that are used for evaluation of gait and assessment of lameness. The trot is usually the easier gait to obtain data. At a walk, dogs are more likely to be distracted and have a less consistent gait. When the animal steps on the plate, the sensitive strain gauges or crystals generate an electrical signal that is sent to a data processor and a computer. Multiple signals, collected over time, define the stance phase of the gait, from the time the foot strikes the plate until the instant the foot leaves the plate. The right side of the equation, the

Figure 24-9 The dog is trotting over the force platform, which is connected to a computer for the measurement of ground reaction forces.

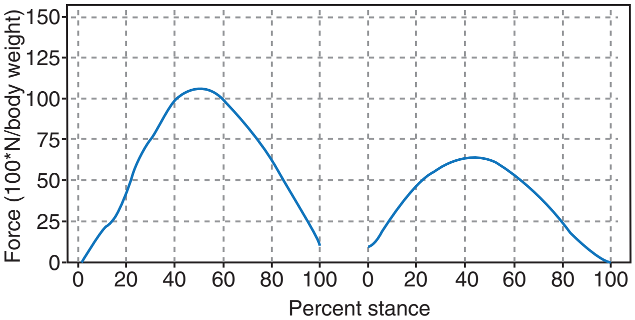


Figure 24-10 Vertical forces of the forelimbs and hindlimbs while trotting. Note that there is only a single smooth curve for the forelimbs and hindlimbs, with the greater forces placed on the forelimb.

displacement side, is kept constant by setting the velocity and acceleration to a well-defined range, for example, trotting at a velocity of 1.7-2 m/s and acceleration of ± 0.4 m/s^2. To generate a readable report, the animal is trotted across the plate such that the front and rear limb, on the same side, fully contact the plate once (Figure 24-9). Often an animal may not lead consistently or may be cautious of the plate by shifting weight, attempting to avoid contact. Therefore a single tracing is not considered reliable by itself. Several identical tracings (usually a minimum of three) are considered more reliable in presenting to the evaluator a more realistic representation of the actual gait routine for that animal.

The force plate is capable of measuring reaction forces in all three dimensions: the vertical (z axis), the horizontal (y axis), and the transverse direction (x axis). Pressure pads or mats have also been developed and have the advantage of being able to evaluate patients too small to participate in force platform analysis and to collect data from multiple limb strikes. This may allow the objective evaluation of small dogs and cats with conditions such as avascular necrosis of the femoral head and medial patella luxation.[2,3] However, they typically only measure relative weight bearing by a limb rather than actual force, and they do not allow the resolution to z, y, and x forces.

Vertical Ground Reaction Forces

Pressure exerted in the vertical direction on the force plate generates an equal but opposite reaction force in the force plate. This vertical GRF represents the animal's weight-bearing performance and is often measured as a percent of body weight. The term *peak force* represents the maximum value of the GRF during the stance phase of gait, whereas *impulse* represents the generation or dissipation of reaction force occurring over time. Normal peak vertical GRF for the front limb at a trot is 110-125% of body weight, and for the rear limb is 65-75% of body weight,[4] and are typically symmetric with a single peak (Figure 24-10). Walking generates a biphasic Z force curve similar to that seen in humans, resulting in a classic m-shaped graph in which the initial peak represents the vertical force component associated with the initial paw strike in the early stance phase and the second peak represents the increase in vertical force at the time of toe off or propulsion (Figure 24-11).[5] Typical peak vertical force (Z_{peak}) measurements for the fore- and rear limbs while walking are lower than those measured while trotting and are typically 60% and 30% of body weight, respectively. Z_{peak} is affected by the gait, velocity, and acceleration of the dog, the dog's body weight, conformation, and musculoskeletal structure.[6-8] In both the walk and the trot, Z_{peak} increases as the forward velocity of the animal increases.[6,9]

Impulse, the area under the force-versus-time curve, is the change in momentum over time. As velocity increases at a trot, the vertical impulse (VI) increases as a result of the increased force.[6,9,10] When comparing only peak vertical GRFs between individuals, how rapidly the limb is loaded and how long the limb bears the load are overlooked. These factors affect the impulse value and are not reflected in the peak force. If the limb bears load longer, for a similar peak load, the impulse will be significantly greater (Figure 24-12). Z_{Peak} of a rear limb is greater while trotting than while walking, but because the stance time is longer at a walk, the impulse is greater while walking than while trotting. Therefore changes in both peak reaction force and impulse should be evaluated when rating gait performance, and repeated evaluations may only be

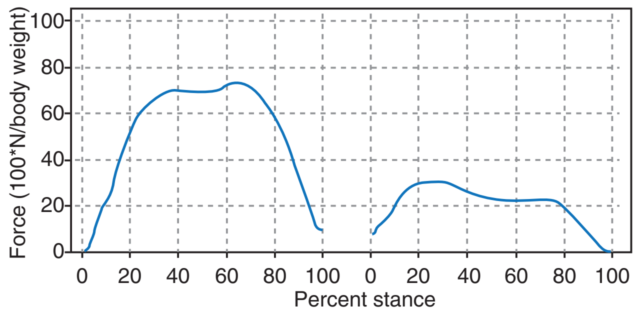
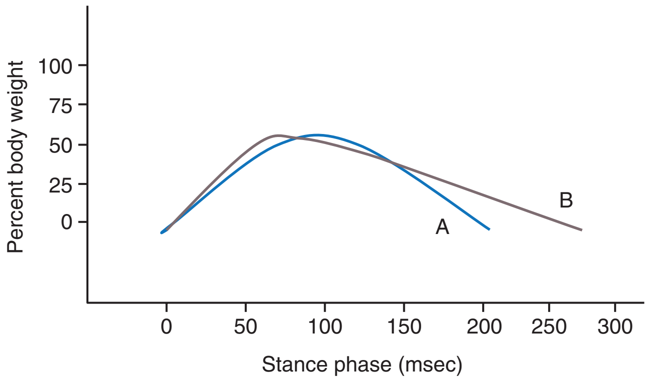


Figure 24-11 Vertical forces of the forelimbs and hindlimbs while walking. Note the biphasic nature of the force curve.



	Peak 100*N/N	Impulse 100*N-sec/N	Time msec	Average rise slope N/msec
A	50.97	5.56	207	1.34
B	52.45	8.49	283	4.77

Figure 24-12 The rate of limb loading and stance time affects the impulse value and is not reflected in the peak force. If the limb bears load longer, for a similar peak load, the impulse is greater.

compared if the dog is ambulating at the same gait, and velocity, and there is no acceleration or deceleration while measuring GRFs.

The animal's COG is particularly important in weight distribution at a stance or while in motion. Traveling uphill tends to shift the balance of forces toward the hindlimbs, and moving downhill shifts the forces to the forelimbs.[8] These factors may affect the forces generated during muscle contraction and be used advantageously during strengthening programs. Further evaluation of kinetic information includes symmetry of the tracing, and comparisons of tracings to the contralateral limb and previous tracings. In cases of lameness, the peak vertical GRF and VI are significantly reduced. Coefficients of variation between trials range from 5% to 9% for Z_{peak} and VI, whereas variation is higher for braking and propulsive forces. Most of the variability is due to dog and trial variation, with little influence on variation by experienced handlers.[11] The symmetry of the tracing can also be altered with lameness. If the shape of the tracing is shifted to the left or right, the animal is quickly loading the limb and peaking early in the stance phase and then slowly unloading the limb throughout the remainder of the stance phase, or slowly loading the limb and quickly unloading it. The clinical significance is not clearly understood regarding this shift, but one study has suggested that a combination of Z_{peak} and the falling slope (unloading the limb) is more valuable in assessing lameness caused by cruciate ligament disease than either measure by itself.[12] Changes in the shape of the tracing could represent lameness because it is reasonable to surmise that a painful limb would not be allowed to increase its load all the way to midstance but would peak early, or the limb would be rapidly unloaded, when the limb pain reaches an intolerant level. However this shift has been seen in contralateral limbs that are compensating for lameness in other limbs. In the compensation mode, loading of the limb peaks early to reduce the load on the painful limb. The difference may be related to the Z_{peak}. In the case of the shift occurring with lameness, the Z_{peak} is less than normal, whereas in the case of compensation, the left shift in the compensating limb is normal or higher than normal.

The stance time may also be assessed. The stance time of the gait cycle is the time when the limb is in contact with the ground and depends largely on the velocity of the subject. As the velocity increases, the stance time decreases.[9,10] In addition, a lame limb typically has a shorter stance time compared with the unaffected limb.[7,8,10,12]

Horizontal Ground Reaction Forces

Pressure exerted in the direction of the animal's motion occurs when the animal places the foot on the plate. This creates a GRF that is equal to the applied pressure but in the opposite direction. This GRF pushes back against the forward motion of the animal resulting in a braking action. As with the vertical force, the intensity of this braking force increases to a peak value then begins to decrease (Figure 24-13). However, there are important differences. Unlike the Z_{peak}, the braking force rises and falls in the first half of the stance phase, such that the horizontal GRF is zero at the midpoint of the stance phase. In addition, the braking force is opposite to the direction of animal motion and is therefore negative and appears below the horizontal axis. After the animal passes the midpoint of the stance phase, the pressure exerted by the foot on the plate causes a GRF that is in the same direction as the animal's motion. This GRF pushes the animal along its motion and is therefore a propulsive force. Similar to braking, the propulsive force increases to a peak value and

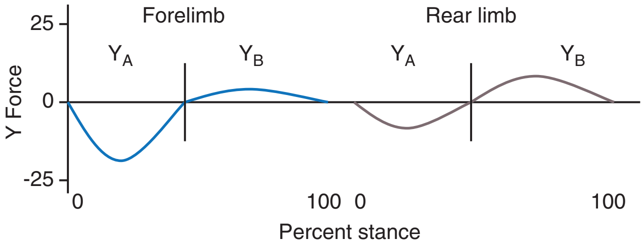


Figure 24-13 Braking and propulsion forces of the forelimbs and hindlimbs while trotting. Note the biphasic nature of the force curve, with the negative deflection representing braking, and the positive deflection representing propulsion.

Figure 24-14 A computerized system may be used to measure weight bearing on each limb while standing. In addition, the distribution of weight to various limbs may be assessed during therapeutic exercises, such as weight shifting.

then decreases. The propulsive force cycles completely within the second half of the stance phase. These forces of braking and propulsion occur along the *y* axis. Normally, the forelimbs have greater braking forces than propulsive forces, and the hindlimbs have greater propulsion as compared with braking.

Transverse Ground Reaction Forces

Some side-to-side pressure is exerted by the foot during stance phase; however, in the case of the quadruped animal moving in a straight line, the intensity is minimal when compared to the *x* and *y* forces. As a result this GRF is seldom used in the evaluation of lameness. The forces may be quite significant, however, in an animal that is turning rapidly, such as negotiating vertical weave poles during an agility competition.

Measurement of Static Weight Bearing

Force plate and motion analysis systems are costly and require a relatively large dedicated workspace. Therefore their use in private practice is limited. Subjective lameness scales have traditionally been used, but have significant limitations. Weight bearing at a stance may be one method to obtain quantitative data regarding the relative amount of force placed on each limb while standing. A computerized system to objectively evaluate weight bearing at a stance in dogs may be useful for this purpose (Figure 24-14). Preliminary studies indicate that there is good correlation between Z_{peak} at a trot determined by force plate analysis and static weight-bearing pressures in dogs with rear-limb lameness.[13] In fact, the difference between the lame and contralateral sound limb may be greater at a stance because there are three other limbs to shift the weight to, and the limbs are in contact for a longer time than at a trot, allowing additional time to shift the weight.

Kinematics

Kinematics is the science of the right side of Newton's fundamental equation of motion. The right side of the equation relates displacement to its cause, the unbalanced force. Typically the equation is stated using acceleration as the unit of displacement. Acceleration is the rate at which the velocity changes with respect to time. Velocity is the rate at which the displacement occurs with respect to time. When viewing an animal in motion, linear motion is easily understood as the animal is seen moving in a straight line from one point to another. However, when the internal mechanics of the animal's muscles and bones are studied, the initiating action is more circular than linear. This is analogous to the motion of an automobile. When viewing the car in motion, the action is linear as the car moves from one point to another. However, when the internal mechanics of the car are studied, the action is circular involving drive shafts, axels, and wheels. In the car, the force is generated by the up-and-down motion of pistons and rods. The action of this force is transmitted to a rotating camshaft to circular gears and rotating drive shafts and ultimately to the wheels, which rotate to propel the car forward in linear motion. In the animal, the force is generated by the up-and-down contraction of actin and myosin elements in the muscle. The contraction of these muscle elements causes rotation of the bones about the joints, propelling the animal forward in linear motion. A car is rated in terms of torque (i.e., its ability to generate force for motion). Similarly, in the analysis of the forces that generate motion in the body, torque or angular moment is important. Because circular motion is important in the generation of body motion, angular displacements must be measured and from these measurements angular momentum can be derived.

Real-time photographic, optoelectric technology is effective in measuring angles during continuous motion.

Optoelectric technology involves the use of infrared cameras positioned for views in three-dimensional space. The animal is prepared with reflective markers secured to the body at predetermined points (Figure 24-15). These points are anatomic features that represent joint anatomy. Infrared light emitted from individual cameras is reflected from these markers back to the camera (Figure 24-16). As the joints move, the cameras record the displacement of the markers. Data points are analyzed with respect to a three-dimensional grid set up and calibrated at the beginning of the test. Such a system can record joint angle excursion throughout a full stride. Typical data acquired include stride length, stance and swing times, joint angles in all planes of motion, and linear and angular joint velocity and acceleration.

A stride includes a single swing and stance phase. The stance time of gait is the period when the foot is in contact with the ground, and the swing phase occurs when the foot is off the ground between stance phases. The swing phase has three distinct movements. The limb initially swings caudally following propulsion. The limb is then pulled cranially, and finally travels briefly caudally toward the ground in preparation for the next stance phase.[14,15] The distance from initial contact of one limb to the point of second contact of the same limb is termed *stride length,* and a *gait cycle* is a series of events that includes one stride for each of the four limbs.[5,14,15]

The joint angle data can be plotted over the period of a single stride (Figure 24-17). These patterns of motion are specific for each joint and gait. Some results have been obtained both for normal gait and for lameness associated with cranial cruciate deficiency[16] and hip dysplasia.

Combining Kinetics and Kinematics

The evaluation of force (kinetics) and of displacement (kinematics) can be combined or connected via Newton's second law of motion. This connection is possible when kinetic and kinematic data are synchronously obtained.

Figure 24-15 The animal is prepared with reflective markers secured to the body over anatomic features that represent joint anatomy.

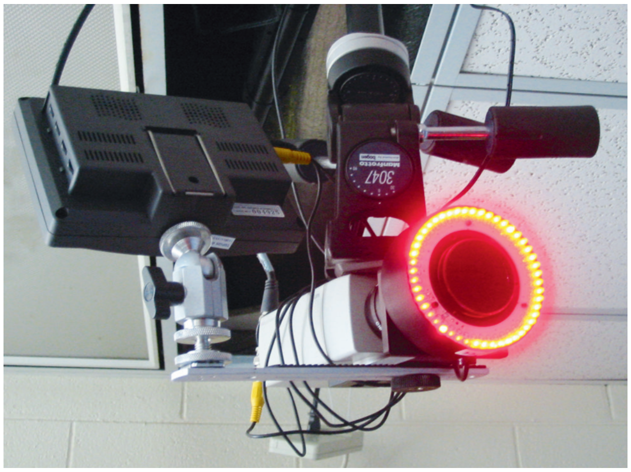


Figure 24-16 Cameras that detect infrared light reflected back to the camera are used to capture the motion of reflective markers as the animal moves through the test area.

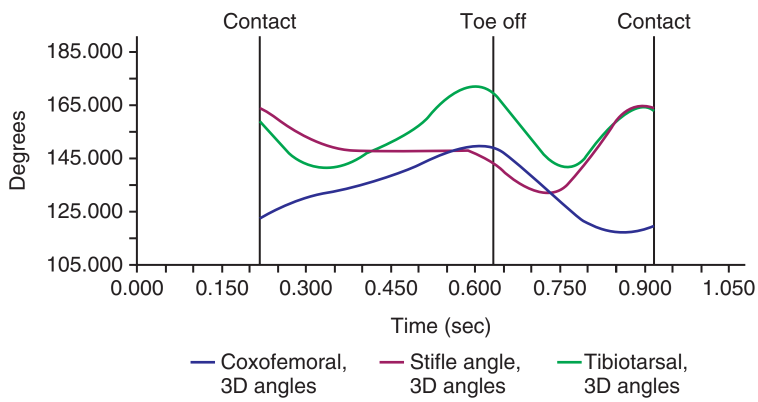


Figure 24-17 Angular data from each joint can be plotted over the period of a single stride. This graph represents motion of the hip, stifle, and hock joints at a walk for a dog.

Such effort requires special technology to coordinate data acquisition. Once this is achieved, the constant in Newton's equation, mass, must be determined for the body segments involved in the analysis. This includes not only the mass value itself, but also the distribution of mass (mass moment of inertia). Therefore Newton's second law may be expressed specifically for linear motion (the study of the entire animal moving in a linear fashion):

$$\vec{F} = m\vec{a}$$

where F is the force (kinetics: evaluation of force), m is mass, and a is acceleration (kinematics: evaluation of linear displacement). Specifically for angular motion (the study of the interior forces of the muscles and the angular movements of the bones):

$$\bar{M} = \vec{I}\bar{\alpha}$$

where $\bar{M}$ is the moment (kinetics: evaluation of moment of force), I is the mass moment of inertia, and $\bar{\alpha}$ is the angular acceleration (kinematics: evaluation of angular displacement).

When kinetics and kinematics are connected mathematically, forces and moments of individual joints can be determined by calculating the values from force plate ground reaction and joint motion values. This approach is referred to as *inverse dynamics.*

Kinetic and Kinematic Gait Analysis Research

Normal Gait

Knowledge of typical weight-bearing patterns in normal dogs and those with orthopedic conditions provides information that may be valuable in the rehabilitation of patients. In a normal standing patient, each forelimb bears approximately 30% of the dog's body weight, and each rear limb bears 20% of body weight. The relative proportion of weight bearing on the fore- and rear limbs is relatively consistent at the walk and trot, which are symmetric gaits. However, because of the relationship of velocity and acceleration to the forces placed on the limbs during the stance phase of gait, significant increases in absolute forces during weight bearing occur with increasing speed at various gaits. For example, a dog may have Z_{peak} measurements of 55% and 30% of body weight at a walk in each fore- and rear limb, respectively. The forces may increase to 100%, 118%, and 125% in the forelimbs and 70%, 80%, and 85% in the rear limbs while trotting at 1.5-1.8, 2.1-2.4, and 2.7-3 meters per second, respectively.

Several studies have evaluated normal GRFs and kinematic motion of dogs at the walk and trot.[4,6,7,9-12,14,17-27] Studies in normal dogs indicate that each joint has a characteristic and consistent pattern of flexion and extension during the walk, but complex joint movements may occur during the swing phase.[14,19,27] Two additional studies described kinematic motion of the joints of normal dogs at a trot.[17,18] The coxofemoral and carpal joints were characterized by a single peak of maximal extension and the femorotibial, tarsal, scapulohumeral, and cubital joints had two peaks of extension, with one peak occurring prior to the stance phase, and a second peak occurring during the stance phase.[17,18] There were insignificant differences between trials and few differences among dogs of similar body type. It was concluded that kinematic gait analysis may provide a reliable description of joint motion in dogs of similar size and conformation. A recent study has evaluated not only the joint angle excursions at a walk, but also defined angular velocity and acceleration rates of those movements. This study indicated that these parameters are consistent and repeatable and helps to further characterize the normal walking gait of hound-type dogs.[27] Additional studies are warranted to determine the sensitivity and specificity of alterations in these parameters as indicators for specific causes of lameness.

Cranial Cruciate Ligament Disease

Following cranial cruciate ligament rupture (CCLR), Z_{peak} may be only 50% of normal at a walk, and dogs may be non–weight bearing at a trot.[28] By 7 months after extracapsular surgical repair, weight bearing is usually equal in both rear limbs at a walk. Experimentally, Z_{peak} at the trot with an extracapsular repair technique was normal by 20 weeks after surgery.[29] Weight bearing in the contralateral rear limb initially increases, likely as a result of redistribution of weight from the affected limb to normal limbs, then returns to normal as weight bearing improves on the affected limb.

Dogs with CCLR have variable degrees of lameness and demonstrate altered movement in the hip, stifle, and hock joints.[16] The stifle joint angle in the cruciate-deficient state was more flexed throughout the stance and early swing phase of stride and failed to fully extend in late stance, when limb propulsion is typically developed. In addition, extension velocity was negligible. The hip and hock joint angles, in contrast, were extended more during the stance phase, perhaps as a result of compensatory changes.[16] Paw velocity and stride length were also significantly reduced in dogs with CCLR walking on a treadmill as compared with the contralateral side, and to normal dogs.[30]

Changes in kinematics were evaluated over a 2-year period in dogs with CCL-deficient stifles.[31] All changes occurred primarily between 6 and 12 months. Peak cranial tibial translation increased by 10 mm and the range of abduction and adduction motion nearly doubled (from 3.3 degrees to 6.1 degrees) after CCL transection and did not improve with time. These changes may overload secondary restraints such as the medial meniscus. This objective information further defines the pathologic gait of CCLR patients.

More sophisticated techniques have been used to quantify net joint moments, joint powers, and joint reaction forces (JRFs) across the hock, stifle, and hip joints in Labrador retrievers with and without CCL disease using inverse dynamics calculations.[32] Kinematic, GRF, and morphometric data were combined in an inverse dynamic approach to compute hock, stifle, and hip net moments, powers, and JRF while trotting. Vertical and braking GRF and JRF were decreased in CCL-deficient limbs. In fact, the braking/propulsion ratio shifted from 50% : 50% in normal limbs to 33% : 66% of the stance phase in CCL-deficient limbs. It is possible that reduced braking may result in decreased cranial tibial thrust. In affected limbs, extensor moments at the hock and hip, flexor moment at the stifle, and power in all three joints were less than normal. Greater joint moment and power of the contralateral limbs compared with normal were also identified. Other compensatory changes included increased time spent with the sound foot on the ground and an increased amount of force applied to all joints of the sound limb, especially during propulsion, compared with normal and CCL-deficient limbs. Propulsion generated around that joint at push-off was three times greater than normal, reflecting increased stifle extensor muscle contraction. Lameness predominantly affected forces during the braking phase and extension during push-off, and a greater contribution of the contralateral limbs to propel the dog forward was identified. These changes may be somewhat protective against damage to the articular cartilage and joint and may be mediated, in part, by the nervous system and a response to pain. In one study using a CCL transection and hindlimb deafferentation model, extension of the hip, knee, and ankle joints of the unstable limb was increased, and braking of the unstable knee was delayed or attenuated.[33] These changes are different than models that have not used deafferentation. The model used in this study results in rapid onset and progression of osteoarthritis with cartilage erosions present by 13 weeks, perhaps because there is no pain sensation or joint proprioception is altered. Stifle hyperextension resulting from limb deafferentation, and knee instability resulting from CCL transection may work synergistically to create increased tibiofemoral motion and changes in the loading of articular surfaces that result in rapid joint breakdown.

Hip Dysplasia

Dogs with hip dysplasia also have reduced weight bearing on the affected limbs. In addition, hindlimb propulsion is reduced in dysplastic limbs. These differences persist 1 month after total hip replacement, and in many instances, weight bearing is initially lower after surgery in the operated limb as compared with presurgical values.[34] By 3 to 6 months after surgery, weight bearing is improved at a trot. In most dogs, hip dysplasia is bilateral, and weight may actually be transferred from the unoperated side to the side with the total hip replacement. In addition, young dogs having triple pelvic osteotomy for treatment of hip dysplasia have weight bearing while walking on the operated side that is similar to preoperative levels by 10 weeks, and have significantly greater weight bearing on the operated side by 28 weeks after surgery as compared with preoperative values.[35]

A recent study reported that dogs with hip dysplasia have complex gait alterations that may not be manifested as overt clinical lameness, making subjective evaluation of lameness difficult. Dynamic flexion and extension angles and angular velocities have been calculated for the coxofemoral, femorotibial, and tarsal joints.[36,37] In the late stance phase of the gait cycle, the hindlimb gait of dogs with hip dysplasia was characterized by a more extended coxofemoral joint, and more flexed femorotibial and tarsal joints throughout stance and early swing phases of the stride. All joints flexed more rapidly in the early swing phase. These changes may be the result of pain in the hip joint early in the stance phase when maximum weightbearing and propulsion would be expected, with the increased extension at the end of the stance phase the result of a relatively passive sudden extension of the hip to complete the gait cycle. Other characteristic changes in the swing phase of the gait cycle, such as an increased stride length and decreased Z_{Peak} measurements, were also evident in dogs with hip dysplasia. Another study determined that dogs with hip dysplasia had a greater degree of coxofemoral joint adduction, greater range of abduction-adduction, and greater lateral pelvic movement, compared with controls.[26] These differences were thought to be indicative of compensation in gait of affected dogs as a result of discomfort or biomechanical effects attributable to hip dysplasia and degenerative joint disease.

Joint motion while walking on a treadmill was evaluated in dogs with borderline radiographic signs of hip dysplasia and normal GRFs. Affected dogs had a later time for maximal flexion of the hip joint and more flexion and ROM of the stifle joint, compared with dogs with no radiographic signs of hip dysplasia. In addition, maximal angular velocity of the stifle and tarsal joints was significantly greater during the swing phase in dogs with borderline hip dysplasia, suggesting that dogs with borderline hip dysplasia have altered joint kinematics.[38]

Elbow Arthritis

The compensatory load redistribution caused by osteoarthritis of the elbow joint was determined by evaluating GRFs in one study.[39] Dogs with osteoarthritis had a compensatory gait pattern that reduced the stress on the affected limb. The load was reduced on the lame limb and increased on the contralateral hindlimb. A symmetry index indicated a weight shift to the contralateral forelimb and diagonal hindlimb, which resulted in a more balanced weight distribution compared with normal dogs. It is possible that forelimb lameness could overload nonaffected extremities and the vertebral spine.

Lumbosacral Disease

Thoracolumbar spinal movement has been evaluated in normal Belgian Malinois dogs and those with subclinical radiographic changes of the lumbosacral region.[40] Markers were placed on the spinal processes of C7, T6, T13, L3, L7, and S3, and dogs were walked on a treadmill. ROM in the transverse and vertical direction and the time of occurrence of the maximal marker position were calculated. Significant kinematic changes were detected between clinically sound dogs with radiographic lumbosacral changes and dogs with no radiographic abnormalities.

Golden Retriever Muscular Dystrophy

Various forms of muscular dystrophy in people cause altered gait kinematics. Stifle and tarsal kinematics of the pathologic gait of dogs with golden retriever muscular dystrophy (GRMD) were evaluated while walking at a self-selected speed and compared with carrier littermates (controls).[41] GRMD dogs walked significantly slower than controls. At the stifle joint, both groups displayed similar ROM, but GRMD dogs walked with the stifle joint relatively more extended. At the hock joint, GRMD dogs displayed less ROM and walked with the joint relatively more extended compared with controls.

Dynamic Gearing

As described previously, the biomechanics of the quadruped animal is similar in a general sense with the mechanics of vehicles such as automobiles and bicycles. These vehicles produce rotary motion, which is transferred to linear movement. The efficient transfer of energy in these machines is modulated by gearing mechanisms. The bicycle has a simple gearing mechanism and can be used as an analogy to the quadruped animal. The cyclist generates motion by pedaling. In this action the cyclist pushes on the pedal, producing a moment of force that turns the sprocket, which turns a second sprocket in the rear wheel. The relationship between the pedal sprocket and the wheel sprocket is described as a *gear ratio.* Sprockets with different diameters can affect the energy transfer, making it more or less difficult for the cyclist to move the bicycle. Gearing mechanisms have been proposed for biologic systems, including people and animals.

In the body, muscles produce forces that function as moments that rotate bones about joint centers of motion. A second set of forces, those that resist gravity, the GRFs, also figure into the production of motion. These muscle moments can be labeled as *internal moments* as opposed to those produced by the GRFs, which can be referred to as *external moments.* Each set of these forces has associated moment arms. With the hip, for example, body weight causes the femur to rotate into flexion while the extensor muscles of the hip cause the femur to rotate into extension. Therefore for every unit of moment causing hip flexion, there is a corresponding moment causing hip extension. Like a bicycle gear, for every unit of turn by the pedal, there is a corresponding turn of the rear wheel. This relationship can be described in terms of a gear ratio. A low ratio means that the turn of the pedal gear generates nearly the same turn in the wheel, whereas a high gear ratio means that for every turn of the pedal gear the wheel gear turns many times. One can conceptualize the interplay of external moments (GRF moment) and internal moments (muscle force moment) as a gearing mechanism with a gear ratio. This ratio is found by dividing the external moment arm by the internal moment arm.

Gear ratios are important especially when acceleration is initiated. In the case of the bicycle the cyclist turns the pedals faster to generate more force for more speed. In the animal it might be assumed that the speed of muscle contraction is increased to generate more velocity. Some have observed that as the speed of muscle contraction increases there is a decrease in the force of contraction; however, it has also been observed that as the speed of contraction increases, the power (rate of work) generated is increased. When the cyclist accelerates the bicycle, there is an upshifting of the gears (i.e., a transition from a low gear to a high gear that reduces the requirement for additional force produced by the cyclist). That is, for a given pedal revolution per minute (a constant force), the bicycle can still accelerate because the gear ratio is increased. It has been theorized and demonstrated that in the dog there is a similar change in gear ratios as the animal accelerates. This is called *dynamic gearing.* For a given velocity of muscle contraction (the optimal velocity), the ratio between the external moment arm and internal moment arms increases, facilitating acceleration. These moment arms change because of the change in the angles of the joints.

In the case of running, dynamic gearing has been demonstrated to occur about the shoulder and stifle joints.[42] Such a phenomenon was not demonstrated when evaluating the gear ratios about the carpus, elbow, hock, and hip. Dynamic gearing facilitates motion, especially acceleration and running, by maximizing the efficiency of muscle contraction and the transfer of energy from one segment to another and eventually to the ground. Alterations in mass and its distribution in limb segments, imbalance in the contraction of muscle groups, disruption of angular motion in the joints, and changes in limb dimensions can potentially influence the relationship between the various joint moments (i.e., the gearing ratios), resulting in suboptimal performance and maybe creating more stress and strain on the elements of the system. For example the effects of patella instability and luxation can be better visualized from the perspective of the efficient or inefficient

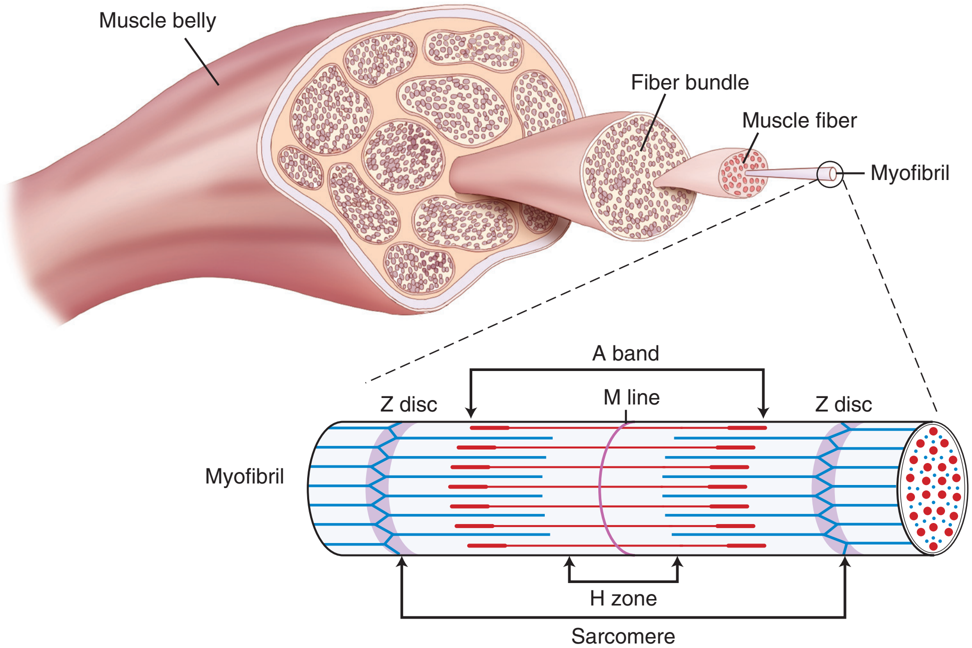


Figure 24-18 Muscle fibers are composed of smaller subunit myofibrils. The sarcomere is the unit of the contractile system of muscles found in the myofibrils and consists of actin and myosin arranged in transverse bands.

transmission of energy across abnormal segments. It has been shown that in the process of running, hock joint excursion is 58 degrees, whereas stifle excursion is 36 degrees; however, the quadriceps is not only responsible for extending the stifle, but it is also partially responsible for extending the hock.[42] This is related to the muscles that cross both joints and to the various moments that these muscles produce. When the moment arm decreases in the stifle because of a dislocated patella, the gear ratios are changed and the quadriceps is less efficient, which could translate to less efficiency in extension of the stifle and hock, resulting in poor performance. Therefore what happens in the stifle has an effect on the hock. Such theories at this stage are assumptions that require more testing by combined kinetic and kinematic measures.

Energetics

Isolating analysis to specific biomechanical phenomena, such as joint moments and GRF, does not reflect the influence of a whole host of variables that are involved in the generation of gait. Another perspective in the study of animal motion is that of energy flow. Throughout motion and gait, energy is transferred from the muscle to bone and then from the bone back to muscle. This transfer of energy depends not only on the forces of weight bearing, and linear and angular motion, but also on the physiologic and metabolic status of the muscles, the integrity of the connective tissue structures, the mobility of the joints, and the neurologic control of gait. Energy flow can be analyzed by assessing the mechanical energy used by the body as work. Work can be assessed mathematically by determining power values for physical activities such as walking, running, jumping, and so on. The core of physical rehabilitation is the use of these activities as a means of therapy. Determining work and power values of the animal's performance can facilitate decisions regarding which activities and their intensity are appropriate for the treatment of specific conditions.

In mechanical systems such as the musculoskeletal system of the body, energy is neither created nor lost but transferred between a stored state (potential energy) and an expended state (kinetic energy). In the body, this transfer physically occurs between muscle and bone, and the result is body movement, such as walking, trotting, running, or swimming.

Biomechanics of Muscle

Muscle is the principal source of the mechanical energy used in the generation of motion. It is important to understand the biomechanics of muscles so that strategies may be designed to restore function in animals with decreased muscle force or movement caused by disease. A brief review of basic muscle contraction is necessary to more completely understand the biomechanics of skeletal muscle. Muscle fibers are composed of smaller subunit myofibrils (Figure 24-18). The sarcomere is the unit of the contractile system of muscles found in the myofibrils, and consists of actin and myosin arranged in transverse bands. The myosin filaments have crossbridges that attach to the actin filaments. During muscle contraction, it is the relationship of the crossbridges to the actin that results in shortening of the myofibrils, similar to an oar that is rowing to advance a boat through the water. This is also known as the *sliding filament theory.*[43] The force of muscle contraction depends on the relative number of crossbridges formed by the interaction of the actin and myosin filaments via the crossbridges.

Muscle not only expends this energy with contraction of myofibrils but also stores and absorbs this energy during the process of weight bearing and gait. The elastic components of muscle (epimysium, perimysium, and endomysium) and tendon store potential energy as the muscle is passively stretched. For example, as the hamstrings contract, the quadriceps is stretched. After contraction in the hamstrings is terminated, the quadriceps releases the stored potential energy and springs back to its resting length. This spring-like function of muscle-tendon units prevents overstretching and damage to the muscle. When an animal stands motionless, the muscles contract in a coordinated and balanced fashion, without the production of motion. Muscle contraction under such conditions is termed *isometric contraction.* In isometric contraction, all the mechanical energy is stored in the muscle as potential energy. Muscle also absorbs energy from impact and dampens vibration, protecting and preserving the moving parts of the skeletal system. Although the function of muscle in expending energy (i.e., kinetic energy in the generation of motion, which is the focus of the physical therapist) is important, the function of muscle in storing energy is equally important. Muscle fibrosis and the loss of elasticity compromise its capacity to store and absorb energy, rendering joint structures and surfaces more vulnerable to additional injury and wear. In treatment plans of osteoarthritis, the focus is naturally on the condition of the articular cartilage, subchondral bone, internal and external ligaments, and the status of the joint capsule; however, the supporting muscles must also be included. The therapist can take advantage of this opportunity to fill the gap in the management of osteoarthritis by focusing attention on the condition of the muscle and applying therapeutics designed to improve muscles' capacity to store, expend, and transfer energy.

The term mechanical energy represents the ability of a mechanical system to do work, such as producing motion.[44] Work represents the flow of mechanical energy between the kinetic and potential states, between the internal parts of the system, and between the system and its environment. The motion of the internal components such as muscle and bone represents internal work, which leads to motion of the body, and represents external work. Work is mathematically calculated by multiplying the force that generates the motion by the distance moved. The rate at which work is performed represents the power of the system. Power, therefore, is the rate of flow of mechanical energy and is mathematically calculated by dividing work by time (W/t), or the rate of work.[44] Power is a sensitive indicator for abnormal gait because it depends on the function of all components of the musculoskeletal system.

Positive Work: Concentric Muscle Contraction

When muscle contracts, it produces a moment that causes a bone to rotate. To calculate the work done, the moment is multiplied by the displacement (angular displacement). The product of angular moment and acceleration is positive when the direction of the angular moment and acceleration is the same. This positive product represents positive work and is associated with concentric muscle contraction, which is defined as the circumstance in which the direction of the motion is the same as the contraction. For example, when a dog jumps a barrier, the quadriceps muscle is producing positive work by extending the stifle joint. The direction of the contraction (i.e., the shortening of the muscle) is in line with the action of extension of the stifle. Co-contraction and positive work are ideal for muscle function and pose less risk for muscle injury as compared with eccentric muscle contraction.

Negative Work: Eccentric Muscle Contraction

Eccentric contraction occurs when the direction of the contraction or shortening is opposite the direction of the motion of the bone to which the muscle is attached. Because the directions are opposite each other, the product of the moment and acceleration is negative, and therefore the work is negative. For example, when a dog is landing on its front limbs after jumping a barrier, the triceps muscle is contracting, trying to extend the elbow in an attempt to cushion the landing. However, the force overpowers the muscle and causes the elbow to flex. The work the muscle does in this circumstance is negative and is more likely to result in muscle injury, such as delayed onset muscle soreness. When formulating and designing physical activities, the therapist should consider the nature of the work the muscles will endure. Positive work scenarios tend to build muscle strength, whereas negative work could inhibit rehabilitation efforts by contributing to muscle soreness.

Zero Work: Isometric Contraction

Isometric contraction is defined as a contraction that is not associated with any movement. Because no displacement takes place, the product of joint moment and displacement is zero, so work is not done. Biomechanics traditionally involves the "inverse solution" to problems, in which displacement (the outcome of applied force) is measured first and the generating force is then calculated. Assessment of the nature of isometric contraction in terms of its causative force is difficult because no displacement occurs except within the muscle itself and at a microscopic level. To understand these forces and energy conditions, direct measurement is required. However, isometric contraction is an important function of muscle and can be used by the therapist to increase strength and elasticity.

Muscle Force Generation

The force generated by the muscle varies with the muscle's length. Muscles generate relatively little tension when a muscle is maintained in a very long or very shortened state. Muscles generate greater force with the muscle at intermediate, or optimal, lengths. For example, the biceps muscle generates the greatest force in humans when the elbow is at an angle of approximately 90 degrees. The tension generated in muscle is a function of the overlap between the actin and myosin filaments. This phenomenon is known as the *sarcomere length-tension relationship.*[45] With a long muscle length (full extension), there is little overlap between actin and myosin filaments, and because there are few cross-bridges, little tension is generated. As muscle length decreases, there is overlap of actin and myosin, and greater tension may be generated. This reaches its maximum effect when the muscle is at its optimal length. As the muscle further shortens, a point is reached at which further decreases in muscle length do not result in greater tension because there is so much actin-myosin overlap that additional cross-bridges cannot be formed. With even further shortening of the muscle, actin filaments begin overlapping with other filaments on the opposite side of the sarcomere, and there is interference with cross-bridge formation, resulting in decreased muscle tension.

The length-tension situation is altered when a muscle is stretched to various lengths without stimulation, or with passive movement. Near the optimal length of muscle, passive tension in the muscle is almost zero. As the muscle is stretched to longer lengths, however, passive tension increases. Therefore passive tension by stretching the muscle can play a role in providing resistive force even in the absence of muscle activity. The phenomenon of passive muscle tension may be due to titin, a large protein within the myofibrils that connects the thick myosin filaments end to end.[46] This protein may also stabilize the myosin lattice so that high muscle forces do not disrupt the sarcomeres.[47]

The effect of passive tension is greater in muscles that cross two joints, such as the hamstring or caudal thigh muscles. In this situation, it is difficult to fully extend the stifle joint while the hip is flexed maximally because the passive tension of the stretched hamstring muscles prevents further stifle extension.

In contrast to the length-tension relationship, the force-velocity relationship seen in muscle contraction does not have a precise anatomically identified basis. The force-velocity relationship establishes that the velocity of muscle contraction is inversely proportional to the load applied to the muscle.[45] For example, as the load applied to a muscle increases, the muscle shortens more slowly. If the external load equals the maximal force of muscle contraction, muscle contraction velocity is zero. If the load is further increased, eccentric muscle contraction and muscle lengthening occur.

The force of muscle contraction is also proportional to the contraction time.[43] The longer the contraction time, the greater the force of contraction. Although maximum force may be developed relatively quickly in the contracting muscle, longer contraction times generate additional forces in the elastic components of the muscle-tendon unit.

The arrangement of the sarcomeres affects muscle contractile properties. Muscle velocity and excursion are proportional to the number of sarcomeres in series, whereas muscle force is proportional to the total cross-sectional area of sarcomeres.[45] Muscles designed for generation of high forces often have muscle fibers slanted at an angle, an arrangement called *pennate muscles.* Various muscle groups appear to be designed for either the production of high forces or velocity and high excursions. The quadriceps muscles seem to be designed more for production of force, whereas the sartorius muscles, with their longer length and lower cross-sectional areas, are more adapted to high excursions. The caudal thigh muscles are also designed for large excursions.

Muscles generate force and transmit this force via tendons to the bones. If the muscles generate adequate force, the bones rotate about joint axes. This activity can be defined as a *torque,* which clinically represents strength. Torque is a mathematical relationship between force and the length of the moment. Torque, or strength, may be changed by changing the magnitude of force (tension-producing capability of muscles), changing the length of the moment (changing the location of the muscle insertion site), or changing the angle between a force, such as a loading force and the magnitude of the moment generated by a muscle. When a joint is fully extended, there is an unfavorable mechanical advantage to the muscle because the moment is relatively small.[45] Much of the force generated by muscles in this situation compress the joint surface rather than rotate the joint. The maximum moment of most joints occurs with the joint flexed to 90 degrees. However, studies of torque generation have demonstrated that the joint angle at which the muscle generates maximal force is not the same angle at which the moment arm is maximum. In reality, both the moment arm and muscle force are constantly changing during normal joint motion, making the measurement of torque very complex. A thorough understanding of the principles regarding generation of force by muscles may allow development of treatments for specific muscle groups that have a rational physiologic basis and that result in measurable improvements. For example, when administering neuromuscular electrical stimulation

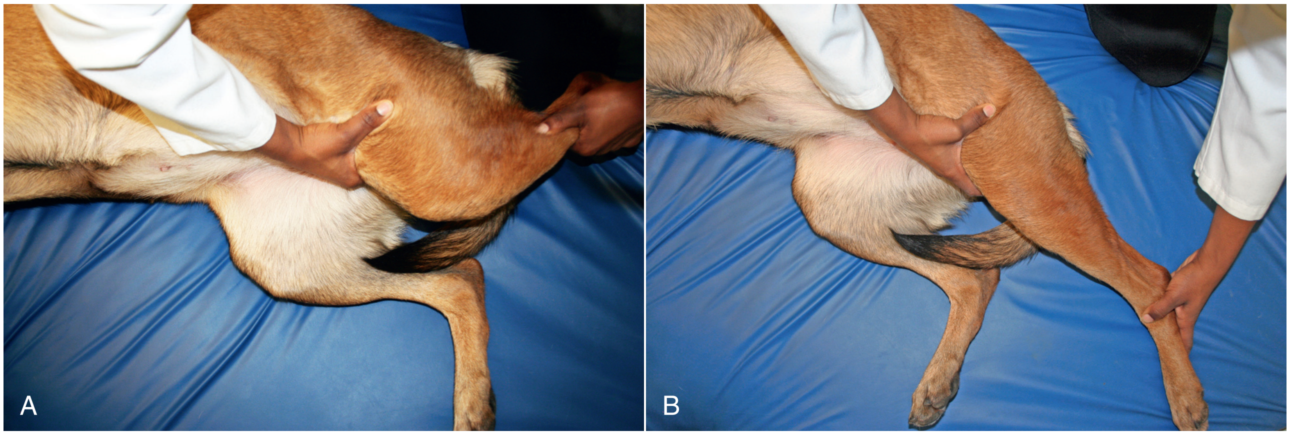


Figure 24-19 If hip joint extension is desired, the contribution of the rectus femoris muscle to resistance may reduce hip extension if the stifle is kept flexed during hip extension (*A*), as compared with extending the stifle (*B*), because of the influence of the rectus femoris muscle on hip extension.

for quadriceps muscle strengthening in a patient with muscle weakness, it may be beneficial to perform the treatment with the stifle in mild flexion as compared with full extension or flexion.

Biomechanics of Exercise Modification

Knowledge of the alterations in body posture or positioning that patients undergo while recovering from surgery or during rehabilitation of chronic conditions helps the therapist to appreciate and take advantage of these factors to improve the effectiveness of therapeutic exercises. In patients with proprioceptive disorders, the size of the base of support affects stability. For example, it is more difficult to balance with the feet close together than it is with the feet spaced wide apart. Similarly, the stability of the ground surface may progress from a static or stable surface, such as the floor, to a more mobile base, such as a balance board or trampoline. The tactile and proprioceptive input may also be altered by standing and walking on various surfaces, such as foam rubber.

Increasing the external load that a limb experiences is important for muscle strengthening. Although it is obvious that increasing the weight that a limb carries (e.g., by using weights or resistive elastic bands) increases the magnitude of resistance, there may also be increased feedback from muscle and joint receptors, enhancing the response. The length of the moment arm undergoing an increased load also affects the resistance that a limb undergoes. Placing the weight or elastic band distally on the limb increases the length of the moment arm and increases the force necessary to move the limb. Therefore if a muscle is relatively weak and a limb is not able to withstand a high load, a leg weight or elastic band should be placed relatively proximal on the limb. As muscle strength improves and additional strengthening is desired, the weight or band may be moved further distally. The speed of the exercise should also be considered. It is usually easier to perform an exercise at a medium rate of speed than a very slow or very rapid rate.

Consideration of limb position is also important when targeting muscles that cross two joints. The tension exerted by a muscle spanning more than one joint depends on the position of the second joint over which it passes, because this determines the total length of the muscle. The semitendinosus and semimembranosus muscles are more effective flexors of the stifle when the hip is also flexed as compared with the hip joint extended. If hip joint extension is desired (with concurrent shortening of the gluteal muscles), the contribution of the rectus femoris muscle (which crosses both the hip and stifle joints) to resistance may be minimized if the stifle is kept extended during hip extension as compared with flexing the stifle (Figure 24-19).

Likewise, hip flexion is limited if the stifle is concurrently extended, but hip flexion increases if the stifle is simultaneously flexed (Figure 24-20). Similarly, hock flexion is greater when the stifle is concurrently flexed, but is limited if the stifle is kept extended (Figure 24-21). These biomechanical considerations are particularly important when planning stretching and ROM exercises.

Sensory facilitation or inhibition may be used to alter muscle responses. For example, compression of a joint may stimulate the joint receptors and facilitate extensor muscle activity and stability around a joint. One method of increasing extensor muscle activity with joint compression is rhythmic stabilization. This activity may be accomplished by placing the dog in a standing position on a compressible surface, such as an air mattress or exercise ball. Weak dogs may be supported with a sling to be certain that they do not collapse. While keeping the dog in a standing position, the dog is gently and rhythmically pushed down (Figure 24-22). As the joints become compressed,

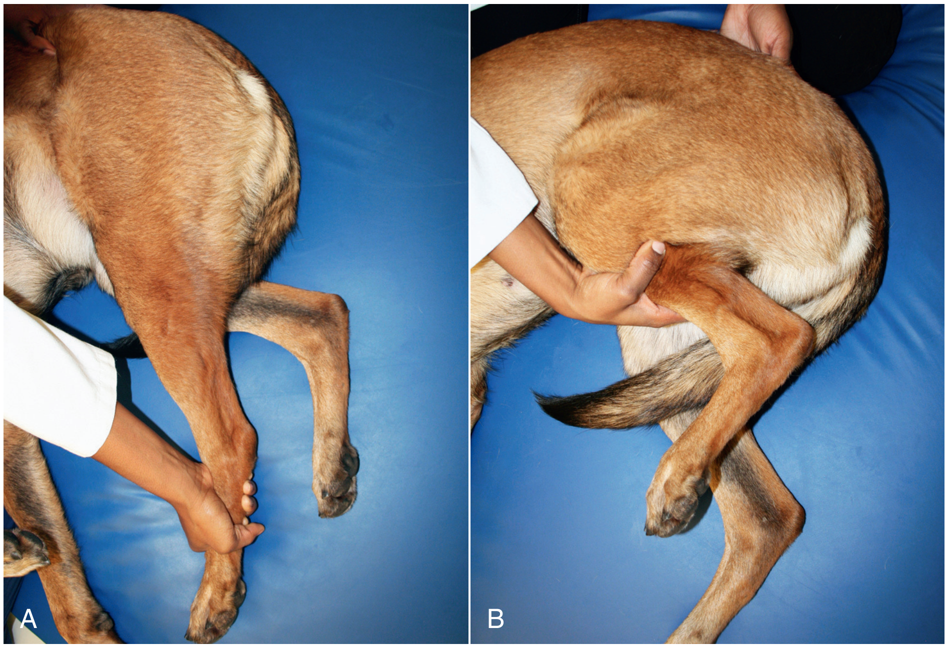


Figure 24-20 Hip flexion is limited if the stifle is concurrently extended (*A*), but hip flexion increases if the stifle is simultaneously flexed (*B*).

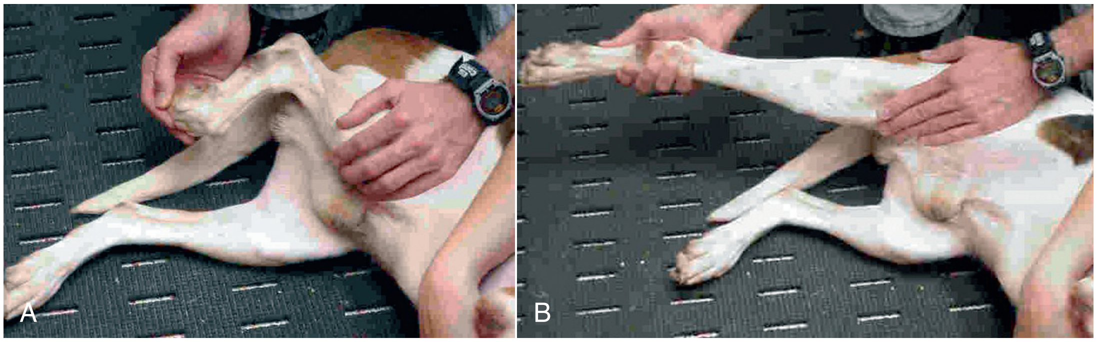


Figure 24-21 Hock flexion is greater when the stifle is concurrently flexed (*A*), but is limited if the stifle is kept extended (*B*).

Figure 24-22 Rhythmic stabilization may be accomplished by supporting the dog in a standing position on an exercise ball and gently pushing the dog down in a rhythmic fashion to stimulate the extensor muscles to contract to maintain body posture.

the extensor muscles responsible for maintaining posture are stimulated to prevent collapse. Conversely, traction separates joint surfaces and is useful if increased ROM is desired.

Biomechanics of Therapeutic Exercises

Walking

Walking is perhaps the most simple therapeutic exercise, but one of the most valuable in the early phases of rehabilitation. Z_{Peak} measurements of the forelimbs and hindlimbs are approximately 54% and 40% of body weight, respectively (Table 24-1). Total motion excursion in the shoulder, elbow, and carpus at a walk is approximately 30, 45, and 90 degrees, respectively (Table 24-2).[14,27] In the rear limb, motion in the hip, stifle, and hock is approximately 35, 40, and 35 degrees, respectively.

Table 24-1 Kinetic Characteristics of Therapeutic Exercises

	Walking Forelimb	Walking Hindlimb	Trotting Forelimb	Trotting Hindlimb	Wheelbarrowing Forelimb	Dancing Forward Hindlimb
Peak vertical force	54%	40%	111%	66%	91%	76%
Vertical impulse	21%	15.2%	20%	12%	17%	16%

Values are a percent of body weight.

Table 24-2 Kinematics of Walking and Trotting in Dogs

Joint	Position	Walk[53]	Walk[14,19]	Trot[54]	Trot[17,18]
Shoulder					
	Extension	148	146	138	137
	Flexion	120	121	119	107
	ROM	28	25	19	30
Elbow					
	Extension	144	143	140	140
	Flexion	94	97	94	88
	ROM	50	46	46	52
Carpus					
	Extension	192	65	199	80
	Flexion	99	−20	84	−20
	ROM	93	85	115	100
Hip					
	Extension	136, 140	132	142	132
	Flexion	103, 103	100	105	100
	ROM	33, 36	32	37	32
Stifle					
	Extension	154, 155	146	152	150
	Flexion	106, 109	111	88	97
	ROM	48, 51	35	63	53
Tarsus					
	Extension	162, 165	158	163	155
	Flexion	125, 123	128	97	120
	ROM	37, 42	30	67	35

ROM, Range of motion.

Trotting

Trotting is frequently used to increase the speed of muscle contraction and the forces placed on the limb, with increased strength of muscle contraction. Z_{Peak} measurements of the forelimbs and hindlimbs at the trot are approximately 115% and 66% of body weight, respectively (see Table 24-1). Increasing the speed to a trot results in little change in shoulder joint motion, increases elbow joint excursion slightly, and increases carpal joint motion approximately 15 to 20 degrees (see Table 24-2).[17] At a trot, hip and hock motion is similar to that at the walk, but motion in the stifle increases to approximately 55 degrees.

Treadmill Walking

Walking on a ground treadmill is a commonly performed exercise for conditioning and to encourage limb use after surgery. One study determined that dogs have an increased stance time and greater stride length when walking on a ground treadmill as compared with walking over ground, but there were no differences in swing time.[48] Furthermore, maximum extension, flexion, and ROM angles are similar for the forelimb and hindlimb joints between ground and treadmill walking (Table 24-3). Maximum joint flexion velocity tended to be lower in dogs walking on ground treadmills, perhaps because of the active assisted nature of

Table 24-3 Joint Motion during Selected Therapeutic Exercises

Joint		Walking	Walking Ground Treadmill	10% Incline Ground Treadmill	Wheelbarrowing	Dancing Forward	Dancing Backward
Shoulder	Ext	148	146†	—	162*	—	—
	Flex	120*	121		131†		
	ROM	28	25†		31*		
Elbow	Ext	144	146*	—	130†	—	—
	Flex	94†	93		80*		
	ROM	50†	53*		50†		
Carpus	Ext	192†	194	—	198*	—	—
	Flex	99*	100		112†		
	ROM	93	94*		86†		
Hip	Ext	136†	138	141	—	140	164*
	Flex	103*	107	103*		120	135†
	ROM	33	31	38*		20†	29
Stifle	Ext	154	156*	153	—	155	129†
	Flex	106	111†	109		110	88*
	ROM	48*	45	44		45	41†
Hock	Ext	162	163	166*	—	163	157†
	Flex	125	128	125		134†	115*
	ROM	37	35	41*		29†	37

Ext, Extension, *Flex,* flexion, *ROM,* range of motion in degrees.
Values represent mean flexion and extension angles, respectively.
For each joint, the most joint extension, flexion, and ROM (*) or least joint extension, flexion, and extension (†) are indicated.

walking on a treadmill. Therefore if increased weightbearing on an affected limb with similar joint angle excursions but less rapid joint motion is desired, walking on a ground treadmill should be considered.

Walking on a treadmill with a 10% incline results in joint motion similar to walking on a level treadmill, probably because dogs can adjust their stride length while propelling the body up an incline to achieve similar joint motion to that obtained with walking over level surfaces.[49] The mean maximum hip extension and total hip ROM were 3 and 7 degrees greater with incline treadmill walking, whereas mean stifle extension was 3 degrees less with incline treadmill walking.

Another study evaluated dogs walking on a treadmill at inclines of 5%, 0%, and –5%.[50] Stance or swing phase durations were not affected by treadmill inclination. When the inclination increased from –5% to 5%, hip joint ROM increased and stifle extension decreased significantly (Table 24-4).

Table 24-4 Joint Motion while Walking on a Treadmill with an Incline at 5%, 0%, or –5%[50]

Joint	Joint Position	Decline	Level	Incline
Hip	Extension	120.9	120.8	121.8
	Flexion	95.8	95	94.5
	ROM	25.1	25.9	27.3
Stifle	Extension	146.2	144.9	143.1
	Flexion	93.7	93.4	91.6
	ROM	52.6	51.5	51.5

ROM, Range of motion.

Incline Ramp Walking

In general, walking up an inclined ramp results in greater ROM of the forelimb joints as compared with ascending stairs or trotting on a level surface.[51] Shoulder extension, ROM, and carpal flexion are greater when walking up a ramp as compared with ascending stairs or trotting over level ground (Table 24-5). Shoulder ROM while ascending the ramp was 66 degrees, compared with 20 degrees while ascending stairs. Elbow flexion and extension are greater when walking up a ramp as compared with trotting over level ground. In another study, incline walking on an 11% slope resulted in increased hip flexion and decreased stifle joint flexion.[52]

Decline Ramp Walking

Walking down a ramp results in less femorotibial flexion and tibiotarsal flexion and extension compared with walking down stairs (see Table 24-5).[53] ROM of the hip, stifle, and hock joints are also less with decline ramp walking as compared with descending stairs. Decline

Table 24-5 Kinematics of Stair and Ramp Ascent and Descent

Joint	Joint Position	Trotting	Stair Ascent	Ramp Ascent	Stair Descent	Ramp Descent
Shoulder	Extension	138	126	152		
	Flexion	119	106	85		
	ROM	19	20	66		
Elbow	Extension	140	149	157		
	Flexion	94	57	46		
	ROM	46	92	111		
Carpus	Extension	199	189	187		
	Flexion	84	68	36		
	ROM	115	121	151		
Hip	Extension	142	147		123	120
	Flexion	105	102		96	97
	ROM	37	45		27	23
Stifle	Extension	152	145		158	157
	Flexion	88	61		62	77
	ROM	63	83		96	80
Tarsus	Extension	163	171		159	152
	Flexion	97	61		61	77
	ROM	67	108		98	76

ROM, range of motion.
Hindlimb trotting and stair ascent from Durant et al.[54]
Stair and ramp decline from Millard et al.[53]
Forelimb trotting, stairs and ramp from Carr et al.[51]

walking at a slope of 11% resulted in decreased hip flexion in another study as compared with overground walking.[52] Therefore ramp descent may be easier than stair descent for dogs with musculoskeletal disease of the pelvic limb and limited motion of pelvic limb joints.

Stair Ascent

ROM of the pelvic limb joints is significantly increased during stair climbing compared with trotting on level ground, with increases of 10, 20, and 40 degrees for the hip, stifle, and hock joints, respectively (see Table 24-5).[54] Extension of the hip and hock joints is increased with stair climbing (approximately 5 and 8 degrees, respectively), whereas stifle extension is decreased by 7 degrees. Flexion is also significantly increased in the stifle and hock by 26 degrees and 35 degrees, respectively.[54] Hip flexion does not contribute to stair ascent, but the stifle and hock joints contribute significantly to raising the limb up to the height of the step.

Another study also found that tarsal flexion, extension, and ROM; stifle flexion and ROM; and hip extension and ROM were greater when ascending stairs as compared with walking on level ground.[55] However, they found that stifle extension increased with stair climbing. The difference may be due to different stair parameters and a convention of joint angle measurement used in people.

Regarding the forelimb joints, there are some differences between stair ascent and trotting.[51] These differences are due to changes in extension and flexion of forelimb joints. Shoulder extension is less with stair climbing as compared with trotting over ground, whereas flexion is greater with stair climbing. Because of these opposite effects, there is no difference in total shoulder ROM. Elbow extension, flexion, and ROM are greater with stair ascent as compared with trotting over ground, and carpal flexion and ROM are greater with stair climbing. Ramp ascent, however gives, greater extension, flexion, and ROM of all forelimb joints as compared with stair ascent.

Stair Descent

Stair descent results in significantly greater stifle flexion and hock flexion and extension compared with walking down a declined surface (see Table 24-5).[53] Significantly greater ROM was also achieved in the coxofemoral, femorotibial, and tibiotarsal joints during stair descent. Another study, using different stair parameters and a convention of joint angle measurement used in people, agreed that tarsal flexion, extension and ROM, stifle flexion and ROM, and hip flexion were greater when descending stairs as compared with walking on level ground.[55]

Wheelbarrowing

Raising the rear limbs off the ground and walking the dog forward is a therapeutic exercise known as *wheelbarrowing* (Figure 24-23). The main objective is to increase the use of the forelimbs and increase weight bearing on them.

Figure 24-23 Wheelbarrowing exercise is performed by supporting the rear limbs of the dog and walking forward on the forelimbs.

Figure 24-24 Dancing exercise is performed by supporting the front limbs of the dog and moving the dog in a forward or backward direction. A muzzle is recommended if the dog is painful or resists the exercise in any way.

However, several specific events occur during wheelbarrowing. The forelimb is a complex structure, with multiple joints continuously undergoing joint angular acceleration and deceleration, with the limb held at changing angles to the ground during the stance phase. During wheelbarrowing, the dog must move its forelimbs to keep from falling and in doing so, the stride length changes as the dog attempts to keep its balance. In one study of hound-type dogs, wheelbarrowing resulted in significantly shorter stride length (0.66 vs. 0.86 m), stance time (0.27 vs. 0.37 s), and swing time (0.22 vs. 0.27 s) as compared with walking at the same speed.[1] Dogs had a mean Z_{Peak} of 91% of body weight, which is intermediate between that expected of walking (approximately 55%) and trotting (approximately 100% to 110%) (see Table 24-1). Mean VI was 17% of body weight, which is similar to walking (approximately 23% of body weight), but greater than trotting (approximately 13% of body weight) because of the shorter stance time while trotting. So although some of the dog's body weight is shifted to the forelimbs, the forces placed on the forelimbs are only intermediate between walking and trotting, likely because of the shorter stride length and stance time. In addition, some of the weight is also transferred to the handler.

Regarding joint kinematics, dogs that wheelbarrowed had significantly more shoulder extension, elbow flexion, and carpal extension, and less shoulder flexion, elbow extension, and carpal flexion as compared with walking (see Table 24-3).[1] These differences are approximately 10 to 15 degrees.

Dancing

Raising the forelimbs off the ground and walking the dog forward and backward is known as *dancing* (Figure 24-24). The main goal is to increase weight bearing on the rear limbs. A comparison of Z_{Peak} measurements and VI of nine hound-type dogs that danced forward and were trotted across a force plate indicated that dancing resulted in greater Z_{Peak} measurements than trotting (75.6% vs. 65.4% body weight) and greater VI (16.1% vs. 12%) (see Table 24-1).[1]

Dancing forward may result in different kinematic characteristics compared with dancing backward or walking over ground. There was significantly less hip flexion, total hip ROM, hock flexion, and hock total ROM with forward dancing compared with walking (see Table 24-3).[1] The stride length (0.56 vs. 0.84 m, respectively) and swing time (0.18 vs. 0.26 s, respectively) were shorter with forward dancing as compared with walking, but the stance times were similar (0.32 s). Dancing backward also differed from walking over ground. Hip extension and stifle flexion were greater with backward dancing as compared with walking, whereas hip flexion, hip ROM, and stifle extension and stifle ROM were less (see Table 24-3). This information may be useful in rehabilitating dogs with various conditions. For example, in the early phases of a therapeutic exercise program for a dog with hip dysplasia that is painful with hip extension, gluteal muscle strengthening may be more comfortable by dancing the dog forward rather than backward.

Cavaletti Rail Walking

Walking over cavaletti rails results in significant increases in flexion of various joints, and the increase is related to the height of the cavaletti rails (Table 24-6).[56] Specifically, the elbow, stifle, and hock have increased joint flexion and

Table 24-6 Joint Motion while Walking over a Cavaletti Rail

		Walking	Low (at Carpus)	Medium	High (mid antebrachium)
Shoulder	Extension	149	152	153	153
	Flexion	115	125	124	124
	ROM	34	26	29	29
Elbow	Extension	147	145	142	143
	Flexion	89	72	60	59
	ROM	58	72	82	86
Carpus	Extension	182	187	187	187
	Flexion	81	86	79	81
	ROM	101	101	108	106
Hip	Extension	140	138	137	138
	Flexion	103	100	97	95
	ROM	36	38	40	43
Stifle	Extension	155	154	153	154
	Flexion	104	83	71	68
	ROM	51	71	82	86
Tarsus	Extension	165	153	154	155
	Flexion	123	84	71	68
	ROM	42	69	84	87

ROM, Range of motion.

subsequent increases in ROM while walking over cavaletti rails as compared with overground walking. There is a mild increase in hip flexion. There are no differences in joint extension, however. The shoulder surprisingly had no changes. The biceps brachii muscle is a flexor of the elbow and extensor of the shoulder, and its dual role may result in no net increased movement of the shoulder. In another study, walking over cavaletti rails at the level of the carpus resulted in increased elbow, carpal, stifle, and tarsal flexion, and increased carpal and stifle joint extension as compared with walking.[52] Stifle and tarsal ROM were also greater walking over cavaletti rails as compared with overground walking.

Sit-to-Stand Exercise

Compared with walking, sit-to-stand exercise results in greater flexion of the tarsus and nearly twice as much flexion of the stifle and hip, while extension of these joints is less[57] (Table 24-7). Others have also found that tarsal and stifle flexion and ROM are increased with sit-to-stand exercise, whereas extension of these joints is less with sit to stand as compared with walking[58] (see Table 24-7). Although the values for hip flexion and extension differed between the two studies, the ROM was similar regarding sit-to-stand and walking exercise.

Jumping

Jumping over obstacles results in significant increases in forces placed on the limbs during landing. Hound-type

Table 24-7 Joint Motion during Walking or Sit-to-Stand Exercise

Joint		Walk	Sit to Stand
Shoulder	Extension	125	119
	Flexion	88	91
	ROM	37	27
Elbow	Extension	146	147
	Flexion	91	109
	ROM	55	37
Carpus	Extension	239	202
	Flexion	128	133
	ROM	111	70
Hip	Extension	147, 140	115, 135
	Flexion	111, 103	49, 104
	ROM	36, 36	66, 31
Stifle	Extension	146, 155	108, 133
	Flexion	111, 104	46, 46
	ROM	35, 51	62, 87
Tarsus	Extension	145, 165	131, 131
	Flexion	111, 123	95, 38
	ROM	34, 42	36, 94

ROM, Range of motion.

Data from Feeney LC, Lin C-F, Marcellin-Little DJ, Tate AR, Queen RM, Yu B: Validation of two-dimensional kinematic analysis of walk and sit-to-stand motions in dogs. Am J Vet Res 2007;68:277-282; and McEachern GL, Millis DL, Headrick J, Hicks DA. Kinematic assessment of sit-to-stand exercise in dogs with chronic extracapsular cranial cruciate repair. Proc 2nd World Veterinary Orthopedic Congress/Veterinary Orthopedic Society, Keystone CO, Feb, 2006.

Table 24-8 Kinetics of Jumping

	Peak Vertical Force	Vertical Impulse	Peak Braking Force	Braking Impulse	Peak Propulsion Force	Propulsion Impulse
HIGH JUMP						
First limb strike	185.8	17.4	–26.0	–0.054	20.9	1.5
Second limb strike	168.1	22.6	–40.2	–2.8	12.2	0.7
LONG JUMP						
First limb strike	192.2	17.3	–29.4	–0.85	17.0	1.0
Second limb strike	155.1	20.5	–41.4	–3.1	9.8	0.5

dogs were jumped across a high jump (61 cm or 24 inches) and a long jump (two 38-cm or 15-inch tall hurdles placed 61 cm or 24 inches apart). Z_{Peak} was significantly higher for the first limb strike than the second limb strike in both high jump and long jump (Table 24-8). VI was lower for the first limb than the second limb for the high jump and long jump because of reduced stance time of the first limb strike. There were no differences in Z_{Peak} or VI between high and long jumping on the first limb strike. However, Z_{Peak} and VI were greater for the high jump than long jump when comparing the second limb strike.

There were also significant differences between braking and propulsion force and impulse for both first and second strike for the high jump when compared with the long jump. There was less peak braking force and braking impulse of the first limb strike as compared with the second limb strike for both high and long jumping. There were no differences in peak braking forces between high and long jumping in either the first or second limb strikes. Braking impulse was greater for the first limb strike with long jumping compared with high jumping, but not with the second limb strike. In addition, there was greater peak propulsion and propulsion impulse in the first limb strike as compared with the second limb strike for high jumping and long jumping. There was greater peak propulsion force and propulsion impulse with the first limb strike for high jumping versus long jumping, and for the second limb strike.

Dogs apparently strike harder with the first limb strike for both high and long jumping, but stay on the second limb strike longer for support, resulting in greater impulse on the second limb. There are minimal differences between high and long jumping regarding the vertical forces placed on the first limb strike, but the second limb strike is slightly greater in dogs with high jumping. Dogs apparently brake primarily with the second limb strike. High jumping results in more vertical drop, perhaps obliterating the need for more active braking because the landing is more vertical. Conversely, there is greater propulsion in the first strike as compared with the second. Dogs that undergo high jumping have greater propulsion than long jumpers for both the first and second limb strikes. This may be a result of a spring-like effect in dogs jumping more vertically, with more potential energy stored in the muscles and soft tissues. As they prepare to move forward, this potential energy may be suddenly released, resulting in greater propulsion. With information about GRFs, trainers may be able to adjust jumping techniques to minimize injury to sporting dogs. More work is needed to study differences in jumping styles and possible predisposition to injury.

The land-based exercises that result in the greatest and least flexion, extension, and ROM are indicated in Table 24-9.

Aquatic Biomechanics and Exercises

One of the principal advantages of exercising in water is the buoyancy that the water exerts, resulting in less force on limbs. Buoyancy is an important force in aquatic therapy because animals in water are lifted upward so that the effect of gravity is partially canceled. Therefore the need for antigravity muscle action is greatly reduced. The amount of unloading is related to the height of the water in relation to the height of the individual. One study evaluated the changes in body weight on ground compared with different water levels while standing.[59] Dogs standing in water to the level of the tarsus, stifle, and greater trochanter weighed 91%, 85%, and 38% of body weight, respectively, as compared with standing on dry ground, indicating that buoyancy can be a significant factor in reducing loads placed on limbs. This may be of particular advantage in animals with lower motor neuron conditions that have difficulty supporting their weight, or in patients with arthritic joints to help reduce the weight-bearing forces on painful joints.

Movement of a body through water is complex and involves forces generated by the individual, and counterforces to the movement by the water.[60] *Thrust* refers to the forces in the direction of velocity of movement, and may be thought of as the force that the body generates to push

Table 24-9 Exercises That Have the Greatest and Least Joint Motion

		Greatest Motion	Least Motion
Shoulder	Extension	Wheelbarrowing	Sit to stand
	Flexion	Ramp ascent	Wheelbarrowing
	ROM	Ramp ascent	Stair ascent
Elbow	Extension	Ramp ascent	Wheelbarrowing
	Flexion	Ramp ascent, stair ascent, cavaletti rail	Sit to stand
	ROM	Ramp ascent, stair ascent, cavaletti rail	Sit to stand
Carpus	Extension	All exercises similar	All exercises similar
	Flexion	Ramp ascent	Sit to stand
	ROM	Ramp ascent	Sit to stand, wheelbarrowing
Hip	Extension	Dancing backward	Sit to stand, ramp descent, ramp ascent, stair descent
	Flexion	Sit to stand	Dancing backward
	ROM	Sit to stand	Dancing forward
Stifle	Extension	Ground treadmill, stair descent, ramp descent, dancing forward	Dancing backward, sit to stand
	Flexion	Sit to stand, stair ascent, stair descent	Dancing forward, ground treadmill, inclined treadmill
	ROM	Stair ascent, stair descent, cavaletti rail	Dancing backward, walking
Tarsus	Extension	Inclined treadmill, stair ascent	Sit to stand
	Flexion	Sit to stand, stair ascent, stair descent	Dancing forward
	ROM	Sit to stand, stair ascent, stair descent	Dancing forward

ROM, Range of motion.

the water backward. This helps to accelerate the body and keep it moving forward.

Drag is a force that is opposite to the velocity of movement and opposes the motion between the body and the fluid.

$$\text{Drag Force} = (\text{Constant}) \times (\text{Area}) \times (\text{Drag Coefficient}) \times (\text{Velocity})^2$$

The constant is determined by the density of water, and area is the cross-sectional area of the body in the direction of the motion. The drag coefficient is affected by the shape of the body and the surface of the body. Of particular importance is the velocity of the body part moving through the water, which creates drag related to an exponential function of the velocity. Therefore if the velocity of the dog moving in water doubles, the amount of drag quadruples. The major resistance to moving through water is the turbulent drag, with formation of eddies and currents behind the body.

Another important source of drag exists at the surface of the water, at the water-air interface. There is high surface tension of the water here, and movement just under the surface tends to create waves. The energy loss with this motion results in an increase in forces that resist movement close to the cube of the velocity rather than the square. Therefore with movement just beneath the surface of the water, nearly eight times as much energy must be expended to double the velocity of the body.

One study determined stifle joint ROM and angular velocities during swimming and walking in healthy dogs, and in dogs that had undergone surgical treatment of a ruptured CCL.[61] For dogs in both groups, swimming resulted in significantly greater overall ROM of the stifle joint than did walking, primarily because of greater joint flexion. However, dogs had significantly less stifle extension while swimming compared with walking.

Two-dimensional kinematic analysis of gait has been used to evaluate joint motion in dogs walking on ground and underwater treadmills. In addition, the effect of different water levels on active joint ROM was evaluated when patients walked underwater.[62] Hip, stifle, hock, shoulder, and elbow flexion are greater in dogs walking in water compared with walking on land (Table 31-3). Joint flexion is generally greatest with water levels at or higher than the joint of interest. Maximum joint extension is similar for underwater and ground treadmill walking. Maximum and minimum joint flexion and extension are related to the level of water. These effects are likely a result of the sudden action of a limb to break the surface tension of the water at different water heights.

Summary

Technology will continue to advance, broadening the options for physical rehabilitation. Understanding how these technologies are applied depends on an understanding of the fundamental concepts of the physics of motion and the transfer of energy. The musculoskeletal system is a machine, based on applied force and motion, with the transfer of energy resulting in work and power. These concepts can be used to predict and evaluate performance of the body and to evaluate the effectiveness and outcomes of therapies.

REFERENCES

1. Millis DL, Schwartz P, Hicks DA et al: Kinematic assessment of selected therapeutic exercises in dogs. *Proc 3rd International Symposium on Physical Therapy and Rehabilitation in Veterinary Medicine,* Raleigh, NC, August, 2004.
2. Besancon MF, Conzemius MG et al: Comparison of vertical forces in normal greyhounds between force platform and pressure walkway measurement systems, *Vet Comp Orthop Traumatology* 16:153-157, 2003.
3. Romans CW, Conzemius MG, Horstman CL et al: Use of pressure platform gait analysis in cats with and without bilateral onychectomy, *Am J Vet Res* 65:1276-1278, 2004.
4. Rumph PF, Steiss JE, West MS: Interday variation in vertical ground reaction force in clinically normal Greyhounds at the trot, *Am J Vet Res* 60:679-683, 1999.
5. DeCamp CE: Kinetic and kinematic gait analysis and the assessment of lameness in the dog, *Vet Clin North Am Small Anim Pract* 27(4):825-840, 1997.
6. Budsberg SC, Verstraete MC, Soutas Little RW: Force plate analysis of the walking gait in healthy dogs, *Am J Vet Res* 48:915-918, 1987.
7. Roush JK, McLaughlin RM Jr: Effects of subject stance time and velocity on ground reaction forces in clinically normal Greyhounds at the walk, *Am J Vet Res* 55:1672-1676, 1994.
8. Millis D, Levine D, Taylor R, editors: *Canine rehabilitation and physical therapy*, St Louis, 2004, WB Saunders.
9. Riggs CM, DeCamp CE, Soutas-Little RW et al: Effects of subject velocity on force plate-measured ground reaction forces in healthy greyhounds at the trot, *Am J Vet Res* 54: 1523-1526, 1993.
10. Renberg WC, Johnston SA, Ye K et al: Comparison of stance time and velocity as control variables in force plate analysis of dogs, *Am J Vet Res* 60:814-819, 1999.
11. Jevens DJ, Hauptman JG, DeCamp CE et al: Contributions to variance in force-plate analysis of gait in dogs, *Am J Vet Res* 54:612-615, 1993.
12. McLaughlin RM Jr, Roush JK: Effects of subject stance time and velocity on ground reaction forces in clinically normal Greyhounds at the trot, *Am J Vet Res* 55:1666-1671, 1994.
13. Hicks DA, Millis D, Arnold GA, Evans MB: Comparison of weightbearing at a stance vs. trotting in dogs with rear limb lameness. *32nd Annual Conference Veterinary Orthopedic Society,* Snowmass, CO, 2005.
14. Hottinger HA, DeCamp CE, Olivier NB et al: Noninvasive kinematic analysis of the walk in healthy large-breed dogs, *Am J Vet Res* 57:381-388, 1996.
15. McLaughlin RM: Kinetic and kinematic gait analysis in dogs, *Vet Clin North Am Small Anim Pract* 31(1):193-201, 2001.
16. DeCamp CE, Riggs CM, Olivier NB et al: Kinematic evaluation of gait in dogs with cranial cruciate ligament rupture, *Am J Vet Res* 57(1):120-126, 1996.
17. DeCamp CE, Soutas-Little RW, Hauptman J et al: Kinematic gait analysis of the trot in healthy Greyhounds, *Am J Vet Res* 54:627-634, 1993.
18. Allen K, DeCamp CE, Braden TD et al: Kinematic gait analysis of the trot in healthy mixed breed dogs, *Vet Comp Orthop Traumatol* 7:148-153, 1994.
19. Hottinger HA, Decamp CE, Olivier NB et al: Kinematic gait analysis of the walk in healthy dogs, *Vet Surg* 23(5):404, 1994.
20. Rumph PF, Lander JE, Kincaid SA et al: Ground reaction force profiles from force platform gait analyses of clinically normal mesomorphic dogs at the trot, *Am J Vet Res* 55:756-761, 1994.
21. Budsberg SC, Verstraete MC, Brown J et al: Vertical loading rates in clinically normal dogs at a trot, *Am J Vet Res* 56:1275-1280, 1995.
22. McLaughlin R Jr, Roush JK: Effects of increasing velocity on braking and propulsion times during force plate gait analysis in Greyhounds, *Am J Vet Res* 56:159-161, 1995.
23. Bertram JEA, Lee DV, Todhunter RJ et al: Multiple force platform analysis of the canine trot: a new approach to assessing basic characteristics of locomotion, *Vet Comp Orthop Traumatol* 10:160-169, 1997.
24. Schaefer SL, DeCamp CE, Hauptman JG et al: Kinematic gait analysis of hind limb symmetry in dogs at the trot, *Am J Vet Res* 59(6):680-685, 1998.
25. Bertram JE, Lee DV, Case HN et al: Comparison of the trotting gaits of Labrador Retrievers and Greyhounds, *Am J Vet Res* 61:832-838, 2000.
26. Poy NSJ, DeCamp CE, Bennett RL et al: Additional kinematic variables to describe differences in the trot between clinically normal dogs and dogs with hip dysplasia, *Am J Vet Res* 61(8):974-978, 2000.
27. Arnold G, Millis D et al: Three dimensional kinematic motion analysis of the dog at a walk. *32nd Annual Conference: Veterinary Orthopedic Society,* Snowmass, CO, 2005.
28. Budsberg SC, Verstraete MC, Soutas-Little RW et al: Force plate analyses before and after stabilization of canine stifles for cruciate injury, *Am J Vet Res* 49:1522-1524, 1988.
29. Jevens DJ, DeCamp CE, Hauptman J et al: Use of force-plate analysis of gait to compare two surgical techniques for treatment of cranial cruciate ligament rupture in dogs, *Am J Vet Res* 57:389-393, 1996.
30. Sanchez-Bustinduy M, De Medeiros MA, Radke H et al: Comparison of Kinematic Variables in Defining Lameness Caused by Naturally Occurring Rupture of the Cranial Cruciate Ligament in Dogs, *Vet Surg* 39:523-530, 2010.
31. Tashman S, Anderst W, Kolowich P et al: Kinematics of the ACL-deficient canine knee during gait: serial changes over two years, *J Orthop Res* 22:931-941, 2004.
32. Ragetly CA, Griffon DJ, Mostafa AA et al: Inverse dynamics analysis of the pelvic limbs in Labrador retrievers with and without cranial cruciate ligament disease, *Vet Surg* 39: 513-522, 2010.

33. Vilensky JA, O'Connor BL, Brandt KD et al: Serial kinematic analysis of the canine hindlimb joints after deafferentation and anterior cruciate ligament transection, *Osteoarthritis Cartilage* 5:173-182, 1997.
34. Budsberg SC, Chambers JN, Vanlue SL et al: Prospective evaluation of ground reaction forces in dogs undergoing unilateral total hip replacement, *Am J Vet Res* 57:1781-1785, 1996.
35. McLaughlin RM, Miller CW, Taves CL et al: Force plate analysis of triple pelvic osteotomy for the treatment of canine hip dysplasia, *Vet Surg* 20:291-297, 1991.
36. Bennett RL, DeCamp CE, Flo GL et al: Kinematic gait analysis in dogs with hip dysplasia, *Am J Vet Res* 57(7):966-971, 1996.
37. Poy NSJ, DeCamp CE, Bennett RL et al: Kinematic gait analysis to describe canine hip dysplasia, *Vet Comp Orthop Traumatol* 10(2):73, 1997.
38. Bockstahler BA, Henninger W, Müller M et al: Influence of borderline hip dysplasia on joint kinematics of clinically sound Belgian Shepherd dogs, *Am J Vet Res* 68:271-276, 2007.
39. Bockstahler BA, Vobornik A, Müller M, Peham C: Compensatory load redistribution in naturally occurring osteoarthritis of the elbow joint and induced weight-bearing lameness of the forelimbs compared with clinically sound dogs, *Vet J* 180:202-212, 2009.
40. Gradner G, Bockstahler B, Peham C et al: Kinematic study of back movement in clinically sound malinois dogs with consideration of the effect of radiographic changes in the lumbosacral junction, *Vet Surg* 36:472-481, 2007.
41. Marsh AP, Eggebeen JD, Kornegay JN et al: Kinematics of gait in golden retriever muscular dystrophy, *Neuromuscul Dis* 20:16-20, 2010.
42. Carrier DR, Gregersen CS, Silverton NA: Dynamic gearing in running dogs, *J Exp Biol* 201:3185-3195, 1998.
43. Pitman MI, Peterson L: Biomechanics of skeletal muscle. In *Basic biomechanics of the musculoskeletal system*, ed 2, Philadelphia, 1989, Lea & Febiger.
44. Winter DA: Biomechanics of human movement with applications to the study of human locomotion, *CRC Critical Reviews In Biomedical Engineering* 9:287-314, 1984.
45. Lieber RL, Bodine-Fowler SC: Skeletal muscle mechanics: implication for rehabilitation, *Phys Ther* 73:844-856, 1993.
46. Funatsu T, Higuchi H, Ishiwata S: Elastic filaments in skeletal muscle revealed by selective removal of thin filaments with plasma gelsolin, *J Cell Biol* 110:53-62, 1990.
47. Horowitz R, Podolsky RJ: The positional stability of thick filaments in activated skeletal muscle depends on sarcomere length: evidence for the role of titin filaments, *J Cell Biol* 105:2217-2223, 1987.
48. Schwartz P, Millis D, Hicks DA, Evans MB: A kinematic comparison of over ground vs. treadmill walking in dogs. *31st Annual Conference, Veterinary Orthopedic Society,* Big Sky, MT, 2004.
49. Gassel AG, Millis DL, Schwartz P et al: Kinematic gait analysis of the pelvic limb; comparison of overground versus incline walking in 10 dogs, *Proc Vet Orthop Soc* 32nd Annual Conference for Veterinary Orthopedic Society, Snowman, Colorado, March 5-12, 2005.
50. Lauer S, Hillman RB, Li L et al: Effects of treadmill inclination on electromyographic activity and hind limb kinematics in healthy hounds at a walk, *Am J Vet Res* 70:658-664, 2009.
51. Carr J, Millis DL, Weng HY: Canine physical rehabilitation: range of motion of the forelimb during stair and ramp ascent, *J Sm Anim Pract* In Press, 2013.
52. Holler PJ, Brazda V, Dal-Bianco B et al: Kinematic motion analysis of the joints of the forelimbs and hind limbs of dogs during walking exercise regimens, *Am J Vet Res* 71:734-740, 2010.
53. Millard RP, Headrick JF, Millis DL: Kinematic analysis of the pelvic limbs of healthy dogs during stair and decline slope walking, *J Small Anim Pract* 51(8):419-422, 2010.
54. Durant AM, Millis DL, Headrick JF: Kinematics of stair ascent in healthy dogs, *Vet Comp Orthop Traumatol* 24:99-105, 2011.
55. Richards J, Holler P, Bockstahler B et al: A comparison of human and canine kinematics during level walking, stair ascent, and stair descent, *Wiener TierarztIiche Monatsschrift* 97:92-100, 2010.
56. Headrick JH, Hicks DA, McEachern GL et al: Kinematics of walking over cavaletti rails compared to overground walking in dogs. Proc 2nd World Veterinary Orthopedic Congress/Veterinary Orthopedic Society, Keystone CO, Feb 2006.
57. Feeney LC, Lin C-F, Marcellin-Little DJ et al: Validation of two-dimensional kinematic analysis of walk and sit-to-stand motions in dogs, *Am J Vet Res* 68:277-282, 2007.
58. McEachern GL, Millis DL, Headrick J et al: Kinematic assessment of sit-to-stand exercise in dogs with chronic extracapsular cranial cruciate repair. *Proc 2nd World Veterinary Orthopedic Congress/Veterinary Orthopedic Society,* Keystone CO, February 2006.
59. Levine D, Marcellin-Little DJ, Millis DL et al: Effects of partial immersion on vertical ground reaction forces and weight distribution in dogs, *Am J Vet Res* 71:1413-1416, 2009.
60. Northrip JW, Logan GA, McKinney WC: Fluid mechanics. In *Introduction to biomechanic analysis of sport*, ed 2, Dubuque, IA, 1979, Wm C Brown Company.
61. Marsolais GS, McLean S, Derrick T et al: Kinematic analysis of the hind limb during swimming and walking in healthy dogs and dogs with surgically corrected cranial cruciate ligament rupture, *J Am Vet Med Assoc* 222:739-743, 2003.
62. Jackson AM, Stevens M, Barnett S: Joint kinematics during underwater treadmill activity. *2nd International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine,* Knoxville, TN, 2002.

BIBLIOGRAPHY

Clayton HM: Instrumentation and techniques in locomotion and lameness, *Vet Clin North Am Equine Pract* 12:337-350, 1996.

Hamill J, Knutzen K: *Biomechanical basis of human movement*, ed 2, Philadelphia, 2003, Lippincott Williams & Wilkins.

Low J, Reed A: *Basic biomechanics explained*, Boston, 1996, Butterworth-Heinemann.

Meriam JL, Kraige LG: *Engineering Mechanics Dynamics Vol 2*, Hoboken, NJ, 1986, John Wiley and Sons.

Ozkaya N, Nordin M: *Fundamentals of biomechanics equilibrium, motion, and deformation*, ed 2, New York, 1991, Springer-Verlag.

Rasch PJ, Burke RK: *Kinesiology and applied anatomy: the science of human movement*, ed 6, Philadelphia, 1978, Lea and Febiger.

Renberg WC: Evaluation of the lame patient, *Vet Clin North Am Small Anim Pract* 31:1-16, 2001.

Straface SF, Newbold PJ, Nade S: The effects of direction and velocity of movement, and intra-articular fluid volume and intra-articular pressure, *Vet Comp Orthop Trauma* 3:113-121, 1988.

Technologies PP: *Discovering peak motus: user's manual*, ed 8, Centennial, CO, 2004.

25 Range-of-Motion and Stretching Exercises

Darryl L. Millis and David Levine

Range of motion (ROM) and stretching exercises are very important to achieve improved motion of joints after acute injury or surgery, or in patients afflicted with chronic conditions. They are also important to help increase flexibility, prevent adhesions between soft tissues and bones, remodel periarticular fibrosis, and improve muscle and other soft tissue extensibility to help prevent further injury to joints, muscles, tendons, and ligaments.

As with many types of rehabilitation activities, it may be beneficial to administer analgesic medication, such as a nonsteroidal antiinflammatory medication, 30 to 60 minutes before beginning any rehabilitation exercises to allow peak absorption of the medication and take advantage of its maximal analgesic effect.

Range of Motion

The full motion that a joint may be moved through is termed the *ROM*. The structure of the joint and the volume, integrity, character, and flexibility of the soft tissues that surround the joint affect joint motion.[1] ROM is commonly measured with a goniometer, and each joint has characteristic angles, such as flexion and extension of the stifle, and flexion, extension, abduction, adduction, and internal and external rotation of the hip (Appendix 3).

Muscles also have a ROM, which is termed the *functional excursion of muscles*.[1] This is the distance that a muscle is able to shorten after it has been maximally elongated. The functional excursion may be affected by the joint it crosses, particularly if joint motion is restricted. Some muscles, such as the biceps brachii and the rectus femoris muscles, cross two joints. In general, muscles that cross multiple joints may lose functional excursion easier than those that cross only one joint.

ROM exercises are useful to diminish the effects of disuse and immobilization.[2] To maintain ROM, the joints and muscles must be periodically moved through their available ranges. Movement may be passive, active assisted, or active. In each situation, a load is produced on the soft tissues that helps to maintain the articular cartilage, muscles, ligaments, and tendons in a healthy state. It is generally appropriate to initiate passive ROM as soon after injury or surgery as possible provided there are no contraindications. Progression through active assisted and active ROM follows, and when appropriate, some resistance to motion may be introduced for strengthening.

Passive Range of Motion

Passive ROM is motion of a joint that is performed without muscle contraction within the available ROM, using an external force to move the joint. In animals, passive ROM is performed by the therapist. Additional force applied at the end of the available ROM is defined as stretching. Passive ROM and stretching can be performed in conjunction with each other to help maintain and improve joint ROM.

The benefits of continuous passive ROM immediately after joint surgery were initially described by Salter and colleagues[3] and include decreased pain and improved rate of recovery. The scar tissue deposited after an injury or surgery is laid down in a random fashion, and ROM helps this scar tissue to align along the lines of stress that the tissue normally undergoes. This results in a stronger scar and may help prevent future injury. However, it is critical that the therapist maintain a ROM that is comfortable to the patient and not injure tissues by exceeding their limits.

Tissues limiting passive ROM may be normal or pathologic and include the joint capsule, periarticular soft tissues, muscles, ligaments, tendons, skin, and scar tissue. For example, open wounds over joints that are allowed to heal secondarily by contraction and epithelialization may result in limitations to passive ROM. Surgical incisions may result in adhesions and fibrosis between skin, subcutaneous tissues, fascia, muscles, and bone, limiting the ability of tissues to glide over one another. Musculotendinous tissue may also be relatively shortened as a result of spasm or contracture. Any restriction of motion may result in resistance to joint movement and pain.

Passive ROM is used whenever a patient is unable to move joints on its own, or if active motion across a joint may be deleterious to the patient, such as with a tenuous articular fracture repair. Passive ROM is sometimes used to help relax an anxious patient. The most common indications for passive ROM exercises are immediately after surgery (before active weight bearing) to help prevent joint contracture and soft tissue adaptive shortening, maintain mobility between soft tissue layers, reduce pain, enhance blood and lymphatic flow, and improve synovial fluid production and diffusion.[2]

Another indication for passive ROM is the prevention of joint contracture during healing and recovery in paralyzed patients. Ideally, exercises should be performed two to six times per day to help maintain normal joint mobility. However, if the patient does not regain functional neuromuscular control (active motion), it is likely that joints will undergo some degree of contracture despite aggressive efforts. Although passive ROM may help slow muscle atrophy, it will not prevent muscle atrophy, increase strength, improve endurance, or be as effective in improving vascular and lymphatic flow as active ROM techniques.

Proper technique for performing passive ROM is important. The patient should be relaxed and comfortable. If active muscle contraction is not desired, it is especially important to be gentle and not create pain or discomfort. The bones proximal and distal to the joint should be supported to avoid excessive varus and valgus stresses on the joint. The therapist should gently hold the limb and avoid painful areas, such as incisions and wounds. The location of the therapist's hands on the limb is also important; the closer the hands are to the joint, the less torque will be produced at the joint. This may decrease pain and the risk of patient injury. The motion should be smooth, slow, and steady with motion generally occurring by moving the distal limb with the proximal limb held steady. The patient should be continually monitored for any discomfort, and the technique altered if necessary to enhance comfort.

Active Assisted Range of Motion

Active assisted ROM occurs when the therapist guides joint motion and the patient's muscle activity assists joint motion to some degree. In animal patients, the amount of muscle activity provided by the patient is difficult to control. In reality, because it is difficult to avoid muscle activation in patients that are not paralyzed, most ROM exercises in small animal patients involve a degree of active assisted ROM. In anxious patients or in those with upper motor neuron conditions, muscle contraction of antagonist muscle groups may oppose joint ROM exercises.

Active assisted ROM exercises are most useful for those patients that are weak or recovering from lower motor neuron conditions. In addition to assisted ROM with the patient in lateral recumbency, exercises may be performed with the patient supported by a sling, with the therapist assisting limb movement and joint ROM during ambulation. This may be performed while the patient is slowly walked over ground, on a ground treadmill, or on an underwater treadmill. Another form of active assisted ROM exercise is performed while swimming. Patients with limited ability to ambulate on the ground may benefit from controlled swimming, during which the therapist assists movement of the animal's limbs and joints while in the water with the patient. The buoyancy of the water helps to support the weight of the limbs while the therapist concentrates on assisting the limb through a normal cycle of movement.

Active assisted ROM helps to combat the negative effects of immobilization on limbs, similar to those achieved with passive ROM. Some degree of active muscle contraction allows muscle strengthening and strengthening of bone at muscle origin and insertion sites. In addition, some neuromuscular reeducation, and proprioceptive and gait training may be achieved.

Active Range of Motion

Active ROM is the motion of a joint that may be achieved by active muscle contraction. In addition to increasing strength, coordination between muscle groups is necessary because the guidance of assisting the patient through ROM is no longer provided. The active ROM may be performed during a regular gait cycle, in which the excursion of joint motion is relatively limited, or under special conditions designed to expand motion and encourage more complete use of the full available ROM. Examples of these activities include swimming; walking in water, tall grass, snow, or sand; climbing stairs; crawling through a tunnel; and negotiating cavaletti rails.

As the patient improves ROM of a joint, it is helpful to continue to perform passive ROM and stretching to achieve as complete ROM as possible, and then perform active ROM through this increased motion to emphasize more complete use of the limb. Greater strength is required for patients to perform active ROM, and some of the special conditions require more muscle strength than normal ambulation during walking or trotting. Therefore for those exercises, a transition between active assisted and active ROM may be necessary. Active ROM exercises may be a prelude to other strengthening activities. Owners may also be involved with active ROM exercises in helping with a home care program for their pet.

Precautions and Contraindications to Range of Motion

All forms of ROM exercises are contraindicated when motion may result in further injury or instability. Such examples include unstable fractures near joints and unstable ligament or tendon injuries. Communication between the surgeon and therapist is critical to be certain that appropriate ROM exercises are performed and that the limits of the range are not exceeded, which may result in damage to the tissues. In most cases, early passive ROM is felt to be beneficial if the therapist stays within a ROM that is reasonable for the patient and condition being treated. In

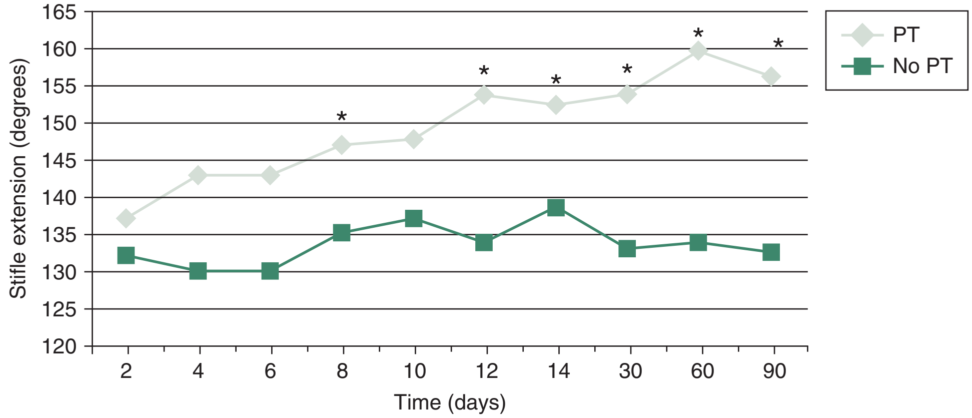


Figure 25-1 Effect of physical rehabilitation on stifle extension. Note the return of normal stifle extension by 2 weeks after surgery, in those dogs receiving rehabilitation and the loss of complete stifle extension in some dogs not receiving postsurgical rehabilitation. *$P < 0.05$.

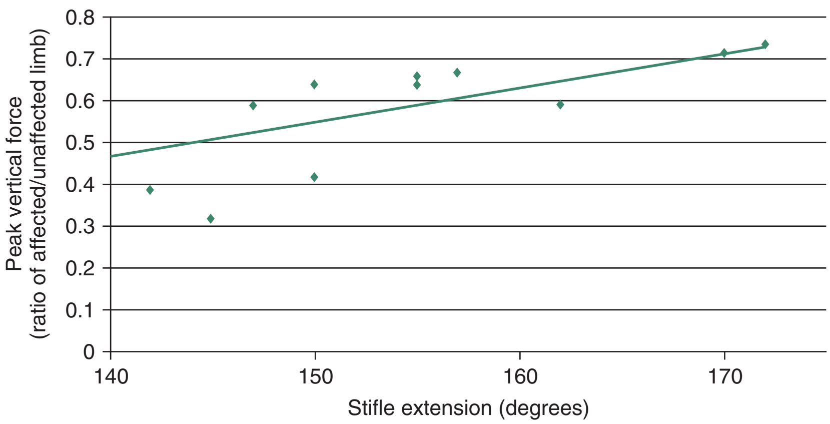


Figure 25-2 Correlation between stifle extension and peak vertical force. Relationship of comfortable angle of stifle extension to peak vertical force in dogs 10 weeks after cranial cruciate ligament transection and stifle stabilization. Dogs with normal stifle extension have greater weight bearing on the limb at a trot than those with limitation to stifle extension. Although a cause-and-effect relationship cannot be definitively identified from these data, there is a significant relationship. $r = 0.78$; $P < 0.001$.

addition, the therapist should perform the exercises at a reasonable speed that is not painful for the patient.

Range of Motion Studies

In some canine conditions, active ROM is a prerequisite to a successful outcome. For example, it is critical that puppies with distal femoral physeal fractures have early passive and active ROM exercises to avoid fracture disease and tiedown of the quadriceps muscle group.[4] Similarly, dogs with condylar fractures of the distal humerus must begin active ROM to prevent permanent joint stiffness.

In one study of dogs following surgery for cranial cruciate ligament rupture, patients not receiving early passive and active ROM exercises had reduced stifle extension (Figure 25-1).[5] This loss of motion appeared to be permanent in some dogs if extension was not achieved within 2 weeks. In contrast, those dogs receiving appropriate rehabilitation had return of near-normal stifle extension. There also appears to be an association of stifle extension with weight bearing, with improved weight bearing in dogs with near-normal stifle extension (Figure 25-2). However, the cause and effect of this relationship are unknown.

The association of stifle ROM with function following surgery for cranial cruciate ligament disease in clinical patients has been further evaluated.[6] In 412 dogs treated with tibial plateau leveling osteotomy (TPLO) for treatment of cranial cruciate ligament rupture, dogs were determined to have no loss of extension or flexion (n = 322), less than 10 degrees loss of extension or flexion (n = 78), or 10 degrees or more loss of extension or flexion (n = 12). Loss of extension or flexion greater than 10 degrees was associated with significantly greater lameness in comparison with no loss, or loss of extension or flexion less than 10 degrees. In addition, osteoarthritis of the stifle joint was significantly correlated with loss of extension, and loss of extension greater than 10 degrees was less tolerable and less amenable to physical rehabilitation than flexion loss. The authors concluded that loss of extension or flexion should be assessed in dogs with persistent lameness after TPLO so that early intervention can occur.

An explanation for the restricted ROM in clinical cases may include physical restrictions, such as joint capsule fibrosis, soft tissue approximation, or bony impingement that may occur with osteophytes as a result of joint disease.

Another possible explanation is an increase in intraarticular pressure with joint motion, resulting in stretch of the joint capsule and creation of pain. One study evaluated the intraarticular pressure and volume relationships in the knee joint of dogs.[7] In general, infusion of fluid into the joint did not result in intraarticular pressure changes between 125-110 degrees of stifle motion. However, increasing flexion from 110 to 50 degrees resulted in an increase in pressure, which was greater with increased intraarticular volume. Increasing the intraarticular volume also resulted in decreased total ROM of the joint.

The functional and structural consequences of remobilization of the shoulder joint after 12 weeks of immobilization were studied in 10 beagle dogs.[8] It was found that after 12 weeks of immobilization, the passive ROM was markedly impaired, intraarticular pressure increased during movement, and the filling volume of the joint cavity was reduced. On histologic examination, the capsule showed hyperplasia of the synovial lining and vascular proliferation in the wall, but there was no increase of collagen in the capsular wall. Both the functional and structural changes were unaltered after 4 weeks of remobilization, but after 8 weeks they began to reverse, and they returned to normal after 12 weeks, indicating that functional and structural changes after immobilization of an uninjured shoulder joint are potentially reversible. The timing and degree of improvement may not be similar in patients with pathologic injury to the shoulder.

Another study of dogs evaluated the effects of joint mobilization treatment on carpal joints of dogs.[9] The right carpal joints of 12 dogs were immobilized for 6 weeks, which resulted in joint hypomobility. The treated group received mobilization therapy daily for 4 weeks following the immobilization period. In control and treated groups, passive ROM, peak angles of extension and flexion of the carpal joint, and the amount of time required in the gait cycle to reach these peak points were evaluated cinematographically during gait before immobilization and once weekly for 4 weeks after immobilization. The treated group had improved passive ROM and motion during gait.

Continuous passive ROM has been assessed in several studies to evaluate its use in the management of cartilage defects. Salter and colleagues[10] demonstrated that young rabbits with a full-thickness defect in the articular cartilage managed with continuous passive motion had healing of 52% of the defects with hyaline-type cartilage. Similar results were obtained in adult rabbits, in which 44% of the defects were repaired with hyaline cartilage. Passive motion may also help to reduce cartilage destruction following infection of a joint.[11,12] The optimal daily dose of passive ROM and the total duration of treatment are unknown. However, one study of rabbits found that at least 8 hours of passive motion per day was necessary to achieve the level of healing obtained by Salter.[13] The effect of active motion on cartilage healing is unknown in animals because of the difficulty in achieving consistent limb use after cartilage injury.

Continuous passive motion has also been recommended to prevent contracture of periarticular tissues and to restore joint function. Immobilization results in shortening of fibrous tissues and loss of motion, depending on the position of immobilization. There is a loss of collagen mass after immobilization, and there is collagen cross-linking of periarticular connective tissues, which may lead to increased stiffness. Passive motion may reduce these changes and prevent joint stiffness. In one study, 16 hours or more of continuous motion prevented increased joint stiffness following articular cartilage damage.[14] Longer durations of passive motion may be necessary to help maintain normal quantities and turnover of collagen and glycosaminoglycans (GAGs), as well as to help maintain the normal alignment of collagen fibers and prevent shortening and cross-linking.

It has been suggested that continuous passive ROM may be somewhat antiinflammatory in patients with inflammatory joint disease. In one study, the effects of continuous passive ROM on early inflammatory events were evaluated in meniscal fibrocartilage of rabbits with antigen-induced arthritis.[15] Rabbit knees injected with bovine serum albumin and Freund's complete adjuvant were treated with 24 or 48 hours of continuous passive ROM or were immobilized. Within 24 hours, immobilized knees had marked degradation of GAGs in meniscal fibrocartilage. Inflammatory mediators, including matrix metalloproteinase 1, cyclooxygenase 2, and interleukin (IL) 1b, increased within 24 hours and continued to increase. Knees undergoing continuous passive ROM had reduced GAG degradation and inflammatory mediators during treatment, and increased synthesis of the antiinflammatory cytokine IL-10. These results suggest that continuous passive ROM suppresses the deleterious effects of inflammatory arthritis more than immobilization and provide information regarding the involved molecular events.

The duration of daily passive motion treatment has varying effects. For example, relatively short daily passive motion may reduce stiffness immediately after treatment, but there may be greater long-term stiffness.[14] In fact, shorter durations of passive motion may be detrimental.[14,16] One study indicated more stiffness with 10 to 30 minutes of passive motion per day than if the joints were immobilized,[14] whereas another indicated increased stiffness with less than 8 hours of daily passive motion.[16] Although somewhat surprising, these results suggest that there may be increased trauma associated with breakdown of soft adhesions that form when motion is not occurring, and that very frequent motion is necessary to keep the tissues mobile.

Both continuous motion and as little as 4 hours of daily passive motion help to preserve muscle mass in rabbits.[14,17]

In fact, 5% to 15% greater limb muscle mass may occur in rabbits with passive motion as compared with those that are immobilized.

The effects of passive motion on bone loss are less clear. It is well documented that immobilization results in a decrease in bone mineral content; however, 16 or 24 hours of passive motion resulted in the loss of greater bone mineral than immobilization.[14] It was suggested by these authors that the increased joint stiffness that resulted from less than 16 hours of motion may have increased the bending stresses and load on the bones, whereas greater duration of passive motion maintained normal joint stiffness and prevented the additional loading on the bones.

In people, the use of active and passive knee motion in the immediate postoperative period and a treatment plan for early postoperative limitations in knee motion has proven highly effective in restoring motion after anterior cruciate ligament reconstruction.[18] In one study of 207 knees, 91% regained full ROM. The remaining patients did not regain motion as rapidly and were placed in an early postoperative phased treatment program of serial extension casts, early gentle manipulation under anesthesia, or arthroscopic treatment of intraarticular adhesions and scar tissue. Most of these patients regained full ROM. Those patients who failed to follow the rehabilitation program had permanent and significant limitation of motion. The incidence of postoperative motion problems was related to the extent of the surgical procedure. Patients who had more extensive surgery, such as a concurrent meniscus repair or a medial collateral ligament repair, had a higher incidence of restricted motion.

Clinical Application of Range of Motion

Passive Range of Motion

Passive ROM treatment should be administered in a quiet and comfortable area, away from distractions, such as loud noises, other pets, and other people who are not involved with the treatment. This will allow the patient to be calm, relaxed, and more receptive to the treatment.

It is recommended that patients have a muzzle applied for the initial treatments, or if they are painful, resistant to treatment, or overly anxious. The patient should be placed in lateral recumbency with the affected limb up (Figure 25-3). Help may be required to restrain the animal and to help keep it quiet and relaxed. In all forms of ROM activities, therapists should be comfortable and use proper body mechanics to avoid injury to themselves.

After the animal has relaxed, gently place both hands on the injured limb and begin a very gentle massage to help further relax the patient. Massage for 2 to 3 minutes, and then gently move one hand to the portion of the limb above the joint. Place the other hand on a portion of the limb below the affected joint. Be certain that the entire limb is supported to avoid any undue valgus, varus, or rotational stresses to the involved joint.

Figure 25-3 Before performing range-of-motion or stretching exercises, the patient should be placed in lateral recumbency with the affected limb up.

After the hands are in the correct position and the limb is supported, begin by slowly and gently flexing the treated joint (Figures 25-4 to 25-7). In general, the other joints of the limb should be allowed to remain in a neutral position (a position as if the animal were standing). Try not to move the other joints while working on the affected joint because some joints may be restricted by the position of the joints above or below the target joint. In some situations, the position of one joint may greatly affect the motion in another joint. In these cases, the other joints should be placed in a position that will allow as complete a ROM as possible to the target joint. For example, maximal hock flexion cannot be obtained while the stifle is maintained in an extended position. In this case, placing the stifle in a flexed position allows more complete flexion of the hock. Slowly continue to flex the joint until the patient shows initial signs of discomfort, such as tensing the limb, moving, turning the head toward the therapist, or trying to pull away, but do not cause undue discomfort, indicated by behaviors such as vocalizing or attempting to bite.

With the hands maintained in the same positions, slowly extend the joint. Again, try to keep the other joints in a neutral position and minimize any movement of the other joints. Slowly continue to extend the joint until the patient shows initial signs of discomfort.

Alternatively, a number of joints may be simultaneously placed through a ROM, a technique sometimes referred to as *ROM through functional patterns* or bicycling movements. This form of ROM exercise may be appropriate as an animal nears active use of a limb. Flexing and extending all of the joints of a limb in a pattern that mimics an exaggerated normal gait pattern may also be beneficial for neuromuscular reeducation.

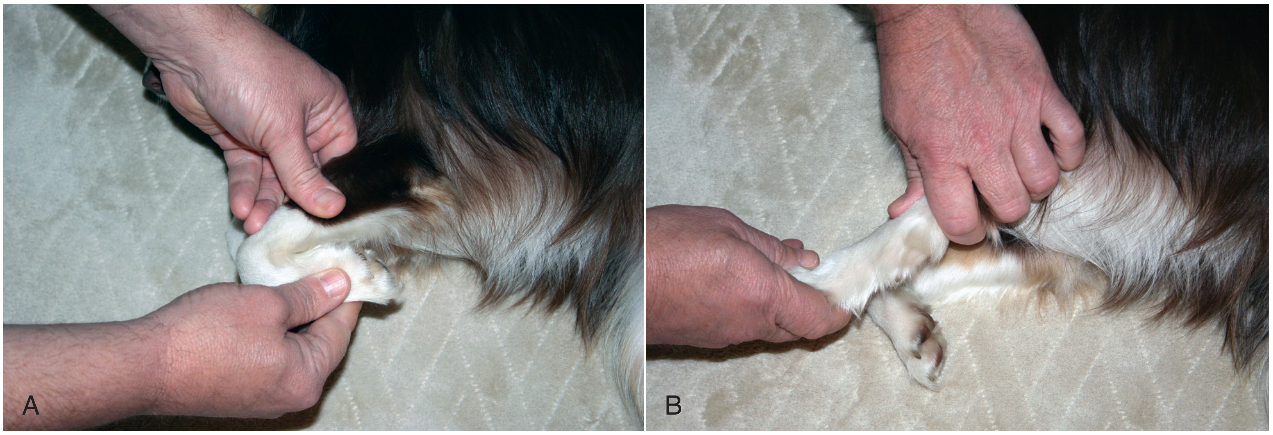


Figure 25-4 **A,** Carpal flexion. While supporting the radius and ulna in one hand and the foot in the other, the carpus is gently flexed. **B,** Carpal extension. While supporting the radius and ulna in one hand and the foot in the other, the carpus is gently extended.

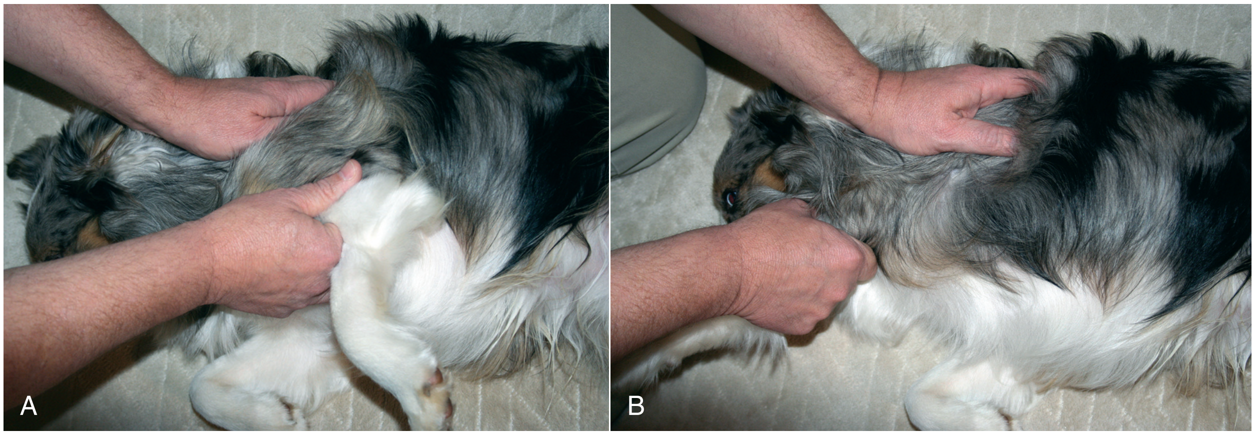


Figure 25-5 **A,** Shoulder flexion. While supporting the distal scapula in one hand and the humerus in the other, the shoulder is gently flexed. **B,** Shoulder extension. While supporting the distal scapula in one hand and the humerus in the other, the shoulder is gently extended.

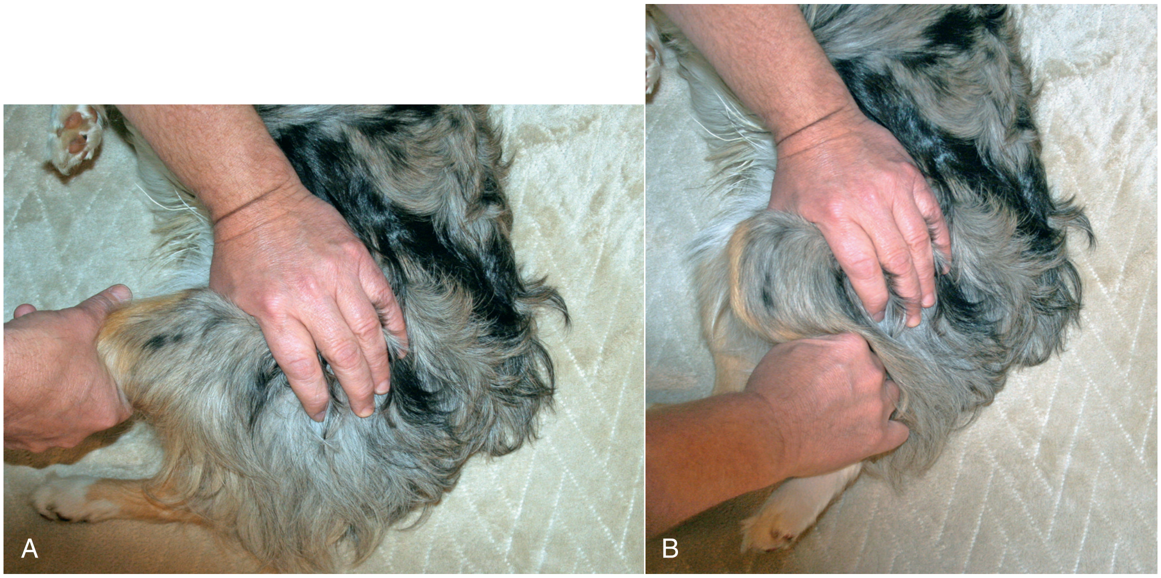


Figure 25-6 **A,** Stifle flexion. While supporting the distal femur in one hand and the tibia in the other, the stifle is gently flexed. **B,** Stifle extension. While supporting the distal femur in one hand and the tibia in the other, the stifle is gently extended.

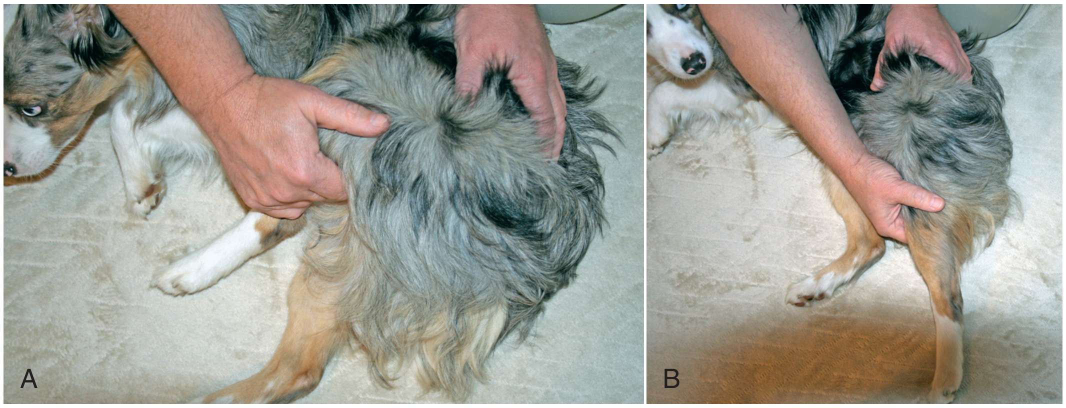


Figure 25-7 **A,** Hip flexion. While supporting the femur in one hand and the pelvis in the other, the hip is gently flexed. **B,** Hip extension. While supporting the femur in one hand and the pelvis in the other, the hip is gently extended.

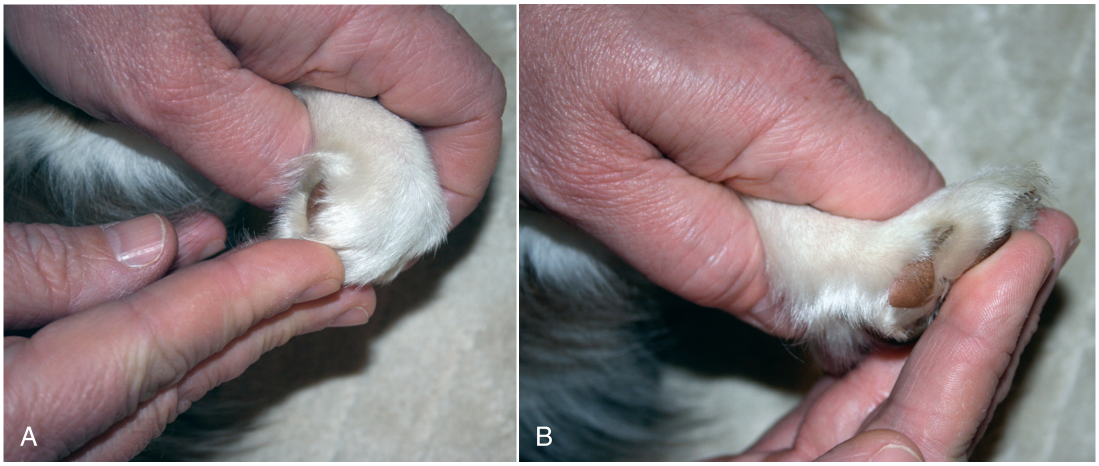


Figure 25-8 **A,** Digit flexion. While supporting the carpus in one hand and the digits in the other, the digits are gently flexed. **B,** Digit extension. While supporting the carpus in one hand and the digits in the other, the digits are gently extended.

The number of ROM repetitions and the frequency of the treatments depend on the condition treated. In general, for most routine postoperative conditions, 15 to 20 repetitions performed two to four times per day are likely adequate. As the ROM returns to normal, the frequency may be reduced. For more challenging conditions, such as articular fractures, physeal fractures in young dogs, joints with contractures, or joints that have been immobilized for a time, an increase in the number and frequency of repetitions may be necessary.

It is also important to maintain normal ROM in the other joints of the affected limb. After completing ROM of the affected joint, keep the limb in a neutral position and slowly move the hands distally to the digits (or if already working on the digits, move proximally). Be certain to keep the injured joint supported. Performing ROM of the digits requires additional consideration because it is difficult to passively flex and extend a single joint at a time because of the proximity of the other joints of each digit. In addition, it is more efficient to perform ROM in all joints simultaneously (Figure 25-8). Place the fingers or palm on the dorsal surface of the digits and slowly flex the digits until the point of initial discomfort. Then place the fingers or palm on the pads of the patient's foot and slowly extend the toes until the point of initial discomfort. After ROM is performed on the digits, ROM exercises are performed on the other joints.

The ROM session may be ended with a gentle massage to the injured limb. Gently and slowly massage for approximately 5 minutes. This helps to maintain a relaxed state. An ice pack may be applied to the injured joint after the ROM and massage.

Active Assisted Range of Motion

Active assisted ROM is the next step in the progression of joint motion during rehabilitation. In most situations, veterinary patients will have some muscle tension during ROM activities, and there is an equally likely chance that an agonist or antagonist muscle group, or likely a combination of both, will contract with joint movement. Patients with neurologic conditions may benefit from assisted movement of limbs while in lateral recumbency, or while standing and generating a gait pattern.

Active assisted ROM may be performed while the patient is ambulating on a ground treadmill or during swimming activity, during which the therapist helps move the limb at the appropriate phase of the gait cycle. Similarly, placing the animal in an ambulatory sling and slowly moving the animal with the sling while assisting the animal to advance the limb are appropriate active assisted ROM activities. Another technique involves the use of an elastic band placed around a limb or the foot. As the animal advances the limb, tension on the elastic band helps to advance the limb.

Active Range of Motion

When the animal is able to ambulate and move the affected limb, active ROM exercises may be initiated. Dogs only use a small amount of their available joint ROM during normal walking and trotting, so the joints do not go through a complete normal ROM during these activities. If joint restriction is present, patients may benefit by performing activities that encourage a more complete ROM. Swimming or walking in water results in greater flexion of joints. Decreased joint extension may occur with swimming, but walking in water maintains relatively normal active joint extension while increasing joint flexion, resulting in greater overall joint ROM. Other activities that may be performed include walking in snow, sand, or tall grass and crawling through a play tunnel. Climbing stairs may increase joint excursion, while also increasing strength. Walking over cavaletti rails is an excellent method of achieving normal limb extension for walking, while increasing joint flexion of certain joints, especially the elbow, stifle, and tarsus as the patient negotiates the rails. In addition, the rails may be raised or lowered to encourage increased or decreased flexion of these joints, based on the needs of the individual patient.

Stretching

Stretching techniques are often performed in conjunction with ROM exercises to improve flexibility of the joints and extensibility of periarticular tissues, muscles, and tendons.[2] Conditions that result in adaptive shortening of tissues, including immobilization, reduced mobility, injury, and fibrosis of periarticular tissues, or neurologic conditions, may respond favorably to stretching.[1] Muscle weakness may occur with adaptive shortening of tissues because peak muscle tension cannot occur, and joint ROM may be decreased as a result of periarticular fibrosis.

Flexibility refers to the ability of tissues, particularly muscle, to relax and respond to an elongation force. Overstretching refers to the elongation of tissues beyond normal limits to allow greater joint ROM than normal. Overstretching may be detrimental if there is inadequate soft tissue support to maintain joint stability and prevent injury. Tightness implies mild shortening of a musculotendinous unit, with no specific pathologic cause. Two-joint muscles, such as the rectus femoris and gastrocnemius muscles, are especially susceptible to tightness. The unit may be lengthened over time with stretching exercises.

Contracture occurs when muscles or other soft tissues that span a joint shorten and limit ROM. Contractures may be defined by the types of soft tissues involved.[1] Myostatic contractures have a musculotendinous junction that has adapted to a shortened position. There is no pathologic condition, but the joint has significant loss of motion. Scar tissue adhesions between normal tissues, such as muscles and fascia, tie down the tissues and prevent normal gliding motion between the tissues, resulting in reduced ROM. Most scar tissue adhesions may be prevented with appropriate stretching, exercise, and massage in the early tissue-healing phase following injury. If the adhesion continues to mature without tissue modification, a fibrotic adhesion may form. These types of adhesions dramatically reduce joint ROM, and the resultant contracture is very difficult to treat and restore normal function to the joint. When there is an excessive amount of fibrous tissue, cartilage, or bone that forms in the area, there may be permanent loss of soft tissue extensibility. Damage to the central nervous system with upper motor neuron signs may result in hypertonic muscles and a pseudomyostatic contracture. In this situation, the muscle seems to be in a chronic state of contraction, resulting in reduced ROM. Appropriate ROM, stretching, and massage may help to improve this form of contracture.

The various tissues that are involved, such as muscle, ligaments, tendons, joint capsule, and skin, respond to stretching differently. Interestingly, in a study of rat denervated gastrocnemius muscle-tendon units, the muscle tissue accounted for 95% of the total increase in length of the unit with loading, whereas the tendon contributed only 5% to the length.[19] Therefore the therapist must consider which tissues are most likely limiting mobility and choose the appropriate techniques to help achieve more normal function. If stretching occurs within the normal physiologic limits of the tissue, elastic deformation occurs in which the tissues return to their normal resting length after the stretch is completed. If the tissue is stretched beyond its limits, permanent deformation of the tissues remains after the stretch is completed, resulting in an increased resting length; this is called *plastic deformation.*

Stretching is a general term that is used to indicate maneuvers to elongate tissues shortened as a result of a pathologic condition and to increase flexibility and joint motion in normal and abnormal tissues. Stretching differs from ROM exercises in that stretching takes tissues beyond the normal ROM. Passive ROM only takes structures through the available range. Passive stretching techniques are most commonly used in veterinary medicine because of the inability to verbally communicate instructions to the patient to selectively relax and contract various muscle groups. It may be possible, in some situations, to apply neuromuscular electrical stimulation to various muscle groups to mimic active muscle contraction of an individual muscle for certain active stretching techniques, but this requires coordination of efforts, and usually two or three people must be involved in the treatment session.

Several stretching techniques have been described, including static stretching, prolonged mechanical passive stretching, ballistic stretching, and proprioceptive neuromuscular facilitation stretching. The acute effect of stretching is an immediate elongation of the elastic component of the musculotendinous unit. A chronic stretch may be applied with the limb immobilized with the target tissues in an elongated position. Chronic stretching over time may result in sarcomeres being added to lengthen muscle tissue at the musculotendinous junction.[20] The number of sarcomeres in series increases, resulting in an increased resting muscle length. This form of stretching is used to lengthen shortened tissue and to decrease muscle stiffness.[2] Immobilization of a muscle in a shortened position results in a decrease in sarcomeres and increased connective tissue; stretching may correct this process.

The muscle spindle monitors the velocity and degree of muscle stretch. When a muscle is stretched very quickly, the muscle spindle is activated, which stimulates the primary afferent fibers and results in increased muscle tension. This is termed the *monosynaptic stretch response.* Caution should therefore be used to avoid stretching too rapidly, which may cause stimulation of the muscle spindle and an increase in muscle contraction instead of relaxation.[1] The Golgi tendon organ senses muscle tension during a contraction and provides a protective function to the muscle by inhibiting contraction of the muscle when tension is applied slowly. When excessive muscle tension develops during contraction, the Golgi tendon organ responds, and the muscle relaxes.

Noncontractile soft tissues, such as tendons and ligaments, are composed primarily of collagen. Initial stretch applied to these tissues results in straightening of the collagen fibers with relatively little force. As the tissue is stretched to the end of a ROM, the tissue remains elastic, and release of the stretch results in return of the collagen to its normal resting position. If the stress on the tissues continues, the bonds between the collagen fibrils and fibers may be damaged, resulting in plastic deformation. Caution must be used during stretching to avoid tissue damage in most instances. Low-magnitude forces placed on the tissues over time allow collagen fibers to rearrange, perhaps resulting in a safer mode of tissue elongation. In people, 15 to 20 minutes of low-intensity sustained stretch, repeated for 5 days, has increased the length of the hamstring muscles.[21] Cyclic submaximal loading may also increase resting tissue length through tissue remodeling. Hamstring length was increased in people following a series of 10-second stretches and 8-second rest periods for 15 minutes per day, for 5 days.[21]

Clinical experience suggests that in uncomplicated cases of muscle tightness, 5 to 10 degrees of ROM may be gained per week, and 3 to 5 degrees may be gained in cases with joint capsule tightness. Other factors may alter these expected responses, including the animal's weight-bearing status, other medical conditions such as Cushing's disease, the extent of trauma to the tissues, and the type of surgical procedure performed.

There is no consensus regarding which stretching technique is most effective, and some controversy exists regarding the efficacy of various stretching techniques for certain conditions. However, if benefits are achieved, the effects of a consistent stretching program may be maintained over time, even if stretching is temporarily discontinued. A stretching program performed three to five times per week may result in measurable increases in flexibility in patients with stiffness. After flexibility improves, the frequency of stretching may be reduced. In fact, stretching too frequently may be harmful in some situations. The natural cycle of stress and adaptive remodeling may be interrupted if adequate time is not allowed for healing between intense stretching sessions. Additional caution is required in aged animals because collagen becomes less elastic, and less stretch is tolerated before plastic deformation.

Static Stretching

Static stretching involves placing the joint or joints in a position so that the muscles and connective tissues are stretched while held in a static position with the tissues at their greatest length. Stretches should be held for 15 to 30 seconds. There is probably little advantage to stretching for longer periods in healthy patients.[22] One advantage of this form of stretching is that less force is applied, reducing the possibility of iatrogenic damage to the tissues. There is also less stimulation of the IA and II spindle afferent fibers, which increase the muscle's resistance to stretch.[2] It is important to be certain that the stretch is gentle, comfortable, and well tolerated by the patient. A low-intensity stretch applied for a longer duration may be more comfortable for some patients. A slow static stretch is less likely to induce contraction in the muscle being stretched.[1] The gains in ROM are relatively transient with static stretching, and the increases are generally attributed to elastic changes in actin-myosin overlap.[1]

When performing this form of stretching, it is important to have the patient as relaxed as possible to allow the maximal stretching of the muscle and tissues with as little resistance as possible. It is important to properly support